

FFNN Final Report

August 18, 2025

```
[1]: # Import necessary libraries
import pandas as pd
import numpy as np
from scipy.sparse import csr_matrix, hstack
from sklearn.preprocessing import OneHotEncoder, MinMaxScaler
from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.layers import LeakyReLU
from tensorflow.keras.layers import ReLU
from tensorflow.keras.optimizers import Adam, RMSprop, SGD
from tensorflow.keras.callbacks import EarlyStopping
import matplotlib.pyplot as plt
import pyhere
```

```
[2]: train_path = pyhere.here('Data', 'train_data.xlsx')
test_path = pyhere.here('Data', 'test_data.xlsx')
train_data = pd.read_excel(train_path)
test_data = pd.read_excel(test_path)
# Use dropna() to remove any rows with missing values
train_data = train_data.dropna()
test_data = test_data.dropna()
print(train_data.head())
```

	CountryName	CountryCode	year	HDI_Index	\
0	Antigua and Barbuda	ATG	2022	0.848	
9	Antigua and Barbuda	ATG	2008	0.830	
10	Antigua and Barbuda	ATG	2009	0.827	
11	Antigua and Barbuda	ATG	2010	0.828	
12	Antigua and Barbuda	ATG	2011	0.828	

	HDI_Category	FoodIndex	ElectricAccess	PopDensity	\
0	Very high human development	89.95	100.0	211.000000	
9	Very high human development	145.98	100.0	188.436364	
10	Very high human development	141.40	98.5	191.302273	
11	Very high human development	102.49	98.7	193.909091	
12	Very high human development	104.99	94.6	196.209091	

	PopInSlums	WaterStress	...	IncomeGroup	YearlyChgInternet	\
0	2.646340	8.461538	...	High income	0.997844	
9	2.646317	9.230769	...	High income	3.164091	
10	2.646311	9.038462	...	High income	3.164091	
11	2.646300	8.846154	...	High income	3.831893	
12	2.646300	8.653846	...	High income	3.831893	

	YearlyChgHDI	Lag1_InternetUsersPct	Lag2_InternetUsersPct	\
0	0.005	8.247411	7.147796	
9	0.003	5.957120	5.079504	
10	-0.003	6.242918	5.336653	
11	0.001	6.508026	5.571669	
12	0.000	6.815164	5.788459	

	Lag1_YearlyChgInternet	Lag2_YearlyChgInternet	\
0	0.993070	0.908060	
9	3.128215	2.432146	
10	3.128215	3.116179	
11	3.128215	3.116179	
12	3.782704	3.116179	

	Cumulative3yrChg_InternetUsersPct	GovernanceScore	HealthSpend
0	3.2	0.180504	853.834335
9	11.0	0.904398	523.176632
10	12.0	0.929469	472.044125
11	13.0	0.914123	573.175838
12	14.0	0.898586	562.880747

[5 rows x 30 columns]

```
[3]: train_target = train_data['InternetUsersPct']
test_target = test_data['InternetUsersPct']

train_predictors = train_data.drop('InternetUsersPct', axis=1)
test_predictors = test_data.drop('InternetUsersPct', axis=1)
```

```
[4]: train_target = train_data['InternetUsersPct']
test_target = test_data['InternetUsersPct']

train_predictors = train_data.drop('InternetUsersPct', axis=1)
test_predictors = test_data.drop('InternetUsersPct', axis=1)

# Combine the predictor data frames to ensure consistent encoding
combined_predictors = pd.concat([train_predictors, test_predictors],
    ↪ ignore_index=True)
```

```

# Identify numerical and categorical columns
numerical_cols = combined_predictors.select_dtypes(include=np.number).columns
categorical_cols = combined_predictors.select_dtypes(include='object').columns

# Initialize MinMaxScaler
scaler = MinMaxScaler()

# Fit the scaler on the combined numerical data to learn the min/max values
combined_predictors[numerical_cols] = scaler.
    ↪fit_transform(combined_predictors[numerical_cols])

# Initialize OneHotEncoder to create sparse output and handle unknown categories
ohe = OneHotEncoder(sparse_output=False, handle_unknown='ignore')

# Fit the encoder on the combined data to learn all possible categories
combined_categorical_encoded = ohe.
    ↪fit_transform(combined_predictors[categorical_cols])

# Combine the encoded categorical data with the numerical data
combined_features = np.concatenate([combined_predictors[numerical_cols].values,
                                    combined_categorical_encoded], axis=1)

# Split the combined feature matrix back into training and test sets
train_features = combined_features[:len(train_predictors)]
test_features = combined_features[len(train_predictors):]

```

```

[5]: print("\n--- Training Model with Tanh and RMSProp---")
model_tanh_rms = Sequential()
model_tanh_rms.add(Dense(256, input_dim=train_features.shape[1],
    ↪activation='tanh'))
model_tanh_rms.add(Dense(128, activation='tanh'))
model_tanh_rms.add(Dense(64, activation='tanh'))
model_tanh_rms.add(Dense(1, activation='linear'))

model_tanh_rms.compile(optimizer=RMSprop(learning_rate=0.001), loss="mse",
    ↪metrics=["mae"])

# Define early stopping
early_stop_tanh = EarlyStopping(monitor='val_loss', patience=50, verbose=1,
    ↪restore_best_weights=True)

model_tanh_rms.fit(train_features, train_target,
                    epochs=1000,
                    batch_size=32,
                    validation_data=(test_features, test_target),
                    callbacks=[early_stop_tanh],
                    verbose=1)

```

```

loss_tanh_rms = model_tanh_rms.evaluate(test_features, test_target, verbose=0)

history_tanh = model_tanh_rms.fit(train_features, train_target,
                                  epochs=1000,
                                  batch_size=32,
                                  validation_split=0.2,
                                  callbacks=[early_stop_tanh],
                                  verbose=1)

tanh_results = model_tanh_rms.evaluate(test_features, test_target, verbose=0)
print("Raw evaluate results:", tanh_results)

# Plot the training and validation MAE
mae_tanh = history_tanh.history['mae']
val_mae_tanh = history_tanh.history['val_mae']
epochs = range(1, len(mae_tanh) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs, mae_tanh, 'b', label='Training MAE')
plt.plot(epochs, val_mae_tanh, 'r', label='Validation MAE')
plt.title('Training and Validation MAE (Tanh + RMSprop)')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()

```

--- Training Model with Tanh and RMSProp---

Epoch 1/1000

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

71/71 1s 5ms/step - loss:

3.2290 - mae: 1.2480

Epoch 2/1000

50/71 0s 2ms/step - loss:

0.6117 - mae: 0.6238

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran out of data; interrupting training. Make sure that your dataset or generator can generate at least `steps\_per\_epoch \* epochs` batches. You may need to use the `.repeat()` function when building your dataset.

```
self._interrupted_warning()
```

C:\Users\barre\my_new_env2\Lib\site-

```
packages\keras\src\callbacks\early_stopping.py:99: UserWarning: Early stopping
conditioned on metric `val_loss` which is not available. Available metrics are:
loss,mae
```

```
    current = self.get_monitor_value(logs)
```

```
71/71          0s 3ms/step - loss:
```

```
0.5789 - mae: 0.6083
```

```
Epoch 3/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.3725 - mae: 0.4755
```

```
Epoch 4/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.3250 - mae: 0.4484
```

```
Epoch 5/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.2873 - mae: 0.4297
```

```
Epoch 6/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.2410 - mae: 0.3821
```

```
Epoch 7/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.2252 - mae: 0.3696
```

```
Epoch 8/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.2123 - mae: 0.3660
```

```
Epoch 9/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.1687 - mae: 0.3201
```

```
Epoch 10/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.1679 - mae: 0.3221
```

```
Epoch 11/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.1515 - mae: 0.3005
```

```
Epoch 12/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.1516 - mae: 0.3101
```

```
Epoch 13/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.1386 - mae: 0.2886
```

```
Epoch 14/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.1241 - mae: 0.2740
```

```
Epoch 15/1000
```

```
71/71          0s 3ms/step - loss:
```

```
0.1353 - mae: 0.2863
```

```
Epoch 16/1000
```

```
71/71          0s 3ms/step - loss:
```

0.1200 - mae: 0.2708
Epoch 17/1000
71/71 0s 3ms/step - loss:
0.1143 - mae: 0.2586
Epoch 18/1000
71/71 0s 3ms/step - loss:
0.1025 - mae: 0.2471
Epoch 19/1000
71/71 0s 3ms/step - loss:
0.0957 - mae: 0.2395
Epoch 20/1000
71/71 0s 3ms/step - loss:
0.1111 - mae: 0.2515
Epoch 21/1000
71/71 0s 3ms/step - loss:
0.0915 - mae: 0.2347
Epoch 22/1000
71/71 0s 3ms/step - loss:
0.0943 - mae: 0.2379
Epoch 23/1000
71/71 0s 3ms/step - loss:
0.0875 - mae: 0.2266
Epoch 24/1000
71/71 0s 3ms/step - loss:
0.0896 - mae: 0.2289
Epoch 25/1000
71/71 0s 4ms/step - loss:
0.0828 - mae: 0.2210
Epoch 26/1000
71/71 0s 3ms/step - loss:
0.0799 - mae: 0.2168
Epoch 27/1000
71/71 0s 3ms/step - loss:
0.0827 - mae: 0.2202
Epoch 28/1000
71/71 0s 3ms/step - loss:
0.0807 - mae: 0.2209
Epoch 29/1000
71/71 0s 3ms/step - loss:
0.0732 - mae: 0.2082
Epoch 30/1000
71/71 0s 3ms/step - loss:
0.0752 - mae: 0.2102
Epoch 31/1000
71/71 0s 3ms/step - loss:
0.0722 - mae: 0.2063
Epoch 32/1000
71/71 0s 3ms/step - loss:

0.0678 - mae: 0.2016
Epoch 33/1000
71/71 0s 3ms/step - loss:
0.0646 - mae: 0.1963
Epoch 34/1000
71/71 0s 3ms/step - loss:
0.0703 - mae: 0.2021
Epoch 35/1000
71/71 0s 3ms/step - loss:
0.0618 - mae: 0.1841
Epoch 36/1000
71/71 0s 3ms/step - loss:
0.0630 - mae: 0.1894
Epoch 37/1000
71/71 0s 3ms/step - loss:
0.0623 - mae: 0.1903
Epoch 38/1000
71/71 0s 3ms/step - loss:
0.0617 - mae: 0.1937
Epoch 39/1000
71/71 0s 3ms/step - loss:
0.0577 - mae: 0.1817
Epoch 40/1000
71/71 0s 4ms/step - loss:
0.0577 - mae: 0.1808
Epoch 41/1000
71/71 0s 3ms/step - loss:
0.0586 - mae: 0.1832
Epoch 42/1000
71/71 0s 3ms/step - loss:
0.0549 - mae: 0.1801
Epoch 43/1000
71/71 0s 3ms/step - loss:
0.0572 - mae: 0.1839
Epoch 44/1000
71/71 0s 3ms/step - loss:
0.0541 - mae: 0.1783
Epoch 45/1000
71/71 0s 3ms/step - loss:
0.0496 - mae: 0.1674
Epoch 46/1000
71/71 0s 3ms/step - loss:
0.0483 - mae: 0.1657
Epoch 47/1000
71/71 0s 3ms/step - loss:
0.0539 - mae: 0.1764
Epoch 48/1000
71/71 0s 3ms/step - loss:

0.0485 - mae: 0.1704
 Epoch 49/1000
 71/71 0s 3ms/step - loss:
 0.0461 - mae: 0.1612
 Epoch 50/1000
 71/71 0s 4ms/step - loss:
 0.0473 - mae: 0.1630
 Epoch 51/1000
 71/71 0s 3ms/step - loss:
 0.0494 - mae: 0.1707
 Epoch 52/1000
 71/71 0s 3ms/step - loss:
 0.0459 - mae: 0.1630
 Epoch 53/1000
 71/71 0s 3ms/step - loss:
 0.0431 - mae: 0.1553
 Epoch 54/1000
 71/71 0s 3ms/step - loss:
 0.0417 - mae: 0.1550
 Epoch 55/1000
 71/71 0s 3ms/step - loss:
 0.0488 - mae: 0.1693
 Epoch 56/1000
 71/71 0s 3ms/step - loss:
 0.0433 - mae: 0.1577
 Epoch 57/1000
 71/71 0s 3ms/step - loss:
 0.0412 - mae: 0.1525
 Epoch 58/1000
 71/71 0s 3ms/step - loss:
 0.0423 - mae: 0.1573
 Epoch 59/1000
 71/71 0s 3ms/step - loss:
 0.0404 - mae: 0.1533
 Epoch 60/1000
 71/71 0s 4ms/step - loss:
 0.0429 - mae: 0.1559
 Epoch 61/1000
 71/71 0s 3ms/step - loss:
 0.0373 - mae: 0.1476
 Epoch 62/1000
 71/71 0s 3ms/step - loss:
 0.0385 - mae: 0.1480
 Epoch 63/1000
 71/71 0s 3ms/step - loss:
 0.0406 - mae: 0.1533
 Epoch 64/1000
 71/71 0s 3ms/step - loss:

0.0374 - mae: 0.1499
 Epoch 65/1000
 71/71 0s 3ms/step - loss:
 0.0391 - mae: 0.1508
 Epoch 66/1000
 71/71 0s 3ms/step - loss:
 0.0353 - mae: 0.1445
 Epoch 67/1000
 71/71 0s 3ms/step - loss:
 0.0332 - mae: 0.1365
 Epoch 68/1000
 71/71 0s 3ms/step - loss:
 0.0383 - mae: 0.1480
 Epoch 69/1000
 71/71 0s 3ms/step - loss:
 0.0377 - mae: 0.1434
 Epoch 70/1000
 71/71 0s 4ms/step - loss:
 0.0354 - mae: 0.1426
 Epoch 71/1000
 71/71 0s 3ms/step - loss:
 0.0348 - mae: 0.1451
 Epoch 72/1000
 71/71 0s 3ms/step - loss:
 0.0356 - mae: 0.1431
 Epoch 73/1000
 71/71 0s 3ms/step - loss:
 0.0344 - mae: 0.1404
 Epoch 74/1000
 71/71 0s 3ms/step - loss:
 0.0346 - mae: 0.1401
 Epoch 75/1000
 71/71 0s 3ms/step - loss:
 0.0344 - mae: 0.1435
 Epoch 76/1000
 71/71 0s 3ms/step - loss:
 0.0315 - mae: 0.1373
 Epoch 77/1000
 71/71 0s 3ms/step - loss:
 0.0340 - mae: 0.1404
 Epoch 78/1000
 71/71 0s 3ms/step - loss:
 0.0331 - mae: 0.1382
 Epoch 79/1000
 71/71 0s 3ms/step - loss:
 0.0308 - mae: 0.1333
 Epoch 80/1000
 71/71 0s 4ms/step - loss:

0.0299 - mae: 0.1311
Epoch 81/1000
71/71 0s 3ms/step - loss:
0.0332 - mae: 0.1378
Epoch 82/1000
71/71 0s 3ms/step - loss:
0.0293 - mae: 0.1301
Epoch 83/1000
71/71 0s 3ms/step - loss:
0.0333 - mae: 0.1384
Epoch 84/1000
71/71 0s 3ms/step - loss:
0.0324 - mae: 0.1321
Epoch 85/1000
71/71 0s 3ms/step - loss:
0.0292 - mae: 0.1308
Epoch 86/1000
71/71 0s 3ms/step - loss:
0.0311 - mae: 0.1322
Epoch 87/1000
71/71 0s 3ms/step - loss:
0.0289 - mae: 0.1280
Epoch 88/1000
71/71 0s 3ms/step - loss:
0.0305 - mae: 0.1319
Epoch 89/1000
71/71 0s 3ms/step - loss:
0.0293 - mae: 0.1309
Epoch 90/1000
71/71 0s 4ms/step - loss:
0.0282 - mae: 0.1282
Epoch 91/1000
71/71 0s 3ms/step - loss:
0.0281 - mae: 0.1257
Epoch 92/1000
71/71 0s 3ms/step - loss:
0.0280 - mae: 0.1222
Epoch 93/1000
71/71 0s 3ms/step - loss:
0.0297 - mae: 0.1311
Epoch 94/1000
71/71 0s 3ms/step - loss:
0.0284 - mae: 0.1254
Epoch 95/1000
71/71 0s 3ms/step - loss:
0.0251 - mae: 0.1186
Epoch 96/1000
71/71 0s 3ms/step - loss:

0.0284 - mae: 0.1260
Epoch 97/1000
71/71 0s 3ms/step - loss:
0.0265 - mae: 0.1251
Epoch 98/1000
71/71 0s 3ms/step - loss:
0.0256 - mae: 0.1178
Epoch 99/1000
71/71 0s 3ms/step - loss:
0.0255 - mae: 0.1221
Epoch 100/1000
71/71 0s 4ms/step - loss:
0.0262 - mae: 0.1213
Epoch 101/1000
71/71 0s 3ms/step - loss:
0.0261 - mae: 0.1224
Epoch 102/1000
71/71 0s 3ms/step - loss:
0.0259 - mae: 0.1230
Epoch 103/1000
71/71 0s 3ms/step - loss:
0.0241 - mae: 0.1193
Epoch 104/1000
71/71 0s 3ms/step - loss:
0.0251 - mae: 0.1196
Epoch 105/1000
71/71 0s 3ms/step - loss:
0.0248 - mae: 0.1201
Epoch 106/1000
71/71 0s 3ms/step - loss:
0.0253 - mae: 0.1225
Epoch 107/1000
71/71 0s 3ms/step - loss:
0.0259 - mae: 0.1189
Epoch 108/1000
71/71 0s 3ms/step - loss:
0.0228 - mae: 0.1154
Epoch 109/1000
71/71 0s 3ms/step - loss:
0.0238 - mae: 0.1173
Epoch 110/1000
71/71 0s 4ms/step - loss:
0.0241 - mae: 0.1138
Epoch 111/1000
71/71 0s 3ms/step - loss:
0.0234 - mae: 0.1187
Epoch 112/1000
71/71 0s 3ms/step - loss:

0.0231 - mae: 0.1152
Epoch 113/1000
71/71 0s 3ms/step - loss:
0.0247 - mae: 0.1184
Epoch 114/1000
71/71 0s 3ms/step - loss:
0.0246 - mae: 0.1179
Epoch 115/1000
71/71 0s 3ms/step - loss:
0.0222 - mae: 0.1121
Epoch 116/1000
71/71 0s 3ms/step - loss:
0.0220 - mae: 0.1122
Epoch 117/1000
71/71 0s 3ms/step - loss:
0.0238 - mae: 0.1172
Epoch 118/1000
71/71 0s 3ms/step - loss:
0.0213 - mae: 0.1084
Epoch 119/1000
71/71 0s 3ms/step - loss:
0.0210 - mae: 0.1072
Epoch 120/1000
71/71 0s 4ms/step - loss:
0.0218 - mae: 0.1116
Epoch 121/1000
71/71 0s 3ms/step - loss:
0.0218 - mae: 0.1108
Epoch 122/1000
71/71 0s 3ms/step - loss:
0.0236 - mae: 0.1183
Epoch 123/1000
71/71 0s 3ms/step - loss:
0.0227 - mae: 0.1120
Epoch 124/1000
71/71 0s 3ms/step - loss:
0.0191 - mae: 0.1053
Epoch 125/1000
71/71 0s 3ms/step - loss:
0.0213 - mae: 0.1077
Epoch 126/1000
71/71 0s 3ms/step - loss:
0.0209 - mae: 0.1129
Epoch 127/1000
71/71 0s 3ms/step - loss:
0.0223 - mae: 0.1133
Epoch 128/1000
71/71 0s 3ms/step - loss:

0.0207 - mae: 0.1077
Epoch 129/1000
71/71 0s 3ms/step - loss:
0.0225 - mae: 0.1104
Epoch 130/1000
71/71 0s 4ms/step - loss:
0.0206 - mae: 0.1081
Epoch 131/1000
71/71 0s 3ms/step - loss:
0.0205 - mae: 0.1076
Epoch 132/1000
71/71 0s 3ms/step - loss:
0.0212 - mae: 0.1089
Epoch 133/1000
71/71 0s 3ms/step - loss:
0.0196 - mae: 0.1070
Epoch 134/1000
71/71 0s 3ms/step - loss:
0.0211 - mae: 0.1112
Epoch 135/1000
71/71 0s 3ms/step - loss:
0.0199 - mae: 0.1049
Epoch 136/1000
71/71 0s 3ms/step - loss:
0.0204 - mae: 0.1089
Epoch 137/1000
71/71 0s 3ms/step - loss:
0.0188 - mae: 0.1066
Epoch 138/1000
71/71 0s 3ms/step - loss:
0.0190 - mae: 0.1057
Epoch 139/1000
71/71 0s 3ms/step - loss:
0.0180 - mae: 0.0983
Epoch 140/1000
71/71 0s 5ms/step - loss:
0.0198 - mae: 0.1037
Epoch 141/1000
71/71 0s 3ms/step - loss:
0.0181 - mae: 0.1019
Epoch 142/1000
71/71 0s 3ms/step - loss:
0.0208 - mae: 0.1099
Epoch 143/1000
71/71 0s 3ms/step - loss:
0.0171 - mae: 0.0993
Epoch 144/1000
71/71 0s 3ms/step - loss:

0.0194 - mae: 0.1047
Epoch 145/1000
71/71 0s 3ms/step - loss:
0.0198 - mae: 0.1066
Epoch 146/1000
71/71 0s 3ms/step - loss:
0.0178 - mae: 0.1004
Epoch 147/1000
71/71 0s 3ms/step - loss:
0.0177 - mae: 0.1011
Epoch 148/1000
71/71 0s 3ms/step - loss:
0.0177 - mae: 0.1018
Epoch 149/1000
71/71 0s 3ms/step - loss:
0.0183 - mae: 0.1017
Epoch 150/1000
71/71 0s 3ms/step - loss:
0.0181 - mae: 0.1006
Epoch 151/1000
71/71 0s 3ms/step - loss:
0.0183 - mae: 0.1014
Epoch 152/1000
71/71 0s 3ms/step - loss:
0.0188 - mae: 0.1042
Epoch 153/1000
71/71 0s 3ms/step - loss:
0.0179 - mae: 0.1016
Epoch 154/1000
71/71 0s 3ms/step - loss:
0.0176 - mae: 0.0979
Epoch 155/1000
71/71 0s 3ms/step - loss:
0.0175 - mae: 0.1002
Epoch 156/1000
71/71 0s 3ms/step - loss:
0.0179 - mae: 0.1012
Epoch 157/1000
71/71 0s 3ms/step - loss:
0.0170 - mae: 0.1001
Epoch 158/1000
71/71 0s 3ms/step - loss:
0.0167 - mae: 0.0963
Epoch 159/1000
71/71 0s 4ms/step - loss:
0.0177 - mae: 0.0993
Epoch 160/1000
71/71 0s 3ms/step - loss:

0.0177 - mae: 0.0998
Epoch 161/1000
71/71 0s 3ms/step - loss:
0.0155 - mae: 0.0930
Epoch 162/1000
71/71 0s 3ms/step - loss:
0.0162 - mae: 0.0970
Epoch 163/1000
71/71 0s 3ms/step - loss:
0.0170 - mae: 0.1009
Epoch 164/1000
71/71 0s 3ms/step - loss:
0.0165 - mae: 0.0967
Epoch 165/1000
71/71 0s 3ms/step - loss:
0.0163 - mae: 0.0947
Epoch 166/1000
71/71 0s 3ms/step - loss:
0.0176 - mae: 0.0989
Epoch 167/1000
71/71 0s 3ms/step - loss:
0.0176 - mae: 0.1019
Epoch 168/1000
71/71 0s 3ms/step - loss:
0.0156 - mae: 0.0952
Epoch 169/1000
71/71 0s 4ms/step - loss:
0.0160 - mae: 0.0962
Epoch 170/1000
71/71 0s 3ms/step - loss:
0.0162 - mae: 0.0964
Epoch 171/1000
71/71 0s 3ms/step - loss:
0.0161 - mae: 0.0965
Epoch 172/1000
71/71 0s 3ms/step - loss:
0.0154 - mae: 0.0907
Epoch 173/1000
71/71 0s 3ms/step - loss:
0.0171 - mae: 0.0988
Epoch 174/1000
71/71 0s 3ms/step - loss:
0.0160 - mae: 0.0960
Epoch 175/1000
71/71 0s 3ms/step - loss:
0.0156 - mae: 0.0931
Epoch 176/1000
71/71 0s 3ms/step - loss:

0.0151 - mae: 0.0917
Epoch 177/1000
71/71 0s 3ms/step - loss:
0.0156 - mae: 0.0945
Epoch 178/1000
71/71 0s 3ms/step - loss:
0.0149 - mae: 0.0910
Epoch 179/1000
71/71 0s 4ms/step - loss:
0.0155 - mae: 0.0935
Epoch 180/1000
71/71 0s 3ms/step - loss:
0.0155 - mae: 0.0945
Epoch 181/1000
71/71 0s 3ms/step - loss:
0.0161 - mae: 0.0976
Epoch 182/1000
71/71 0s 3ms/step - loss:
0.0142 - mae: 0.0887
Epoch 183/1000
71/71 0s 3ms/step - loss:
0.0140 - mae: 0.0894
Epoch 184/1000
71/71 0s 3ms/step - loss:
0.0151 - mae: 0.0921
Epoch 185/1000
71/71 0s 3ms/step - loss:
0.0155 - mae: 0.0949
Epoch 186/1000
71/71 0s 3ms/step - loss:
0.0153 - mae: 0.0947
Epoch 187/1000
71/71 0s 3ms/step - loss:
0.0142 - mae: 0.0885
Epoch 188/1000
71/71 0s 3ms/step - loss:
0.0144 - mae: 0.0903
Epoch 189/1000
71/71 0s 4ms/step - loss:
0.0137 - mae: 0.0880
Epoch 190/1000
71/71 0s 3ms/step - loss:
0.0158 - mae: 0.0960
Epoch 191/1000
71/71 0s 3ms/step - loss:
0.0145 - mae: 0.0922
Epoch 192/1000
71/71 0s 3ms/step - loss:

0.0147 - mae: 0.0948
Epoch 193/1000
71/71 0s 3ms/step - loss:
0.0139 - mae: 0.0870
Epoch 194/1000
71/71 0s 3ms/step - loss:
0.0141 - mae: 0.0898
Epoch 195/1000
71/71 0s 3ms/step - loss:
0.0158 - mae: 0.0944
Epoch 196/1000
71/71 0s 3ms/step - loss:
0.0141 - mae: 0.0905
Epoch 197/1000
71/71 0s 3ms/step - loss:
0.0148 - mae: 0.0905
Epoch 198/1000
71/71 0s 3ms/step - loss:
0.0138 - mae: 0.0902
Epoch 199/1000
71/71 0s 4ms/step - loss:
0.0137 - mae: 0.0881
Epoch 200/1000
71/71 0s 3ms/step - loss:
0.0143 - mae: 0.0915
Epoch 201/1000
71/71 0s 3ms/step - loss:
0.0138 - mae: 0.0887
Epoch 202/1000
71/71 0s 3ms/step - loss:
0.0145 - mae: 0.0914
Epoch 203/1000
71/71 0s 3ms/step - loss:
0.0118 - mae: 0.0821
Epoch 204/1000
71/71 0s 3ms/step - loss:
0.0138 - mae: 0.0881
Epoch 205/1000
71/71 0s 3ms/step - loss:
0.0142 - mae: 0.0913
Epoch 206/1000
71/71 0s 3ms/step - loss:
0.0136 - mae: 0.0867
Epoch 207/1000
71/71 0s 3ms/step - loss:
0.0132 - mae: 0.0863
Epoch 208/1000
71/71 0s 3ms/step - loss:

0.0138 - mae: 0.0895
Epoch 209/1000
71/71 0s 4ms/step - loss:
0.0131 - mae: 0.0863
Epoch 210/1000
71/71 0s 3ms/step - loss:
0.0136 - mae: 0.0878
Epoch 211/1000
71/71 0s 3ms/step - loss:
0.0148 - mae: 0.0924
Epoch 212/1000
71/71 0s 3ms/step - loss:
0.0121 - mae: 0.0851
Epoch 213/1000
71/71 0s 3ms/step - loss:
0.0131 - mae: 0.0849
Epoch 214/1000
71/71 0s 3ms/step - loss:
0.0138 - mae: 0.0894
Epoch 215/1000
71/71 0s 3ms/step - loss:
0.0134 - mae: 0.0880
Epoch 216/1000
71/71 0s 3ms/step - loss:
0.0125 - mae: 0.0840
Epoch 217/1000
71/71 0s 3ms/step - loss:
0.0137 - mae: 0.0902
Epoch 218/1000
71/71 0s 3ms/step - loss:
0.0121 - mae: 0.0853
Epoch 219/1000
71/71 0s 4ms/step - loss:
0.0136 - mae: 0.0879
Epoch 220/1000
71/71 0s 3ms/step - loss:
0.0131 - mae: 0.0891
Epoch 221/1000
71/71 0s 3ms/step - loss:
0.0128 - mae: 0.0875
Epoch 222/1000
71/71 0s 3ms/step - loss:
0.0134 - mae: 0.0845
Epoch 223/1000
71/71 0s 3ms/step - loss:
0.0119 - mae: 0.0827
Epoch 224/1000
71/71 0s 3ms/step - loss:

0.0137 - mae: 0.0878
Epoch 225/1000
71/71 0s 3ms/step - loss:
0.0120 - mae: 0.0815
Epoch 226/1000
71/71 0s 3ms/step - loss:
0.0127 - mae: 0.0837
Epoch 227/1000
71/71 0s 3ms/step - loss:
0.0135 - mae: 0.0867
Epoch 228/1000
71/71 0s 3ms/step - loss:
0.0119 - mae: 0.0818
Epoch 229/1000
71/71 0s 4ms/step - loss:
0.0123 - mae: 0.0869
Epoch 230/1000
71/71 0s 3ms/step - loss:
0.0126 - mae: 0.0846
Epoch 231/1000
71/71 0s 3ms/step - loss:
0.0125 - mae: 0.0841
Epoch 232/1000
71/71 0s 3ms/step - loss:
0.0118 - mae: 0.0823
Epoch 233/1000
71/71 0s 3ms/step - loss:
0.0115 - mae: 0.0834
Epoch 234/1000
71/71 0s 3ms/step - loss:
0.0116 - mae: 0.0825
Epoch 235/1000
71/71 0s 3ms/step - loss:
0.0128 - mae: 0.0861
Epoch 236/1000
71/71 0s 3ms/step - loss:
0.0110 - mae: 0.0808
Epoch 237/1000
71/71 0s 3ms/step - loss:
0.0117 - mae: 0.0824
Epoch 238/1000
71/71 0s 3ms/step - loss:
0.0115 - mae: 0.0822
Epoch 239/1000
71/71 0s 4ms/step - loss:
0.0116 - mae: 0.0831
Epoch 240/1000
71/71 0s 3ms/step - loss:

0.0123 - mae: 0.0848
Epoch 241/1000
71/71 0s 3ms/step - loss:
0.0106 - mae: 0.0787
Epoch 242/1000
71/71 0s 3ms/step - loss:
0.0118 - mae: 0.0821
Epoch 243/1000
71/71 0s 3ms/step - loss:
0.0119 - mae: 0.0844
Epoch 244/1000
71/71 0s 3ms/step - loss:
0.0123 - mae: 0.0864
Epoch 245/1000
71/71 0s 3ms/step - loss:
0.0116 - mae: 0.0786
Epoch 246/1000
71/71 0s 3ms/step - loss:
0.0103 - mae: 0.0754
Epoch 247/1000
71/71 0s 3ms/step - loss:
0.0117 - mae: 0.0832
Epoch 248/1000
71/71 0s 3ms/step - loss:
0.0121 - mae: 0.0830
Epoch 249/1000
71/71 0s 3ms/step - loss:
0.0119 - mae: 0.0842
Epoch 250/1000
71/71 0s 4ms/step - loss:
0.0114 - mae: 0.0793
Epoch 251/1000
71/71 0s 3ms/step - loss:
0.0114 - mae: 0.0825
Epoch 252/1000
71/71 0s 3ms/step - loss:
0.0109 - mae: 0.0786
Epoch 253/1000
71/71 0s 3ms/step - loss:
0.0102 - mae: 0.0746
Epoch 254/1000
71/71 0s 3ms/step - loss:
0.0113 - mae: 0.0805
Epoch 255/1000
71/71 0s 3ms/step - loss:
0.0107 - mae: 0.0770
Epoch 256/1000
71/71 0s 3ms/step - loss:

0.0120 - mae: 0.0829
Epoch 257/1000
71/71 0s 3ms/step - loss:
0.0110 - mae: 0.0803
Epoch 258/1000
71/71 0s 3ms/step - loss:
0.0121 - mae: 0.0837
Epoch 259/1000
71/71 0s 3ms/step - loss:
0.0109 - mae: 0.0793
Epoch 260/1000
71/71 0s 3ms/step - loss:
0.0105 - mae: 0.0783
Epoch 261/1000
71/71 0s 5ms/step - loss:
0.0112 - mae: 0.0820
Epoch 262/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0776
Epoch 263/1000
71/71 0s 3ms/step - loss:
0.0101 - mae: 0.0764
Epoch 264/1000
71/71 0s 3ms/step - loss:
0.0107 - mae: 0.0787
Epoch 265/1000
71/71 0s 3ms/step - loss:
0.0113 - mae: 0.0801
Epoch 266/1000
71/71 0s 3ms/step - loss:
0.0098 - mae: 0.0764
Epoch 267/1000
71/71 0s 3ms/step - loss:
0.0104 - mae: 0.0779
Epoch 268/1000
71/71 0s 3ms/step - loss:
0.0105 - mae: 0.0771
Epoch 269/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0797
Epoch 270/1000
71/71 0s 3ms/step - loss:
0.0107 - mae: 0.0771
Epoch 271/1000
71/71 0s 3ms/step - loss:
0.0097 - mae: 0.0750
Epoch 272/1000
71/71 0s 4ms/step - loss:

0.0114 - mae: 0.0824
Epoch 273/1000
71/71 0s 3ms/step - loss:
0.0101 - mae: 0.0763
Epoch 274/1000
71/71 0s 3ms/step - loss:
0.0106 - mae: 0.0781
Epoch 275/1000
71/71 0s 3ms/step - loss:
0.0088 - mae: 0.0712
Epoch 276/1000
71/71 0s 3ms/step - loss:
0.0114 - mae: 0.0787
Epoch 277/1000
71/71 0s 3ms/step - loss:
0.0103 - mae: 0.0775
Epoch 278/1000
71/71 0s 3ms/step - loss:
0.0103 - mae: 0.0788
Epoch 279/1000
71/71 0s 3ms/step - loss:
0.0098 - mae: 0.0763
Epoch 280/1000
71/71 0s 3ms/step - loss:
0.0104 - mae: 0.0788
Epoch 281/1000
71/71 0s 3ms/step - loss:
0.0097 - mae: 0.0747
Epoch 282/1000
71/71 0s 3ms/step - loss:
0.0104 - mae: 0.0770
Epoch 283/1000
71/71 0s 4ms/step - loss:
0.0097 - mae: 0.0750
Epoch 284/1000
71/71 0s 3ms/step - loss:
0.0104 - mae: 0.0784
Epoch 285/1000
71/71 0s 3ms/step - loss:
0.0096 - mae: 0.0773
Epoch 286/1000
71/71 0s 3ms/step - loss:
0.0099 - mae: 0.0753
Epoch 287/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0808
Epoch 288/1000
71/71 0s 3ms/step - loss:

0.0093 - mae: 0.0722
Epoch 289/1000
71/71 0s 3ms/step - loss:
0.0107 - mae: 0.0789
Epoch 290/1000
71/71 0s 3ms/step - loss:
0.0101 - mae: 0.0779
Epoch 291/1000
71/71 0s 3ms/step - loss:
0.0095 - mae: 0.0758
Epoch 292/1000
71/71 0s 3ms/step - loss:
0.0089 - mae: 0.0726
Epoch 293/1000
71/71 0s 3ms/step - loss:
0.0100 - mae: 0.0766
Epoch 294/1000
71/71 0s 4ms/step - loss:
0.0090 - mae: 0.0707
Epoch 295/1000
71/71 0s 3ms/step - loss:
0.0102 - mae: 0.0756
Epoch 296/1000
71/71 0s 3ms/step - loss:
0.0102 - mae: 0.0780
Epoch 297/1000
71/71 0s 3ms/step - loss:
0.0091 - mae: 0.0715
Epoch 298/1000
71/71 0s 3ms/step - loss:
0.0097 - mae: 0.0756
Epoch 299/1000
71/71 0s 3ms/step - loss:
0.0096 - mae: 0.0761
Epoch 300/1000
71/71 0s 3ms/step - loss:
0.0099 - mae: 0.0771
Epoch 301/1000
71/71 0s 3ms/step - loss:
0.0097 - mae: 0.0725
Epoch 302/1000
71/71 0s 3ms/step - loss:
0.0084 - mae: 0.0687
Epoch 303/1000
71/71 0s 3ms/step - loss:
0.0103 - mae: 0.0782
Epoch 304/1000
71/71 0s 3ms/step - loss:

0.0096 - mae: 0.0748
Epoch 305/1000
71/71 0s 4ms/step - loss:
0.0088 - mae: 0.0696
Epoch 306/1000
71/71 0s 3ms/step - loss:
0.0092 - mae: 0.0730
Epoch 307/1000
71/71 0s 3ms/step - loss:
0.0104 - mae: 0.0781
Epoch 308/1000
71/71 0s 3ms/step - loss:
0.0085 - mae: 0.0710
Epoch 309/1000
71/71 0s 3ms/step - loss:
0.0093 - mae: 0.0721
Epoch 310/1000
71/71 0s 3ms/step - loss:
0.0094 - mae: 0.0737
Epoch 311/1000
71/71 0s 3ms/step - loss:
0.0092 - mae: 0.0753
Epoch 312/1000
71/71 0s 3ms/step - loss:
0.0094 - mae: 0.0747
Epoch 313/1000
71/71 0s 3ms/step - loss:
0.0090 - mae: 0.0716
Epoch 314/1000
71/71 0s 3ms/step - loss:
0.0088 - mae: 0.0722
Epoch 315/1000
71/71 0s 3ms/step - loss:
0.0095 - mae: 0.0722
Epoch 316/1000
71/71 0s 3ms/step - loss:
0.0084 - mae: 0.0706
Epoch 317/1000
71/71 0s 3ms/step - loss:
0.0100 - mae: 0.0758
Epoch 318/1000
71/71 0s 3ms/step - loss:
0.0079 - mae: 0.0663
Epoch 319/1000
71/71 0s 3ms/step - loss:
0.0093 - mae: 0.0727
Epoch 320/1000
71/71 0s 3ms/step - loss:

0.0096 - mae: 0.0767
Epoch 321/1000
71/71 0s 3ms/step - loss:
0.0092 - mae: 0.0716
Epoch 322/1000
71/71 0s 3ms/step - loss:
0.0089 - mae: 0.0722
Epoch 323/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0643
Epoch 324/1000
71/71 0s 3ms/step - loss:
0.0088 - mae: 0.0711
Epoch 325/1000
71/71 0s 3ms/step - loss:
0.0091 - mae: 0.0725
Epoch 326/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0664
Epoch 327/1000
71/71 0s 3ms/step - loss:
0.0090 - mae: 0.0715
Epoch 328/1000
71/71 0s 4ms/step - loss:
0.0090 - mae: 0.0730
Epoch 329/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0694
Epoch 330/1000
71/71 0s 3ms/step - loss:
0.0085 - mae: 0.0721
Epoch 331/1000
71/71 0s 3ms/step - loss:
0.0086 - mae: 0.0701
Epoch 332/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0677
Epoch 333/1000
71/71 0s 3ms/step - loss:
0.0091 - mae: 0.0735
Epoch 334/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0653
Epoch 335/1000
71/71 0s 3ms/step - loss:
0.0095 - mae: 0.0727
Epoch 336/1000
71/71 0s 3ms/step - loss:

0.0075 - mae: 0.0656
Epoch 337/1000
71/71 0s 3ms/step - loss:
0.0091 - mae: 0.0729
Epoch 338/1000
71/71 0s 3ms/step - loss:
0.0090 - mae: 0.0700
Epoch 339/1000
71/71 0s 4ms/step - loss:
0.0077 - mae: 0.0663
Epoch 340/1000
71/71 0s 3ms/step - loss:
0.0085 - mae: 0.0710
Epoch 341/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0693
Epoch 342/1000
71/71 0s 3ms/step - loss:
0.0085 - mae: 0.0705
Epoch 343/1000
71/71 0s 3ms/step - loss:
0.0082 - mae: 0.0680
Epoch 344/1000
71/71 0s 3ms/step - loss:
0.0082 - mae: 0.0676
Epoch 345/1000
71/71 0s 3ms/step - loss:
0.0083 - mae: 0.0705
Epoch 346/1000
71/71 0s 3ms/step - loss:
0.0083 - mae: 0.0701
Epoch 347/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0675
Epoch 348/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0675
Epoch 349/1000
71/71 0s 3ms/step - loss:
0.0085 - mae: 0.0705
Epoch 350/1000
71/71 0s 4ms/step - loss:
0.0078 - mae: 0.0684
Epoch 351/1000
71/71 0s 3ms/step - loss:
0.0084 - mae: 0.0698
Epoch 352/1000
71/71 0s 3ms/step - loss:

0.0086 - mae: 0.0709
Epoch 353/1000
71/71 0s 3ms/step - loss:
0.0083 - mae: 0.0688
Epoch 354/1000
71/71 0s 3ms/step - loss:
0.0081 - mae: 0.0663
Epoch 355/1000
71/71 0s 3ms/step - loss:
0.0082 - mae: 0.0695
Epoch 356/1000
71/71 0s 3ms/step - loss:
0.0081 - mae: 0.0675
Epoch 357/1000
71/71 0s 3ms/step - loss:
0.0081 - mae: 0.0694
Epoch 358/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0651
Epoch 359/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0668
Epoch 360/1000
71/71 0s 3ms/step - loss:
0.0087 - mae: 0.0707
Epoch 361/1000
71/71 0s 4ms/step - loss:
0.0082 - mae: 0.0681
Epoch 362/1000
71/71 0s 3ms/step - loss:
0.0076 - mae: 0.0658
Epoch 363/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0686
Epoch 364/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0640
Epoch 365/1000
71/71 0s 3ms/step - loss:
0.0086 - mae: 0.0717
Epoch 366/1000
71/71 0s 3ms/step - loss:
0.0069 - mae: 0.0626
Epoch 367/1000
71/71 0s 3ms/step - loss:
0.0081 - mae: 0.0707
Epoch 368/1000
71/71 0s 3ms/step - loss:

0.0081 - mae: 0.0701
Epoch 369/1000
71/71 0s 3ms/step - loss:
0.0075 - mae: 0.0647
Epoch 370/1000
71/71 0s 3ms/step - loss:
0.0077 - mae: 0.0673
Epoch 371/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0648
Epoch 372/1000
71/71 0s 4ms/step - loss:
0.0076 - mae: 0.0684
Epoch 373/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0697
Epoch 374/1000
71/71 0s 3ms/step - loss:
0.0075 - mae: 0.0642
Epoch 375/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0641
Epoch 376/1000
71/71 0s 3ms/step - loss:
0.0077 - mae: 0.0672
Epoch 377/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0640
Epoch 378/1000
71/71 0s 3ms/step - loss:
0.0075 - mae: 0.0657
Epoch 379/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0641
Epoch 380/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0631
Epoch 381/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0681
Epoch 382/1000
71/71 0s 3ms/step - loss:
0.0077 - mae: 0.0661
Epoch 383/1000
71/71 0s 4ms/step - loss:
0.0074 - mae: 0.0635
Epoch 384/1000
71/71 0s 3ms/step - loss:

0.0071 - mae: 0.0634
Epoch 385/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0663
Epoch 386/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0646
Epoch 387/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0663
Epoch 388/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0644
Epoch 389/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0636
Epoch 390/1000
71/71 0s 3ms/step - loss:
0.0066 - mae: 0.0622
Epoch 391/1000
71/71 0s 3ms/step - loss:
0.0075 - mae: 0.0663
Epoch 392/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0663
Epoch 393/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0637
Epoch 394/1000
71/71 0s 4ms/step - loss:
0.0069 - mae: 0.0636
Epoch 395/1000
71/71 0s 3ms/step - loss:
0.0077 - mae: 0.0645
Epoch 396/1000
71/71 0s 3ms/step - loss:
0.0065 - mae: 0.0618
Epoch 397/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0641
Epoch 398/1000
71/71 0s 3ms/step - loss:
0.0070 - mae: 0.0634
Epoch 399/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0668
Epoch 400/1000
71/71 0s 3ms/step - loss:

0.0066 - mae: 0.0611
Epoch 401/1000
71/71 0s 3ms/step - loss:
0.0066 - mae: 0.0624
Epoch 402/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0644
Epoch 403/1000
71/71 0s 3ms/step - loss:
0.0076 - mae: 0.0662
Epoch 404/1000
71/71 0s 3ms/step - loss:
0.0069 - mae: 0.0634
Epoch 405/1000
71/71 0s 4ms/step - loss:
0.0074 - mae: 0.0656
Epoch 406/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0648
Epoch 407/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0620
Epoch 408/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0628
Epoch 409/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0643
Epoch 410/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0639
Epoch 411/1000
71/71 0s 3ms/step - loss:
0.0077 - mae: 0.0662
Epoch 412/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0643
Epoch 413/1000
71/71 0s 3ms/step - loss:
0.0075 - mae: 0.0669
Epoch 414/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0625
Epoch 415/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0643
Epoch 416/1000
71/71 0s 4ms/step - loss:

0.0068 - mae: 0.0636
Epoch 417/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0651
Epoch 418/1000
71/71 0s 3ms/step - loss:
0.0066 - mae: 0.0627
Epoch 419/1000
71/71 0s 3ms/step - loss:
0.0070 - mae: 0.0633
Epoch 420/1000
71/71 0s 3ms/step - loss:
0.0069 - mae: 0.0641
Epoch 421/1000
71/71 0s 3ms/step - loss:
0.0069 - mae: 0.0628
Epoch 422/1000
71/71 0s 3ms/step - loss:
0.0065 - mae: 0.0612
Epoch 423/1000
71/71 0s 3ms/step - loss:
0.0077 - mae: 0.0674
Epoch 424/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0619
Epoch 425/1000
71/71 0s 3ms/step - loss:
0.0064 - mae: 0.0612
Epoch 426/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0598
Epoch 427/1000
71/71 0s 4ms/step - loss:
0.0067 - mae: 0.0627
Epoch 428/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0647
Epoch 429/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0543
Epoch 430/1000
71/71 0s 3ms/step - loss:
0.0069 - mae: 0.0638
Epoch 431/1000
71/71 0s 3ms/step - loss:
0.0064 - mae: 0.0594
Epoch 432/1000
71/71 0s 3ms/step - loss:

0.0071 - mae: 0.0633
Epoch 433/1000
71/71 0s 3ms/step - loss:
0.0062 - mae: 0.0592
Epoch 434/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0626
Epoch 435/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0636
Epoch 436/1000
71/71 0s 3ms/step - loss:
0.0070 - mae: 0.0607
Epoch 437/1000
71/71 0s 3ms/step - loss:
0.0066 - mae: 0.0609
Epoch 438/1000
71/71 0s 4ms/step - loss:
0.0067 - mae: 0.0622
Epoch 439/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0579
Epoch 440/1000
71/71 0s 3ms/step - loss:
0.0066 - mae: 0.0629
Epoch 441/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0634
Epoch 442/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0587
Epoch 443/1000
71/71 0s 3ms/step - loss:
0.0062 - mae: 0.0586
Epoch 444/1000
71/71 0s 3ms/step - loss:
0.0065 - mae: 0.0619
Epoch 445/1000
71/71 0s 3ms/step - loss:
0.0065 - mae: 0.0616
Epoch 446/1000
71/71 0s 3ms/step - loss:
0.0065 - mae: 0.0604
Epoch 447/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0606
Epoch 448/1000
71/71 0s 3ms/step - loss:

0.0069 - mae: 0.0613
Epoch 449/1000
71/71 0s 4ms/step - loss:
0.0060 - mae: 0.0582
Epoch 450/1000
71/71 0s 3ms/step - loss:
0.0062 - mae: 0.0586
Epoch 451/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0547
Epoch 452/1000
71/71 0s 3ms/step - loss:
0.0064 - mae: 0.0613
Epoch 453/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0629
Epoch 454/1000
71/71 0s 3ms/step - loss:
0.0062 - mae: 0.0595
Epoch 455/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0576
Epoch 456/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0651
Epoch 457/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0585
Epoch 458/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0608
Epoch 459/1000
71/71 0s 3ms/step - loss:
0.0064 - mae: 0.0614
Epoch 460/1000
71/71 0s 4ms/step - loss:
0.0064 - mae: 0.0619
Epoch 461/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0593
Epoch 462/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0617
Epoch 463/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0553
Epoch 464/1000
71/71 0s 3ms/step - loss:

0.0061 - mae: 0.0595
Epoch 465/1000
71/71 0s 3ms/step - loss:
0.0062 - mae: 0.0600
Epoch 466/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0592
Epoch 467/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0603
Epoch 468/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0591
Epoch 469/1000
71/71 0s 3ms/step - loss:
0.0065 - mae: 0.0614
Epoch 470/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0584
Epoch 471/1000
71/71 0s 4ms/step - loss:
0.0061 - mae: 0.0586
Epoch 472/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0595
Epoch 473/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0565
Epoch 474/1000
71/71 0s 3ms/step - loss:
0.0065 - mae: 0.0630
Epoch 475/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0581
Epoch 476/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0580
Epoch 477/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0600
Epoch 478/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0571
Epoch 479/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0616
Epoch 480/1000
71/71 0s 3ms/step - loss:

0.0058 - mae: 0.0580
Epoch 481/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0580
Epoch 482/1000
71/71 0s 4ms/step - loss:
0.0061 - mae: 0.0597
Epoch 483/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0603
Epoch 484/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0546
Epoch 485/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0637
Epoch 486/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0569
Epoch 487/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0570
Epoch 488/1000
71/71 0s 3ms/step - loss:
0.0062 - mae: 0.0587
Epoch 489/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0575
Epoch 490/1000
71/71 0s 3ms/step - loss:
0.0064 - mae: 0.0589
Epoch 491/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0565
Epoch 492/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0567
Epoch 493/1000
71/71 0s 4ms/step - loss:
0.0061 - mae: 0.0596
Epoch 494/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0547
Epoch 495/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0574
Epoch 496/1000
71/71 0s 3ms/step - loss:

0.0054 - mae: 0.0532
Epoch 497/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0560
Epoch 498/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0567
Epoch 499/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0585
Epoch 500/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0588
Epoch 501/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0567
Epoch 502/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0566
Epoch 503/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0579
Epoch 504/1000
71/71 0s 4ms/step - loss:
0.0060 - mae: 0.0591
Epoch 505/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0550
Epoch 506/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0533
Epoch 507/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0602
Epoch 508/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0572
Epoch 509/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0579
Epoch 510/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0578
Epoch 511/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0572
Epoch 512/1000
71/71 0s 3ms/step - loss:

0.0053 - mae: 0.0550
Epoch 513/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0582
Epoch 514/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0538
Epoch 515/1000
71/71 0s 4ms/step - loss:
0.0055 - mae: 0.0557
Epoch 516/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0563
Epoch 517/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0547
Epoch 518/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0545
Epoch 519/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0597
Epoch 520/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0582
Epoch 521/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0529
Epoch 522/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0514
Epoch 523/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0565
Epoch 524/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0577
Epoch 525/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0523
Epoch 526/1000
71/71 0s 4ms/step - loss:
0.0069 - mae: 0.0621
Epoch 527/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0543
Epoch 528/1000
71/71 0s 3ms/step - loss:

```

0.0056 - mae: 0.0570
Epoch 529/1000
71/71          0s 3ms/step - loss:
0.0054 - mae: 0.0543
Epoch 530/1000
71/71          0s 3ms/step - loss:
0.0053 - mae: 0.0551
Epoch 531/1000
71/71          0s 3ms/step - loss:
0.0054 - mae: 0.0561
Epoch 532/1000
71/71          0s 3ms/step - loss:
0.0054 - mae: 0.0574
Epoch 533/1000
71/71          0s 3ms/step - loss:
0.0055 - mae: 0.0552
Epoch 534/1000
71/71          0s 3ms/step - loss:
0.0057 - mae: 0.0566
Epoch 535/1000
71/71          0s 3ms/step - loss:
0.0058 - mae: 0.0583
Epoch 536/1000
71/71          0s 3ms/step - loss:
0.0051 - mae: 0.0531
Epoch 537/1000
71/71          0s 4ms/step - loss:
0.0054 - mae: 0.0576
Epoch 538/1000
71/71          0s 3ms/step - loss:
0.0053 - mae: 0.0549
Epoch 539/1000
71/71          0s 3ms/step - loss:
0.0051 - mae: 0.0556
Epoch 540/1000
71/71          0s 3ms/step - loss:
0.0056 - mae: 0.0575
Epoch 541/1000
71/71          0s 3ms/step - loss:
0.0054 - mae: 0.0548
Epoch 542/1000
71/71          0s 3ms/step - loss:
0.0055 - mae: 0.0561
Epoch 543/1000
71/71          0s 3ms/step - loss:
0.0051 - mae: 0.0508
Epoch 544/1000
71/71          0s 3ms/step - loss:

```

0.0049 - mae: 0.0535
Epoch 545/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0557
Epoch 546/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0557
Epoch 547/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0554
Epoch 548/1000
71/71 0s 4ms/step - loss:
0.0051 - mae: 0.0550
Epoch 549/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0517
Epoch 550/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0542
Epoch 551/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0556
Epoch 552/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0556
Epoch 553/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0546
Epoch 554/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0563
Epoch 555/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0533
Epoch 556/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0553
Epoch 557/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0518
Epoch 558/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0549
Epoch 559/1000
71/71 0s 4ms/step - loss:
0.0048 - mae: 0.0521
Epoch 560/1000
71/71 0s 3ms/step - loss:

0.0051 - mae: 0.0532
Epoch 561/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0547
Epoch 562/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0536
Epoch 563/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0556
Epoch 564/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0567
Epoch 565/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0581
Epoch 566/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0530
Epoch 567/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0532
Epoch 568/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0523
Epoch 569/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0563
Epoch 570/1000
71/71 0s 4ms/step - loss:
0.0049 - mae: 0.0547
Epoch 571/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0546
Epoch 572/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0523
Epoch 573/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0563
Epoch 574/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0540
Epoch 575/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0547
Epoch 576/1000
71/71 0s 3ms/step - loss:

0.0052 - mae: 0.0548
Epoch 577/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0536
Epoch 578/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0514
Epoch 579/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0536
Epoch 580/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0540
Epoch 581/1000
71/71 0s 4ms/step - loss:
0.0052 - mae: 0.0539
Epoch 582/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0508
Epoch 583/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0528
Epoch 584/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0538
Epoch 585/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0518
Epoch 586/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0522
Epoch 587/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0550
Epoch 588/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0520
Epoch 589/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0519
Epoch 590/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0547
Epoch 591/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0554
Epoch 592/1000
71/71 0s 4ms/step - loss:

0.0047 - mae: 0.0518
Epoch 593/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0509
Epoch 594/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0549
Epoch 595/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0517
Epoch 596/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0545
Epoch 597/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0553
Epoch 598/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0490
Epoch 599/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0550
Epoch 600/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0528
Epoch 601/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0491
Epoch 602/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0550
Epoch 603/1000
71/71 0s 4ms/step - loss:
0.0047 - mae: 0.0506
Epoch 604/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0520
Epoch 605/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0558
Epoch 606/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0540
Epoch 607/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0515
Epoch 608/1000
71/71 0s 3ms/step - loss:

0.0049 - mae: 0.0527
Epoch 609/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0519
Epoch 610/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0534
Epoch 611/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0513
Epoch 612/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0478
Epoch 613/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0524
Epoch 614/1000
71/71 0s 4ms/step - loss:
0.0047 - mae: 0.0503
Epoch 615/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0519
Epoch 616/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0525
Epoch 617/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0509
Epoch 618/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0467
Epoch 619/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0538
Epoch 620/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0509
Epoch 621/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0528
Epoch 622/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0504
Epoch 623/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0491
Epoch 624/1000
71/71 0s 3ms/step - loss:

0.0048 - mae: 0.0519
Epoch 625/1000
71/71 0s 4ms/step - loss:
0.0047 - mae: 0.0517
Epoch 626/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0489
Epoch 627/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0495
Epoch 628/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0515
Epoch 629/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0476
Epoch 630/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0519
Epoch 631/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0503
Epoch 632/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0511
Epoch 633/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0492
Epoch 634/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0488
Epoch 635/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0512
Epoch 636/1000
71/71 0s 4ms/step - loss:
0.0044 - mae: 0.0507
Epoch 637/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0506
Epoch 638/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0511
Epoch 639/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0529
Epoch 640/1000
71/71 0s 3ms/step - loss:

0.0044 - mae: 0.0499
Epoch 641/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0518
Epoch 642/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0517
Epoch 643/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0487
Epoch 644/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0535
Epoch 645/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0483
Epoch 646/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0513
Epoch 647/1000
71/71 0s 4ms/step - loss:
0.0045 - mae: 0.0502
Epoch 648/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0500
Epoch 649/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0519
Epoch 650/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0503
Epoch 651/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0476
Epoch 652/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0513
Epoch 653/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0504
Epoch 654/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0497
Epoch 655/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0510
Epoch 656/1000
71/71 0s 3ms/step - loss:

0.0043 - mae: 0.0496
Epoch 657/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0514
Epoch 658/1000
71/71 0s 4ms/step - loss:
0.0041 - mae: 0.0472
Epoch 659/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0510
Epoch 660/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0513
Epoch 661/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0482
Epoch 662/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0489
Epoch 663/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0507
Epoch 664/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0503
Epoch 665/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0513
Epoch 666/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0509
Epoch 667/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0493
Epoch 668/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0510
Epoch 669/1000
71/71 0s 4ms/step - loss:
0.0044 - mae: 0.0511
Epoch 670/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0525
Epoch 671/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0481
Epoch 672/1000
71/71 0s 3ms/step - loss:

0.0040 - mae: 0.0467
Epoch 673/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0524
Epoch 674/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0500
Epoch 675/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0454
Epoch 676/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0519
Epoch 677/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0485
Epoch 678/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0501
Epoch 679/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0472
Epoch 680/1000
71/71 0s 4ms/step - loss:
0.0048 - mae: 0.0521
Epoch 681/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0478
Epoch 682/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0484
Epoch 683/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0498
Epoch 684/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0490
Epoch 685/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0510
Epoch 686/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0472
Epoch 687/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0515
Epoch 688/1000
71/71 0s 3ms/step - loss:

0.0041 - mae: 0.0476
Epoch 689/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0493
Epoch 690/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0451
Epoch 691/1000
71/71 0s 4ms/step - loss:
0.0048 - mae: 0.0531
Epoch 692/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0469
Epoch 693/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0499
Epoch 694/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0490
Epoch 695/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0474
Epoch 696/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0487
Epoch 697/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0510
Epoch 698/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0504
Epoch 699/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0485
Epoch 700/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0492
Epoch 701/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0496
Epoch 702/1000
71/71 0s 4ms/step - loss:
0.0041 - mae: 0.0484
Epoch 703/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0488
Epoch 704/1000
71/71 0s 3ms/step - loss:

0.0042 - mae: 0.0479
Epoch 705/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0478
Epoch 706/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0484
Epoch 707/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0481
Epoch 708/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0468
Epoch 709/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0476
Epoch 710/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0469
Epoch 711/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0484
Epoch 712/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0493
Epoch 713/1000
71/71 0s 4ms/step - loss:
0.0043 - mae: 0.0511
Epoch 714/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0478
Epoch 715/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0445
Epoch 716/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0466
Epoch 717/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0477
Epoch 718/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0462
Epoch 719/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0470
Epoch 720/1000
71/71 0s 3ms/step - loss:

0.0045 - mae: 0.0515
Epoch 721/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0469
Epoch 722/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0479
Epoch 723/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0464
Epoch 724/1000
71/71 0s 4ms/step - loss:
0.0040 - mae: 0.0486
Epoch 725/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0503
Epoch 726/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0465
Epoch 727/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0480
Epoch 728/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0444
Epoch 729/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0476
Epoch 730/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0488
Epoch 731/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0462
Epoch 732/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0465
Epoch 733/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0472
Epoch 734/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0485
Epoch 735/1000
71/71 0s 4ms/step - loss:
0.0038 - mae: 0.0479
Epoch 736/1000
71/71 0s 3ms/step - loss:

0.0042 - mae: 0.0484
Epoch 737/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0471
Epoch 738/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0478
Epoch 739/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0477
Epoch 740/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0458
Epoch 741/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0485
Epoch 742/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0459
Epoch 743/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0500
Epoch 744/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0442
Epoch 745/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0478
Epoch 746/1000
71/71 0s 4ms/step - loss:
0.0036 - mae: 0.0449
Epoch 747/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0465
Epoch 748/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0477
Epoch 749/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0482
Epoch 750/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0462
Epoch 751/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0466
Epoch 752/1000
71/71 0s 3ms/step - loss:

0.0039 - mae: 0.0489
Epoch 753/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0461
Epoch 754/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0459
Epoch 755/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0478
Epoch 756/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0475
Epoch 757/1000
71/71 0s 4ms/step - loss:
0.0038 - mae: 0.0463
Epoch 758/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0469
Epoch 759/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0503
Epoch 760/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0457
Epoch 761/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0468
Epoch 762/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0479
Epoch 763/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0465
Epoch 764/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0459
Epoch 765/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0473
Epoch 766/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0481
Epoch 767/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0437
Epoch 768/1000
71/71 0s 4ms/step - loss:

0.0036 - mae: 0.0462
Epoch 769/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0466
Epoch 770/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0491
Epoch 771/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0456
Epoch 772/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0478
Epoch 773/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0472
Epoch 774/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0492
Epoch 775/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0441
Epoch 776/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0460
Epoch 777/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0456
Epoch 778/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0475
Epoch 779/1000
71/71 0s 4ms/step - loss:
0.0035 - mae: 0.0427
Epoch 780/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0480
Epoch 781/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0478
Epoch 782/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0467
Epoch 783/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0458
Epoch 784/1000
71/71 0s 3ms/step - loss:

0.0038 - mae: 0.0462
Epoch 785/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0442
Epoch 786/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0489
Epoch 787/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0475
Epoch 788/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0438
Epoch 789/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0462
Epoch 790/1000
71/71 0s 4ms/step - loss:
0.0038 - mae: 0.0470
Epoch 791/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0452
Epoch 792/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0460
Epoch 793/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0448
Epoch 794/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0470
Epoch 795/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0439
Epoch 796/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0449
Epoch 797/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0447
Epoch 798/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0445
Epoch 799/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0447
Epoch 800/1000
71/71 0s 3ms/step - loss:

0.0035 - mae: 0.0458
Epoch 801/1000
71/71 0s 4ms/step - loss:
0.0039 - mae: 0.0471
Epoch 802/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0447
Epoch 803/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0475
Epoch 804/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0458
Epoch 805/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0462
Epoch 806/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0477
Epoch 807/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0450
Epoch 808/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0447
Epoch 809/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0471
Epoch 810/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0442
Epoch 811/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0460
Epoch 812/1000
71/71 0s 4ms/step - loss:
0.0037 - mae: 0.0462
Epoch 813/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0477
Epoch 814/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0452
Epoch 815/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0471
Epoch 816/1000
71/71 0s 3ms/step - loss:

0.0033 - mae: 0.0441
Epoch 817/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0456
Epoch 818/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0448
Epoch 819/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0472
Epoch 820/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0461
Epoch 821/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0462
Epoch 822/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0458
Epoch 823/1000
71/71 0s 4ms/step - loss:
0.0035 - mae: 0.0438
Epoch 824/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0454
Epoch 825/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0431
Epoch 826/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0462
Epoch 827/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0445
Epoch 828/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0453
Epoch 829/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0467
Epoch 830/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0453
Epoch 831/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0433
Epoch 832/1000
71/71 0s 3ms/step - loss:

0.0039 - mae: 0.0458
Epoch 833/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0463
Epoch 834/1000
71/71 0s 4ms/step - loss:
0.0035 - mae: 0.0453
Epoch 835/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0457
Epoch 836/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0443
Epoch 837/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0472
Epoch 838/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0464
Epoch 839/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0461
Epoch 840/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0457
Epoch 841/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0447
Epoch 842/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0460
Epoch 843/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0431
Epoch 844/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0440
Epoch 845/1000
71/71 0s 4ms/step - loss:
0.0037 - mae: 0.0477
Epoch 846/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0413
Epoch 847/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0462
Epoch 848/1000
71/71 0s 3ms/step - loss:

0.0034 - mae: 0.0425
Epoch 849/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0465
Epoch 850/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0426
Epoch 851/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0449
Epoch 852/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0442
Epoch 853/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0450
Epoch 854/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0444
Epoch 855/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0451
Epoch 856/1000
71/71 0s 4ms/step - loss:
0.0032 - mae: 0.0438
Epoch 857/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0435
Epoch 858/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0445
Epoch 859/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0445
Epoch 860/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0466
Epoch 861/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0424
Epoch 862/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0445
Epoch 863/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0426
Epoch 864/1000
71/71 0s 3ms/step - loss:

0.0032 - mae: 0.0423
Epoch 865/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0440
Epoch 866/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0436
Epoch 867/1000
71/71 0s 4ms/step - loss:
0.0033 - mae: 0.0420
Epoch 868/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0429
Epoch 869/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0450
Epoch 870/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0420
Epoch 871/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0447
Epoch 872/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0422
Epoch 873/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0450
Epoch 874/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0434
Epoch 875/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0440
Epoch 876/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0449
Epoch 877/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0443
Epoch 878/1000
71/71 0s 4ms/step - loss:
0.0036 - mae: 0.0448
Epoch 879/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0422
Epoch 880/1000
71/71 0s 3ms/step - loss:

0.0033 - mae: 0.0439
Epoch 881/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0463
Epoch 882/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0441
Epoch 883/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0440
Epoch 884/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0443
Epoch 885/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0433
Epoch 886/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0446
Epoch 887/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0396
Epoch 888/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0441
Epoch 889/1000
71/71 0s 4ms/step - loss:
0.0031 - mae: 0.0415
Epoch 890/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0438
Epoch 891/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0444
Epoch 892/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0424
Epoch 893/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0444
Epoch 894/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0420
Epoch 895/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0446
Epoch 896/1000
71/71 0s 3ms/step - loss:

0.0032 - mae: 0.0431
Epoch 897/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0452
Epoch 898/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0426
Epoch 899/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0426
Epoch 900/1000
71/71 0s 4ms/step - loss:
0.0031 - mae: 0.0435
Epoch 901/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0439
Epoch 902/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0434
Epoch 903/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0448
Epoch 904/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0423
Epoch 905/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0454
Epoch 906/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0414
Epoch 907/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0449
Epoch 908/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0435
Epoch 909/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0427
Epoch 910/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0421
Epoch 911/1000
71/71 0s 4ms/step - loss:
0.0032 - mae: 0.0411
Epoch 912/1000
71/71 0s 3ms/step - loss:

0.0035 - mae: 0.0448
Epoch 913/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0418
Epoch 914/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0433
Epoch 915/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0421
Epoch 916/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0443
Epoch 917/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0422
Epoch 918/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0433
Epoch 919/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0436
Epoch 920/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0408
Epoch 921/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0399
Epoch 922/1000
71/71 0s 4ms/step - loss:
0.0036 - mae: 0.0456
Epoch 923/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0441
Epoch 924/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0429
Epoch 925/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0433
Epoch 926/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0423
Epoch 927/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0447
Epoch 928/1000
71/71 0s 3ms/step - loss:

0.0031 - mae: 0.0426
Epoch 929/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0438
Epoch 930/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0432
Epoch 931/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0431
Epoch 932/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0436
Epoch 933/1000
71/71 0s 4ms/step - loss:
0.0031 - mae: 0.0419
Epoch 934/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0443
Epoch 935/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0430
Epoch 936/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0445
Epoch 937/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0428
Epoch 938/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0430
Epoch 939/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0433
Epoch 940/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0424
Epoch 941/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0408
Epoch 942/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0440
Epoch 943/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0416
Epoch 944/1000
71/71 0s 4ms/step - loss:

0.0032 - mae: 0.0417
Epoch 945/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0422
Epoch 946/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0417
Epoch 947/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0442
Epoch 948/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0409
Epoch 949/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0434
Epoch 950/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0417
Epoch 951/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0430
Epoch 952/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0441
Epoch 953/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0412
Epoch 954/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0420
Epoch 955/1000
71/71 0s 4ms/step - loss:
0.0031 - mae: 0.0416
Epoch 956/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0416
Epoch 957/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0404
Epoch 958/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0423
Epoch 959/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0386
Epoch 960/1000
71/71 0s 3ms/step - loss:

0.0033 - mae: 0.0439
Epoch 961/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0413
Epoch 962/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0410
Epoch 963/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0426
Epoch 964/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0438
Epoch 965/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0430
Epoch 966/1000
71/71 0s 4ms/step - loss:
0.0030 - mae: 0.0410
Epoch 967/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0397
Epoch 968/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0438
Epoch 969/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0417
Epoch 970/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0418
Epoch 971/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0417
Epoch 972/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0418
Epoch 973/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0430
Epoch 974/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0404
Epoch 975/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0417
Epoch 976/1000
71/71 0s 3ms/step - loss:

0.0032 - mae: 0.0428
Epoch 977/1000
71/71 0s 4ms/step - loss:
0.0030 - mae: 0.0391
Epoch 978/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0417
Epoch 979/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0401
Epoch 980/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0427
Epoch 981/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0366
Epoch 982/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0428
Epoch 983/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0397
Epoch 984/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0419
Epoch 985/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0423
Epoch 986/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0421
Epoch 987/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0423
Epoch 988/1000
71/71 0s 4ms/step - loss:
0.0028 - mae: 0.0394
Epoch 989/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0431
Epoch 990/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0400
Epoch 991/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0426
Epoch 992/1000
71/71 0s 3ms/step - loss:

```

0.0028 - mae: 0.0383
Epoch 993/1000
71/71          0s 3ms/step - loss:
0.0029 - mae: 0.0403
Epoch 994/1000
71/71          0s 3ms/step - loss:
0.0034 - mae: 0.0454
Epoch 995/1000
71/71          0s 3ms/step - loss:
0.0031 - mae: 0.0406
Epoch 996/1000
71/71          0s 3ms/step - loss:
0.0028 - mae: 0.0402
Epoch 997/1000
71/71          0s 3ms/step - loss:
0.0032 - mae: 0.0408
Epoch 998/1000
71/71          0s 3ms/step - loss:
0.0030 - mae: 0.0412
Epoch 999/1000
71/71          0s 4ms/step - loss:
0.0028 - mae: 0.0394
Epoch 1000/1000
71/71          0s 3ms/step - loss:
0.0027 - mae: 0.0395
Epoch 1/1000
49/57          0s 2ms/step - loss:
0.0037 - mae: 0.0456

```

```

C:\Users\barre\my_new_env2\Lib\site-
packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran
out of data; interrupting training. Make sure that your dataset or generator can
generate at least `steps_per_epoch * epochs` batches. You may need to use the
`.repeat()` function when building your dataset.

```

```

    self._interrupted_warning()

57/57          0s 7ms/step - loss:
0.0031 - mae: 0.0421 - val_loss: 0.0035 - val_mae: 0.0492
Epoch 2/1000
57/57          0s 4ms/step - loss:
0.0029 - mae: 0.0414 - val_loss: 8.0025e-04 - val_mae: 0.0222
Epoch 3/1000
57/57          0s 4ms/step - loss:
0.0033 - mae: 0.0428 - val_loss: 0.0010 - val_mae: 0.0249
Epoch 4/1000
57/57          0s 4ms/step - loss:
0.0028 - mae: 0.0395 - val_loss: 0.0011 - val_mae: 0.0267
Epoch 5/1000
57/57          0s 4ms/step - loss:

```

0.0028 - mae: 0.0402 - val_loss: 0.0040 - val_mae: 0.0535
 Epoch 6/1000
 57/57 0s 4ms/step - loss:
 0.0030 - mae: 0.0414 - val_loss: 0.0024 - val_mae: 0.0409
 Epoch 7/1000
 57/57 0s 4ms/step - loss:
 0.0026 - mae: 0.0364 - val_loss: 0.0010 - val_mae: 0.0253
 Epoch 8/1000
 57/57 0s 4ms/step - loss:
 0.0032 - mae: 0.0418 - val_loss: 0.0018 - val_mae: 0.0340
 Epoch 9/1000
 57/57 0s 4ms/step - loss:
 0.0031 - mae: 0.0415 - val_loss: 0.0012 - val_mae: 0.0276
 Epoch 10/1000
 57/57 0s 4ms/step - loss:
 0.0025 - mae: 0.0361 - val_loss: 0.0077 - val_mae: 0.0783
 Epoch 11/1000
 57/57 0s 4ms/step - loss:
 0.0030 - mae: 0.0418 - val_loss: 8.5291e-04 - val_mae: 0.0221
 Epoch 12/1000
 57/57 0s 4ms/step - loss:
 0.0028 - mae: 0.0401 - val_loss: 0.0073 - val_mae: 0.0782
 Epoch 13/1000
 57/57 0s 4ms/step - loss:
 0.0030 - mae: 0.0406 - val_loss: 0.0057 - val_mae: 0.0614
 Epoch 14/1000
 57/57 0s 4ms/step - loss:
 0.0034 - mae: 0.0417 - val_loss: 9.7202e-04 - val_mae: 0.0242
 Epoch 15/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0413 - val_loss: 0.0015 - val_mae: 0.0305
 Epoch 16/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0418 - val_loss: 7.3737e-04 - val_mae: 0.0206
 Epoch 17/1000
 57/57 0s 4ms/step - loss:
 0.0031 - mae: 0.0437 - val_loss: 0.0020 - val_mae: 0.0354
 Epoch 18/1000
 57/57 0s 4ms/step - loss:
 0.0032 - mae: 0.0401 - val_loss: 0.0018 - val_mae: 0.0332
 Epoch 19/1000
 57/57 0s 4ms/step - loss:
 0.0026 - mae: 0.0388 - val_loss: 0.0039 - val_mae: 0.0469
 Epoch 20/1000
 57/57 0s 4ms/step - loss:
 0.0030 - mae: 0.0402 - val_loss: 0.0021 - val_mae: 0.0386
 Epoch 21/1000
 57/57 0s 4ms/step - loss:

0.0029 - mae: 0.0406 - val_loss: 0.0031 - val_mae: 0.0467
 Epoch 22/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0416 - val_loss: 0.0015 - val_mae: 0.0304
 Epoch 23/1000
 57/57 0s 4ms/step - loss:
 0.0031 - mae: 0.0423 - val_loss: 9.0634e-04 - val_mae: 0.0238
 Epoch 24/1000
 57/57 0s 4ms/step - loss:
 0.0031 - mae: 0.0411 - val_loss: 9.7169e-04 - val_mae: 0.0248
 Epoch 25/1000
 57/57 0s 4ms/step - loss:
 0.0032 - mae: 0.0406 - val_loss: 8.3120e-04 - val_mae: 0.0221
 Epoch 26/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0388 - val_loss: 0.0080 - val_mae: 0.0801
 Epoch 27/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0400 - val_loss: 0.0024 - val_mae: 0.0400
 Epoch 28/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0427 - val_loss: 0.0044 - val_mae: 0.0548
 Epoch 29/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0411 - val_loss: 0.0010 - val_mae: 0.0255
 Epoch 30/1000
 57/57 0s 4ms/step - loss:
 0.0030 - mae: 0.0427 - val_loss: 0.0031 - val_mae: 0.0454
 Epoch 31/1000
 57/57 0s 4ms/step - loss:
 0.0030 - mae: 0.0425 - val_loss: 9.5527e-04 - val_mae: 0.0239
 Epoch 32/1000
 57/57 0s 4ms/step - loss:
 0.0027 - mae: 0.0378 - val_loss: 0.0059 - val_mae: 0.0627
 Epoch 33/1000
 57/57 0s 4ms/step - loss:
 0.0031 - mae: 0.0406 - val_loss: 0.0050 - val_mae: 0.0570
 Epoch 34/1000
 57/57 0s 4ms/step - loss:
 0.0026 - mae: 0.0382 - val_loss: 0.0135 - val_mae: 0.0952
 Epoch 35/1000
 57/57 0s 4ms/step - loss:
 0.0036 - mae: 0.0448 - val_loss: 0.0012 - val_mae: 0.0280
 Epoch 36/1000
 57/57 0s 4ms/step - loss:
 0.0026 - mae: 0.0383 - val_loss: 0.0020 - val_mae: 0.0366
 Epoch 37/1000
 57/57 0s 4ms/step - loss:

0.0030 - mae: 0.0419 - val_loss: 0.0024 - val_mae: 0.0372
 Epoch 38/1000
 57/57 0s 4ms/step - loss:
 0.0027 - mae: 0.0390 - val_loss: 0.0014 - val_mae: 0.0299
 Epoch 39/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0402 - val_loss: 9.2060e-04 - val_mae: 0.0239
 Epoch 40/1000
 57/57 0s 4ms/step - loss:
 0.0026 - mae: 0.0381 - val_loss: 0.0017 - val_mae: 0.0323
 Epoch 41/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0413 - val_loss: 0.0062 - val_mae: 0.0635
 Epoch 42/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0394 - val_loss: 0.0010 - val_mae: 0.0253
 Epoch 43/1000
 57/57 0s 4ms/step - loss:
 0.0032 - mae: 0.0429 - val_loss: 0.0042 - val_mae: 0.0563
 Epoch 44/1000
 57/57 0s 4ms/step - loss:
 0.0028 - mae: 0.0386 - val_loss: 0.0069 - val_mae: 0.0649
 Epoch 45/1000
 57/57 0s 4ms/step - loss:
 0.0027 - mae: 0.0386 - val_loss: 0.0033 - val_mae: 0.0444
 Epoch 46/1000
 57/57 0s 4ms/step - loss:
 0.0028 - mae: 0.0398 - val_loss: 0.0022 - val_mae: 0.0383
 Epoch 47/1000
 57/57 0s 4ms/step - loss:
 0.0026 - mae: 0.0387 - val_loss: 0.0056 - val_mae: 0.0609
 Epoch 48/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0399 - val_loss: 0.0026 - val_mae: 0.0441
 Epoch 49/1000
 57/57 0s 4ms/step - loss:
 0.0033 - mae: 0.0412 - val_loss: 9.2856e-04 - val_mae: 0.0245
 Epoch 50/1000
 57/57 0s 4ms/step - loss:
 0.0027 - mae: 0.0378 - val_loss: 0.0013 - val_mae: 0.0275
 Epoch 51/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0411 - val_loss: 0.0021 - val_mae: 0.0383
 Epoch 52/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0384 - val_loss: 0.0037 - val_mae: 0.0502
 Epoch 53/1000
 57/57 0s 4ms/step - loss:

```

0.0026 - mae: 0.0382 - val_loss: 0.0015 - val_mae: 0.0302
Epoch 54/1000
57/57          0s 4ms/step - loss:
0.0027 - mae: 0.0402 - val_loss: 0.0017 - val_mae: 0.0330
Epoch 55/1000
57/57          0s 4ms/step - loss:
0.0028 - mae: 0.0394 - val_loss: 0.0076 - val_mae: 0.0700
Epoch 56/1000
57/57          0s 4ms/step - loss:
0.0028 - mae: 0.0407 - val_loss: 0.0059 - val_mae: 0.0660
Epoch 57/1000
57/57          0s 4ms/step - loss:
0.0029 - mae: 0.0415 - val_loss: 0.0027 - val_mae: 0.0410
Epoch 58/1000
57/57          0s 4ms/step - loss:
0.0028 - mae: 0.0392 - val_loss: 0.0013 - val_mae: 0.0290
Epoch 59/1000
57/57          0s 4ms/step - loss:
0.0030 - mae: 0.0427 - val_loss: 0.0025 - val_mae: 0.0396
Epoch 60/1000
57/57          0s 4ms/step - loss:
0.0028 - mae: 0.0389 - val_loss: 0.0011 - val_mae: 0.0255
Epoch 61/1000
57/57          0s 4ms/step - loss:
0.0030 - mae: 0.0415 - val_loss: 0.0025 - val_mae: 0.0406
Epoch 62/1000
57/57          0s 4ms/step - loss:
0.0027 - mae: 0.0376 - val_loss: 0.0050 - val_mae: 0.0577
Epoch 63/1000
57/57          0s 4ms/step - loss:
0.0027 - mae: 0.0393 - val_loss: 0.0030 - val_mae: 0.0423
Epoch 64/1000
57/57          0s 4ms/step - loss:
0.0028 - mae: 0.0398 - val_loss: 0.0024 - val_mae: 0.0379
Epoch 65/1000
57/57          0s 4ms/step - loss:
0.0030 - mae: 0.0390 - val_loss: 0.0042 - val_mae: 0.0492
Epoch 66/1000
57/57          0s 4ms/step - loss:
0.0027 - mae: 0.0385 - val_loss: 0.0032 - val_mae: 0.0458
Epoch 66: early stopping
Restoring model weights from the end of the best epoch: 16.
Raw evaluate results: []

```

```

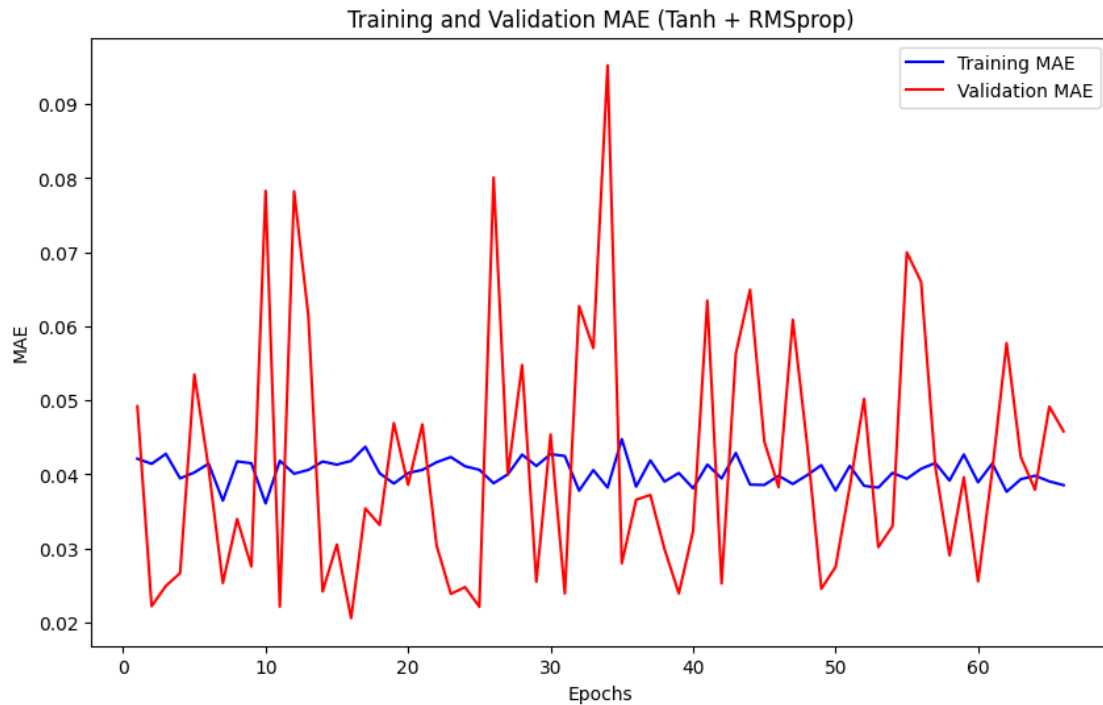
C:\Users\barre\my_new_env2\Lib\site-
packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran
out of data; interrupting training. Make sure that your dataset or generator can
generate at least `steps_per_epoch * epochs` batches. You may need to use the

```

```

.repeat()` function when building your dataset.
self._interrupted_warning()

```



```

[6]: print("\n--- Training Model with Sigmoid and SGD ---")
model_sig = Sequential()
model_sig.add(Dense(256, input_dim=train_features.shape[1], activation='tanh'))
model_sig.add(Dense(128, activation='sigmoid'))
model_sig.add(Dense(64, activation='sigmoid'))
model_sig.add(Dense(1, activation='linear'))

model_sig.compile(optimizer=SGD(learning_rate=0.001), loss="mse",
    ↪metrics=["mae"])

# Define early stopping
early_stop_sig = EarlyStopping(monitor='val_loss', patience=50, verbose=1,
    ↪restore_best_weights=True)

model_sig.fit(train_features, train_target,
              epochs=1000,
              batch_size=32,
              validation_data=(test_features, test_target),
              callbacks=[early_stop_tanh],
              verbose=1)

```



```

loss_sig = model_sig.evaluate(test_features, test_target, verbose=0)

history_sig = model_sig.fit(train_features, train_target,
                             epochs=1000,
                             batch_size=32,
                             validation_split=0.2,
                             callbacks=[early_stop_tanh],
                             verbose=1)

sig_results = model_sig.evaluate(test_features, test_target, verbose=0)
print("Raw evaluate results:", sig_results)

# Plot the training and validation MAE
mae_sig = history_sig.history['mae']
val_mae_sig = history_sig.history['val_mae']
epochs = range(1, len(mae_sig) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs, mae_sig, 'b', label='Training MAE')
plt.plot(epochs, val_mae_sig, 'r', label='Validation MAE')
plt.title('Training and Validation MAE (Sigmoid + SGD)')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()

```

--- Training Model with Sigmoid and SGD ---

Epoch 1/1000

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

71/71 1s 5ms/step - loss:

15.1651 - mae: 3.3292

Epoch 2/1000

55/71 0s 2ms/step - loss:

10.3761 - mae: 2.7964

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran out of data; interrupting training. Make sure that your dataset or generator can generate at least `steps\_per\_epoch \* epochs` batches. You may need to use the `.repeat()` function when building your dataset.

```
self._interrupted_warning()
```

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\callbacks\early_stopping.py:99: UserWarning: Early stopping

conditioned on metric `val\_loss` which is not available. Available metrics are:
loss,mae

```
current = self.get_monitor_value(logs)
```

71/71 0s 3ms/step - loss:

10.6324 - mae: 2.8550

Epoch 3/1000

71/71 0s 3ms/step - loss:

10.5323 - mae: 2.8404

Epoch 4/1000

71/71 0s 3ms/step - loss:

10.4174 - mae: 2.8201

Epoch 5/1000

71/71 0s 3ms/step - loss:

10.3060 - mae: 2.8103

Epoch 6/1000

71/71 0s 3ms/step - loss:

10.1772 - mae: 2.7919

Epoch 7/1000

71/71 0s 3ms/step - loss:

10.0349 - mae: 2.7676

Epoch 8/1000

71/71 0s 3ms/step - loss:

9.8616 - mae: 2.7496

Epoch 9/1000

71/71 0s 3ms/step - loss:

9.6770 - mae: 2.7188

Epoch 10/1000

71/71 0s 4ms/step - loss:

9.4447 - mae: 2.6902

Epoch 11/1000

71/71 0s 3ms/step - loss:

9.1716 - mae: 2.6470

Epoch 12/1000

71/71 0s 3ms/step - loss:

8.8422 - mae: 2.5994

Epoch 13/1000

71/71 0s 3ms/step - loss:

8.4373 - mae: 2.5440

Epoch 14/1000

71/71 0s 3ms/step - loss:

7.9621 - mae: 2.4643

Epoch 15/1000

71/71 0s 3ms/step - loss:

7.3966 - mae: 2.3809

Epoch 16/1000

71/71 0s 3ms/step - loss:

6.7276 - mae: 2.2700

Epoch 17/1000
71/71 0s 3ms/step - loss:
5.9772 - mae: 2.1374
Epoch 18/1000
71/71 0s 3ms/step - loss:
5.1824 - mae: 1.9917
Epoch 19/1000
71/71 0s 4ms/step - loss:
4.3967 - mae: 1.8312
Epoch 20/1000
71/71 0s 3ms/step - loss:
3.6809 - mae: 1.6685
Epoch 21/1000
71/71 0s 3ms/step - loss:
3.0927 - mae: 1.5168
Epoch 22/1000
71/71 0s 3ms/step - loss:
2.6271 - mae: 1.3809
Epoch 23/1000
71/71 0s 3ms/step - loss:
2.2782 - mae: 1.2651
Epoch 24/1000
71/71 0s 3ms/step - loss:
2.0107 - mae: 1.1700
Epoch 25/1000
71/71 0s 3ms/step - loss:
1.8014 - mae: 1.0911
Epoch 26/1000
71/71 0s 3ms/step - loss:
1.6277 - mae: 1.0252
Epoch 27/1000
71/71 0s 3ms/step - loss:
1.4735 - mae: 0.9645
Epoch 28/1000
71/71 0s 3ms/step - loss:
1.3355 - mae: 0.9108
Epoch 29/1000
71/71 0s 3ms/step - loss:
1.2083 - mae: 0.8621
Epoch 30/1000
71/71 0s 4ms/step - loss:
1.0956 - mae: 0.8170
Epoch 31/1000
71/71 0s 3ms/step - loss:
0.9901 - mae: 0.7735
Epoch 32/1000
71/71 0s 3ms/step - loss:
0.8951 - mae: 0.7324

Epoch 33/1000
71/71 0s 3ms/step - loss:
0.8081 - mae: 0.6953
Epoch 34/1000
71/71 0s 3ms/step - loss:
0.7307 - mae: 0.6595
Epoch 35/1000
71/71 0s 3ms/step - loss:
0.6615 - mae: 0.6253
Epoch 36/1000
71/71 0s 3ms/step - loss:
0.5995 - mae: 0.5935
Epoch 37/1000
71/71 0s 3ms/step - loss:
0.5460 - mae: 0.5662
Epoch 38/1000
71/71 0s 3ms/step - loss:
0.4983 - mae: 0.5393
Epoch 39/1000
71/71 0s 3ms/step - loss:
0.4580 - mae: 0.5165
Epoch 40/1000
71/71 0s 3ms/step - loss:
0.4220 - mae: 0.4949
Epoch 41/1000
71/71 0s 4ms/step - loss:
0.3912 - mae: 0.4750
Epoch 42/1000
71/71 0s 3ms/step - loss:
0.3653 - mae: 0.4589
Epoch 43/1000
71/71 0s 3ms/step - loss:
0.3428 - mae: 0.4440
Epoch 44/1000
71/71 0s 3ms/step - loss:
0.3238 - mae: 0.4321
Epoch 45/1000
71/71 0s 3ms/step - loss:
0.3072 - mae: 0.4204
Epoch 46/1000
71/71 0s 3ms/step - loss:
0.2929 - mae: 0.4100
Epoch 47/1000
71/71 0s 3ms/step - loss:
0.2817 - mae: 0.4025
Epoch 48/1000
71/71 0s 3ms/step - loss:
0.2709 - mae: 0.3947

Epoch 49/1000
71/71 0s 3ms/step - loss:
0.2619 - mae: 0.3882
Epoch 50/1000
71/71 0s 3ms/step - loss:
0.2544 - mae: 0.3823
Epoch 51/1000
71/71 0s 3ms/step - loss:
0.2474 - mae: 0.3776
Epoch 52/1000
71/71 0s 4ms/step - loss:
0.2417 - mae: 0.3725
Epoch 53/1000
71/71 0s 3ms/step - loss:
0.2356 - mae: 0.3683
Epoch 54/1000
71/71 0s 3ms/step - loss:
0.2308 - mae: 0.3647
Epoch 55/1000
71/71 0s 3ms/step - loss:
0.2262 - mae: 0.3615
Epoch 56/1000
71/71 0s 3ms/step - loss:
0.2226 - mae: 0.3585
Epoch 57/1000
71/71 0s 3ms/step - loss:
0.2184 - mae: 0.3555
Epoch 58/1000
71/71 0s 3ms/step - loss:
0.2144 - mae: 0.3525
Epoch 59/1000
71/71 0s 3ms/step - loss:
0.2114 - mae: 0.3500
Epoch 60/1000
71/71 0s 3ms/step - loss:
0.2080 - mae: 0.3474
Epoch 61/1000
71/71 0s 3ms/step - loss:
0.2048 - mae: 0.3446
Epoch 62/1000
71/71 0s 3ms/step - loss:
0.2019 - mae: 0.3421
Epoch 63/1000
71/71 0s 4ms/step - loss:
0.1990 - mae: 0.3399
Epoch 64/1000
71/71 0s 3ms/step - loss:
0.1958 - mae: 0.3375

Epoch 65/1000
71/71 0s 3ms/step - loss:
0.1937 - mae: 0.3352
Epoch 66/1000
71/71 0s 3ms/step - loss:
0.1912 - mae: 0.3332
Epoch 67/1000
71/71 0s 3ms/step - loss:
0.1886 - mae: 0.3312
Epoch 68/1000
71/71 0s 3ms/step - loss:
0.1861 - mae: 0.3294
Epoch 69/1000
71/71 0s 3ms/step - loss:
0.1834 - mae: 0.3271
Epoch 70/1000
71/71 0s 3ms/step - loss:
0.1812 - mae: 0.3247
Epoch 71/1000
71/71 0s 3ms/step - loss:
0.1790 - mae: 0.3226
Epoch 72/1000
71/71 0s 3ms/step - loss:
0.1765 - mae: 0.3202
Epoch 73/1000
71/71 0s 3ms/step - loss:
0.1747 - mae: 0.3188
Epoch 74/1000
71/71 0s 4ms/step - loss:
0.1721 - mae: 0.3165
Epoch 75/1000
71/71 0s 3ms/step - loss:
0.1706 - mae: 0.3153
Epoch 76/1000
71/71 0s 3ms/step - loss:
0.1686 - mae: 0.3133
Epoch 77/1000
71/71 0s 3ms/step - loss:
0.1664 - mae: 0.3117
Epoch 78/1000
71/71 0s 3ms/step - loss:
0.1644 - mae: 0.3096
Epoch 79/1000
71/71 0s 3ms/step - loss:
0.1627 - mae: 0.3081
Epoch 80/1000
71/71 0s 3ms/step - loss:
0.1608 - mae: 0.3064

Epoch 81/1000
71/71 0s 3ms/step - loss:
0.1585 - mae: 0.3038
Epoch 82/1000
71/71 0s 3ms/step - loss:
0.1575 - mae: 0.3033
Epoch 83/1000
71/71 0s 3ms/step - loss:
0.1556 - mae: 0.3012
Epoch 84/1000
71/71 0s 3ms/step - loss:
0.1541 - mae: 0.3001
Epoch 85/1000
71/71 0s 4ms/step - loss:
0.1521 - mae: 0.2982
Epoch 86/1000
71/71 0s 3ms/step - loss:
0.1506 - mae: 0.2963
Epoch 87/1000
71/71 0s 3ms/step - loss:
0.1488 - mae: 0.2949
Epoch 88/1000
71/71 0s 3ms/step - loss:
0.1473 - mae: 0.2935
Epoch 89/1000
71/71 0s 3ms/step - loss:
0.1456 - mae: 0.2919
Epoch 90/1000
71/71 0s 3ms/step - loss:
0.1444 - mae: 0.2906
Epoch 91/1000
71/71 0s 3ms/step - loss:
0.1428 - mae: 0.2888
Epoch 92/1000
71/71 0s 3ms/step - loss:
0.1412 - mae: 0.2874
Epoch 93/1000
71/71 0s 3ms/step - loss:
0.1400 - mae: 0.2863
Epoch 94/1000
71/71 0s 3ms/step - loss:
0.1386 - mae: 0.2847
Epoch 95/1000
71/71 0s 3ms/step - loss:
0.1371 - mae: 0.2835
Epoch 96/1000
71/71 0s 5ms/step - loss:
0.1361 - mae: 0.2822

Epoch 97/1000
71/71 0s 3ms/step - loss:
0.1347 - mae: 0.2804
Epoch 98/1000
71/71 0s 3ms/step - loss:
0.1332 - mae: 0.2795
Epoch 99/1000
71/71 0s 3ms/step - loss:
0.1322 - mae: 0.2787
Epoch 100/1000
71/71 0s 3ms/step - loss:
0.1309 - mae: 0.2766
Epoch 101/1000
71/71 0s 3ms/step - loss:
0.1297 - mae: 0.2758
Epoch 102/1000
71/71 0s 3ms/step - loss:
0.1285 - mae: 0.2747
Epoch 103/1000
71/71 0s 3ms/step - loss:
0.1273 - mae: 0.2736
Epoch 104/1000
71/71 0s 3ms/step - loss:
0.1262 - mae: 0.2719
Epoch 105/1000
71/71 0s 3ms/step - loss:
0.1247 - mae: 0.2707
Epoch 106/1000
71/71 0s 3ms/step - loss:
0.1239 - mae: 0.2699
Epoch 107/1000
71/71 0s 4ms/step - loss:
0.1230 - mae: 0.2686
Epoch 108/1000
71/71 0s 3ms/step - loss:
0.1217 - mae: 0.2674
Epoch 109/1000
71/71 0s 3ms/step - loss:
0.1209 - mae: 0.2663
Epoch 110/1000
71/71 0s 3ms/step - loss:
0.1200 - mae: 0.2652
Epoch 111/1000
71/71 0s 3ms/step - loss:
0.1189 - mae: 0.2644
Epoch 112/1000
71/71 0s 3ms/step - loss:
0.1177 - mae: 0.2629

Epoch 113/1000
71/71 0s 3ms/step - loss:
0.1170 - mae: 0.2621
Epoch 114/1000
71/71 0s 3ms/step - loss:
0.1159 - mae: 0.2612
Epoch 115/1000
71/71 0s 3ms/step - loss:
0.1151 - mae: 0.2598
Epoch 116/1000
71/71 0s 3ms/step - loss:
0.1141 - mae: 0.2589
Epoch 117/1000
71/71 0s 3ms/step - loss:
0.1134 - mae: 0.2580
Epoch 118/1000
71/71 0s 4ms/step - loss:
0.1124 - mae: 0.2570
Epoch 119/1000
71/71 0s 3ms/step - loss:
0.1114 - mae: 0.2561
Epoch 120/1000
71/71 0s 3ms/step - loss:
0.1105 - mae: 0.2551
Epoch 121/1000
71/71 0s 3ms/step - loss:
0.1100 - mae: 0.2542
Epoch 122/1000
71/71 0s 3ms/step - loss:
0.1091 - mae: 0.2530
Epoch 123/1000
71/71 0s 3ms/step - loss:
0.1085 - mae: 0.2522
Epoch 124/1000
71/71 0s 3ms/step - loss:
0.1076 - mae: 0.2512
Epoch 125/1000
71/71 0s 3ms/step - loss:
0.1069 - mae: 0.2505
Epoch 126/1000
71/71 0s 3ms/step - loss:
0.1062 - mae: 0.2495
Epoch 127/1000
71/71 0s 3ms/step - loss:
0.1053 - mae: 0.2484
Epoch 128/1000
71/71 0s 3ms/step - loss:
0.1046 - mae: 0.2479

Epoch 129/1000
71/71 0s 4ms/step - loss:
0.1040 - mae: 0.2473
Epoch 130/1000
71/71 0s 3ms/step - loss:
0.1031 - mae: 0.2456
Epoch 131/1000
71/71 0s 3ms/step - loss:
0.1026 - mae: 0.2451
Epoch 132/1000
71/71 0s 3ms/step - loss:
0.1019 - mae: 0.2443
Epoch 133/1000
71/71 0s 3ms/step - loss:
0.1012 - mae: 0.2435
Epoch 134/1000
71/71 0s 3ms/step - loss:
0.1007 - mae: 0.2427
Epoch 135/1000
71/71 0s 3ms/step - loss:
0.1000 - mae: 0.2418
Epoch 136/1000
71/71 0s 3ms/step - loss:
0.0995 - mae: 0.2408
Epoch 137/1000
71/71 0s 3ms/step - loss:
0.0988 - mae: 0.2405
Epoch 138/1000
71/71 0s 3ms/step - loss:
0.0982 - mae: 0.2397
Epoch 139/1000
71/71 0s 3ms/step - loss:
0.0977 - mae: 0.2390
Epoch 140/1000
71/71 0s 4ms/step - loss:
0.0970 - mae: 0.2381
Epoch 141/1000
71/71 0s 3ms/step - loss:
0.0965 - mae: 0.2371
Epoch 142/1000
71/71 0s 3ms/step - loss:
0.0960 - mae: 0.2365
Epoch 143/1000
71/71 0s 3ms/step - loss:
0.0953 - mae: 0.2353
Epoch 144/1000
71/71 0s 3ms/step - loss:
0.0949 - mae: 0.2350

Epoch 145/1000
71/71 0s 3ms/step - loss:
0.0943 - mae: 0.2348
Epoch 146/1000
71/71 0s 3ms/step - loss:
0.0938 - mae: 0.2338
Epoch 147/1000
71/71 0s 3ms/step - loss:
0.0934 - mae: 0.2333
Epoch 148/1000
71/71 0s 3ms/step - loss:
0.0928 - mae: 0.2323
Epoch 149/1000
71/71 0s 3ms/step - loss:
0.0924 - mae: 0.2319
Epoch 150/1000
71/71 0s 3ms/step - loss:
0.0918 - mae: 0.2308
Epoch 151/1000
71/71 0s 4ms/step - loss:
0.0914 - mae: 0.2301
Epoch 152/1000
71/71 0s 3ms/step - loss:
0.0907 - mae: 0.2297
Epoch 153/1000
71/71 0s 3ms/step - loss:
0.0905 - mae: 0.2291
Epoch 154/1000
71/71 0s 3ms/step - loss:
0.0901 - mae: 0.2282
Epoch 155/1000
71/71 0s 3ms/step - loss:
0.0896 - mae: 0.2279
Epoch 156/1000
71/71 0s 3ms/step - loss:
0.0893 - mae: 0.2277
Epoch 157/1000
71/71 0s 3ms/step - loss:
0.0885 - mae: 0.2263
Epoch 158/1000
71/71 0s 3ms/step - loss:
0.0881 - mae: 0.2259
Epoch 159/1000
71/71 0s 3ms/step - loss:
0.0876 - mae: 0.2248
Epoch 160/1000
71/71 0s 3ms/step - loss:
0.0875 - mae: 0.2247

Epoch 161/1000
71/71 0s 3ms/step - loss:
0.0873 - mae: 0.2244
Epoch 162/1000
71/71 0s 4ms/step - loss:
0.0867 - mae: 0.2237
Epoch 163/1000
71/71 0s 3ms/step - loss:
0.0862 - mae: 0.2229
Epoch 164/1000
71/71 0s 3ms/step - loss:
0.0860 - mae: 0.2224
Epoch 165/1000
71/71 0s 3ms/step - loss:
0.0855 - mae: 0.2222
Epoch 166/1000
71/71 0s 3ms/step - loss:
0.0851 - mae: 0.2213
Epoch 167/1000
71/71 0s 3ms/step - loss:
0.0847 - mae: 0.2207
Epoch 168/1000
71/71 0s 3ms/step - loss:
0.0846 - mae: 0.2206
Epoch 169/1000
71/71 0s 3ms/step - loss:
0.0843 - mae: 0.2203
Epoch 170/1000
71/71 0s 3ms/step - loss:
0.0839 - mae: 0.2196
Epoch 171/1000
71/71 0s 3ms/step - loss:
0.0835 - mae: 0.2188
Epoch 172/1000
71/71 0s 3ms/step - loss:
0.0831 - mae: 0.2184
Epoch 173/1000
71/71 0s 4ms/step - loss:
0.0828 - mae: 0.2181
Epoch 174/1000
71/71 0s 3ms/step - loss:
0.0825 - mae: 0.2178
Epoch 175/1000
71/71 0s 3ms/step - loss:
0.0823 - mae: 0.2174
Epoch 176/1000
71/71 0s 3ms/step - loss:
0.0819 - mae: 0.2166

Epoch 177/1000
71/71 0s 3ms/step - loss:
0.0815 - mae: 0.2160
Epoch 178/1000
71/71 0s 3ms/step - loss:
0.0813 - mae: 0.2160
Epoch 179/1000
71/71 0s 3ms/step - loss:
0.0810 - mae: 0.2154
Epoch 180/1000
71/71 0s 3ms/step - loss:
0.0808 - mae: 0.2151
Epoch 181/1000
71/71 0s 3ms/step - loss:
0.0805 - mae: 0.2148
Epoch 182/1000
71/71 0s 3ms/step - loss:
0.0802 - mae: 0.2143
Epoch 183/1000
71/71 0s 3ms/step - loss:
0.0800 - mae: 0.2140
Epoch 184/1000
71/71 0s 4ms/step - loss:
0.0798 - mae: 0.2139
Epoch 185/1000
71/71 0s 3ms/step - loss:
0.0793 - mae: 0.2130
Epoch 186/1000
71/71 0s 3ms/step - loss:
0.0792 - mae: 0.2131
Epoch 187/1000
71/71 0s 3ms/step - loss:
0.0788 - mae: 0.2123
Epoch 188/1000
71/71 0s 3ms/step - loss:
0.0785 - mae: 0.2114
Epoch 189/1000
71/71 0s 3ms/step - loss:
0.0785 - mae: 0.2118
Epoch 190/1000
71/71 0s 3ms/step - loss:
0.0782 - mae: 0.2112
Epoch 191/1000
71/71 0s 3ms/step - loss:
0.0778 - mae: 0.2106
Epoch 192/1000
71/71 0s 3ms/step - loss:
0.0776 - mae: 0.2105

Epoch 193/1000
71/71 0s 3ms/step - loss:
0.0773 - mae: 0.2099
Epoch 194/1000
71/71 0s 3ms/step - loss:
0.0772 - mae: 0.2099
Epoch 195/1000
71/71 0s 4ms/step - loss:
0.0769 - mae: 0.2091
Epoch 196/1000
71/71 0s 3ms/step - loss:
0.0766 - mae: 0.2088
Epoch 197/1000
71/71 0s 3ms/step - loss:
0.0764 - mae: 0.2088
Epoch 198/1000
71/71 0s 3ms/step - loss:
0.0763 - mae: 0.2085
Epoch 199/1000
71/71 0s 3ms/step - loss:
0.0758 - mae: 0.2080
Epoch 200/1000
71/71 0s 3ms/step - loss:
0.0759 - mae: 0.2078
Epoch 201/1000
71/71 0s 3ms/step - loss:
0.0755 - mae: 0.2072
Epoch 202/1000
71/71 0s 3ms/step - loss:
0.0754 - mae: 0.2071
Epoch 203/1000
71/71 0s 3ms/step - loss:
0.0751 - mae: 0.2065
Epoch 204/1000
71/71 0s 3ms/step - loss:
0.0749 - mae: 0.2064
Epoch 205/1000
71/71 0s 3ms/step - loss:
0.0747 - mae: 0.2058
Epoch 206/1000
71/71 0s 4ms/step - loss:
0.0747 - mae: 0.2058
Epoch 207/1000
71/71 0s 3ms/step - loss:
0.0743 - mae: 0.2055
Epoch 208/1000
71/71 0s 3ms/step - loss:
0.0742 - mae: 0.2052

Epoch 209/1000
71/71 0s 3ms/step - loss:
0.0741 - mae: 0.2048
Epoch 210/1000
71/71 0s 3ms/step - loss:
0.0739 - mae: 0.2045
Epoch 211/1000
71/71 0s 3ms/step - loss:
0.0736 - mae: 0.2043
Epoch 212/1000
71/71 0s 3ms/step - loss:
0.0734 - mae: 0.2039
Epoch 213/1000
71/71 0s 3ms/step - loss:
0.0731 - mae: 0.2032
Epoch 214/1000
71/71 0s 3ms/step - loss:
0.0731 - mae: 0.2035
Epoch 215/1000
71/71 0s 3ms/step - loss:
0.0729 - mae: 0.2032
Epoch 216/1000
71/71 0s 3ms/step - loss:
0.0726 - mae: 0.2029
Epoch 217/1000
71/71 0s 4ms/step - loss:
0.0726 - mae: 0.2027
Epoch 218/1000
71/71 0s 3ms/step - loss:
0.0724 - mae: 0.2021
Epoch 219/1000
71/71 0s 3ms/step - loss:
0.0721 - mae: 0.2019
Epoch 220/1000
71/71 0s 3ms/step - loss:
0.0720 - mae: 0.2018
Epoch 221/1000
71/71 0s 3ms/step - loss:
0.0718 - mae: 0.2017
Epoch 222/1000
71/71 0s 3ms/step - loss:
0.0718 - mae: 0.2013
Epoch 223/1000
71/71 0s 3ms/step - loss:
0.0715 - mae: 0.2011
Epoch 224/1000
71/71 0s 3ms/step - loss:
0.0712 - mae: 0.2006

Epoch 225/1000
71/71 0s 3ms/step - loss:
0.0713 - mae: 0.2006
Epoch 226/1000
71/71 0s 3ms/step - loss:
0.0713 - mae: 0.2009
Epoch 227/1000
71/71 0s 3ms/step - loss:
0.0709 - mae: 0.2003
Epoch 228/1000
71/71 0s 5ms/step - loss:
0.0708 - mae: 0.1997
Epoch 229/1000
71/71 0s 3ms/step - loss:
0.0706 - mae: 0.1994
Epoch 230/1000
71/71 0s 3ms/step - loss:
0.0704 - mae: 0.1996
Epoch 231/1000
71/71 0s 3ms/step - loss:
0.0705 - mae: 0.1995
Epoch 232/1000
71/71 0s 3ms/step - loss:
0.0702 - mae: 0.1988
Epoch 233/1000
71/71 0s 3ms/step - loss:
0.0701 - mae: 0.1988
Epoch 234/1000
71/71 0s 3ms/step - loss:
0.0699 - mae: 0.1985
Epoch 235/1000
71/71 0s 3ms/step - loss:
0.0697 - mae: 0.1982
Epoch 236/1000
71/71 0s 3ms/step - loss:
0.0696 - mae: 0.1979
Epoch 237/1000
71/71 0s 3ms/step - loss:
0.0695 - mae: 0.1981
Epoch 238/1000
71/71 0s 3ms/step - loss:
0.0693 - mae: 0.1974
Epoch 239/1000
71/71 0s 4ms/step - loss:
0.0692 - mae: 0.1972
Epoch 240/1000
71/71 0s 3ms/step - loss:
0.0692 - mae: 0.1970

Epoch 241/1000
71/71 0s 3ms/step - loss:
0.0690 - mae: 0.1970
Epoch 242/1000
71/71 0s 3ms/step - loss:
0.0688 - mae: 0.1970
Epoch 243/1000
71/71 0s 3ms/step - loss:
0.0687 - mae: 0.1965
Epoch 244/1000
71/71 0s 3ms/step - loss:
0.0687 - mae: 0.1966
Epoch 245/1000
71/71 0s 3ms/step - loss:
0.0684 - mae: 0.1962
Epoch 246/1000
71/71 0s 3ms/step - loss:
0.0685 - mae: 0.1962
Epoch 247/1000
71/71 0s 3ms/step - loss:
0.0682 - mae: 0.1957
Epoch 248/1000
71/71 0s 3ms/step - loss:
0.0681 - mae: 0.1955
Epoch 249/1000
71/71 0s 3ms/step - loss:
0.0678 - mae: 0.1956
Epoch 250/1000
71/71 0s 4ms/step - loss:
0.0679 - mae: 0.1954
Epoch 251/1000
71/71 0s 3ms/step - loss:
0.0678 - mae: 0.1952
Epoch 252/1000
71/71 0s 3ms/step - loss:
0.0677 - mae: 0.1949
Epoch 253/1000
71/71 0s 3ms/step - loss:
0.0676 - mae: 0.1948
Epoch 254/1000
71/71 0s 3ms/step - loss:
0.0674 - mae: 0.1947
Epoch 255/1000
71/71 0s 3ms/step - loss:
0.0674 - mae: 0.1944
Epoch 256/1000
71/71 0s 3ms/step - loss:
0.0672 - mae: 0.1944

Epoch 257/1000
71/71 0s 3ms/step - loss:
0.0670 - mae: 0.1939
Epoch 258/1000
71/71 0s 3ms/step - loss:
0.0671 - mae: 0.1939
Epoch 259/1000
71/71 0s 3ms/step - loss:
0.0669 - mae: 0.1941
Epoch 260/1000
71/71 0s 3ms/step - loss:
0.0667 - mae: 0.1935
Epoch 261/1000
71/71 0s 4ms/step - loss:
0.0667 - mae: 0.1935
Epoch 262/1000
71/71 0s 3ms/step - loss:
0.0665 - mae: 0.1931
Epoch 263/1000
71/71 0s 3ms/step - loss:
0.0664 - mae: 0.1931
Epoch 264/1000
71/71 0s 3ms/step - loss:
0.0663 - mae: 0.1928
Epoch 265/1000
71/71 0s 3ms/step - loss:
0.0661 - mae: 0.1926
Epoch 266/1000
71/71 0s 3ms/step - loss:
0.0661 - mae: 0.1925
Epoch 267/1000
71/71 0s 3ms/step - loss:
0.0661 - mae: 0.1926
Epoch 268/1000
71/71 0s 3ms/step - loss:
0.0659 - mae: 0.1921
Epoch 269/1000
71/71 0s 3ms/step - loss:
0.0658 - mae: 0.1922
Epoch 270/1000
71/71 0s 3ms/step - loss:
0.0656 - mae: 0.1917
Epoch 271/1000
71/71 0s 3ms/step - loss:
0.0656 - mae: 0.1920
Epoch 272/1000
71/71 0s 4ms/step - loss:
0.0655 - mae: 0.1916

Epoch 273/1000
71/71 0s 3ms/step - loss:
0.0652 - mae: 0.1910
Epoch 274/1000
71/71 0s 3ms/step - loss:
0.0654 - mae: 0.1914
Epoch 275/1000
71/71 0s 3ms/step - loss:
0.0652 - mae: 0.1914
Epoch 276/1000
71/71 0s 3ms/step - loss:
0.0652 - mae: 0.1910
Epoch 277/1000
71/71 0s 3ms/step - loss:
0.0651 - mae: 0.1912
Epoch 278/1000
71/71 0s 3ms/step - loss:
0.0649 - mae: 0.1906
Epoch 279/1000
71/71 0s 3ms/step - loss:
0.0651 - mae: 0.1909
Epoch 280/1000
71/71 0s 3ms/step - loss:
0.0648 - mae: 0.1906
Epoch 281/1000
71/71 0s 3ms/step - loss:
0.0648 - mae: 0.1903
Epoch 282/1000
71/71 0s 3ms/step - loss:
0.0646 - mae: 0.1904
Epoch 283/1000
71/71 0s 4ms/step - loss:
0.0645 - mae: 0.1898
Epoch 284/1000
71/71 0s 3ms/step - loss:
0.0645 - mae: 0.1900
Epoch 285/1000
71/71 0s 3ms/step - loss:
0.0644 - mae: 0.1898
Epoch 286/1000
71/71 0s 3ms/step - loss:
0.0643 - mae: 0.1896
Epoch 287/1000
71/71 0s 3ms/step - loss:
0.0641 - mae: 0.1894
Epoch 288/1000
71/71 0s 3ms/step - loss:
0.0640 - mae: 0.1892

Epoch 289/1000
71/71 0s 3ms/step - loss:
0.0641 - mae: 0.1893
Epoch 290/1000
71/71 0s 3ms/step - loss:
0.0638 - mae: 0.1889
Epoch 291/1000
71/71 0s 3ms/step - loss:
0.0638 - mae: 0.1890
Epoch 292/1000
71/71 0s 3ms/step - loss:
0.0638 - mae: 0.1893
Epoch 293/1000
71/71 0s 3ms/step - loss:
0.0637 - mae: 0.1888
Epoch 294/1000
71/71 0s 4ms/step - loss:
0.0636 - mae: 0.1888
Epoch 295/1000
71/71 0s 3ms/step - loss:
0.0636 - mae: 0.1887
Epoch 296/1000
71/71 0s 3ms/step - loss:
0.0634 - mae: 0.1878
Epoch 297/1000
71/71 0s 3ms/step - loss:
0.0635 - mae: 0.1886
Epoch 298/1000
71/71 0s 3ms/step - loss:
0.0632 - mae: 0.1881
Epoch 299/1000
71/71 0s 3ms/step - loss:
0.0633 - mae: 0.1882
Epoch 300/1000
71/71 0s 3ms/step - loss:
0.0632 - mae: 0.1881
Epoch 301/1000
71/71 0s 3ms/step - loss:
0.0631 - mae: 0.1878
Epoch 302/1000
71/71 0s 3ms/step - loss:
0.0629 - mae: 0.1878
Epoch 303/1000
71/71 0s 3ms/step - loss:
0.0630 - mae: 0.1877
Epoch 304/1000
71/71 0s 3ms/step - loss:
0.0629 - mae: 0.1874

Epoch 305/1000
71/71 0s 4ms/step - loss:
0.0627 - mae: 0.1871
Epoch 306/1000
71/71 0s 3ms/step - loss:
0.0627 - mae: 0.1870
Epoch 307/1000
71/71 0s 3ms/step - loss:
0.0625 - mae: 0.1872
Epoch 308/1000
71/71 0s 3ms/step - loss:
0.0625 - mae: 0.1870
Epoch 309/1000
71/71 0s 3ms/step - loss:
0.0624 - mae: 0.1869
Epoch 310/1000
71/71 0s 3ms/step - loss:
0.0625 - mae: 0.1871
Epoch 311/1000
71/71 0s 3ms/step - loss:
0.0624 - mae: 0.1868
Epoch 312/1000
71/71 0s 3ms/step - loss:
0.0624 - mae: 0.1866
Epoch 313/1000
71/71 0s 3ms/step - loss:
0.0622 - mae: 0.1864
Epoch 314/1000
71/71 0s 3ms/step - loss:
0.0622 - mae: 0.1865
Epoch 315/1000
71/71 0s 3ms/step - loss:
0.0620 - mae: 0.1861
Epoch 316/1000
71/71 0s 4ms/step - loss:
0.0620 - mae: 0.1862
Epoch 317/1000
71/71 0s 3ms/step - loss:
0.0617 - mae: 0.1859
Epoch 318/1000
71/71 0s 3ms/step - loss:
0.0619 - mae: 0.1859
Epoch 319/1000
71/71 0s 3ms/step - loss:
0.0618 - mae: 0.1859
Epoch 320/1000
71/71 0s 3ms/step - loss:
0.0617 - mae: 0.1858

Epoch 321/1000
71/71 0s 3ms/step - loss:
0.0616 - mae: 0.1855
Epoch 322/1000
71/71 0s 3ms/step - loss:
0.0615 - mae: 0.1856
Epoch 323/1000
71/71 0s 3ms/step - loss:
0.0615 - mae: 0.1852
Epoch 324/1000
71/71 0s 3ms/step - loss:
0.0614 - mae: 0.1853
Epoch 325/1000
71/71 0s 3ms/step - loss:
0.0613 - mae: 0.1851
Epoch 326/1000
71/71 0s 3ms/step - loss:
0.0613 - mae: 0.1852
Epoch 327/1000
71/71 0s 4ms/step - loss:
0.0612 - mae: 0.1848
Epoch 328/1000
71/71 0s 3ms/step - loss:
0.0611 - mae: 0.1849
Epoch 329/1000
71/71 0s 3ms/step - loss:
0.0611 - mae: 0.1849
Epoch 330/1000
71/71 0s 3ms/step - loss:
0.0610 - mae: 0.1842
Epoch 331/1000
71/71 0s 3ms/step - loss:
0.0609 - mae: 0.1846
Epoch 332/1000
71/71 0s 3ms/step - loss:
0.0609 - mae: 0.1841
Epoch 333/1000
71/71 0s 3ms/step - loss:
0.0609 - mae: 0.1846
Epoch 334/1000
71/71 0s 3ms/step - loss:
0.0608 - mae: 0.1844
Epoch 335/1000
71/71 0s 3ms/step - loss:
0.0607 - mae: 0.1844
Epoch 336/1000
71/71 0s 3ms/step - loss:
0.0606 - mae: 0.1842

Epoch 337/1000
71/71 0s 3ms/step - loss:
0.0606 - mae: 0.1840
Epoch 338/1000
71/71 0s 4ms/step - loss:
0.0605 - mae: 0.1835
Epoch 339/1000
71/71 0s 3ms/step - loss:
0.0605 - mae: 0.1838
Epoch 340/1000
71/71 0s 3ms/step - loss:
0.0605 - mae: 0.1836
Epoch 341/1000
71/71 0s 3ms/step - loss:
0.0604 - mae: 0.1835
Epoch 342/1000
71/71 0s 3ms/step - loss:
0.0602 - mae: 0.1832
Epoch 343/1000
71/71 0s 3ms/step - loss:
0.0602 - mae: 0.1834
Epoch 344/1000
71/71 0s 3ms/step - loss:
0.0602 - mae: 0.1832
Epoch 345/1000
71/71 0s 3ms/step - loss:
0.0602 - mae: 0.1834
Epoch 346/1000
71/71 0s 3ms/step - loss:
0.0601 - mae: 0.1828
Epoch 347/1000
71/71 0s 3ms/step - loss:
0.0600 - mae: 0.1829
Epoch 348/1000
71/71 0s 3ms/step - loss:
0.0600 - mae: 0.1825
Epoch 349/1000
71/71 0s 4ms/step - loss:
0.0598 - mae: 0.1827
Epoch 350/1000
71/71 0s 3ms/step - loss:
0.0597 - mae: 0.1820
Epoch 351/1000
71/71 0s 3ms/step - loss:
0.0598 - mae: 0.1824
Epoch 352/1000
71/71 0s 3ms/step - loss:
0.0598 - mae: 0.1827

Epoch 353/1000
71/71 0s 3ms/step - loss:
0.0597 - mae: 0.1825
Epoch 354/1000
71/71 0s 3ms/step - loss:
0.0596 - mae: 0.1823
Epoch 355/1000
71/71 0s 3ms/step - loss:
0.0594 - mae: 0.1823
Epoch 356/1000
71/71 0s 3ms/step - loss:
0.0595 - mae: 0.1822
Epoch 357/1000
71/71 0s 3ms/step - loss:
0.0593 - mae: 0.1821
Epoch 358/1000
71/71 0s 3ms/step - loss:
0.0595 - mae: 0.1820
Epoch 359/1000
71/71 0s 3ms/step - loss:
0.0593 - mae: 0.1818
Epoch 360/1000
71/71 0s 5ms/step - loss:
0.0592 - mae: 0.1818
Epoch 361/1000
71/71 0s 3ms/step - loss:
0.0592 - mae: 0.1815
Epoch 362/1000
71/71 0s 3ms/step - loss:
0.0592 - mae: 0.1817
Epoch 363/1000
71/71 0s 3ms/step - loss:
0.0593 - mae: 0.1817
Epoch 364/1000
71/71 0s 3ms/step - loss:
0.0590 - mae: 0.1813
Epoch 365/1000
71/71 0s 3ms/step - loss:
0.0590 - mae: 0.1813
Epoch 366/1000
71/71 0s 3ms/step - loss:
0.0589 - mae: 0.1811
Epoch 367/1000
71/71 0s 3ms/step - loss:
0.0589 - mae: 0.1810
Epoch 368/1000
71/71 0s 3ms/step - loss:
0.0588 - mae: 0.1810

Epoch 369/1000
71/71 0s 3ms/step - loss:
0.0589 - mae: 0.1811
Epoch 370/1000
71/71 0s 3ms/step - loss:
0.0588 - mae: 0.1809
Epoch 371/1000
71/71 0s 4ms/step - loss:
0.0587 - mae: 0.1808
Epoch 372/1000
71/71 0s 3ms/step - loss:
0.0586 - mae: 0.1807
Epoch 373/1000
71/71 0s 3ms/step - loss:
0.0586 - mae: 0.1806
Epoch 374/1000
71/71 0s 3ms/step - loss:
0.0584 - mae: 0.1802
Epoch 375/1000
71/71 0s 3ms/step - loss:
0.0585 - mae: 0.1805
Epoch 376/1000
71/71 0s 3ms/step - loss:
0.0585 - mae: 0.1804
Epoch 377/1000
71/71 0s 3ms/step - loss:
0.0583 - mae: 0.1798
Epoch 378/1000
71/71 0s 3ms/step - loss:
0.0583 - mae: 0.1806
Epoch 379/1000
71/71 0s 3ms/step - loss:
0.0584 - mae: 0.1802
Epoch 380/1000
71/71 0s 3ms/step - loss:
0.0581 - mae: 0.1801
Epoch 381/1000
71/71 0s 3ms/step - loss:
0.0581 - mae: 0.1798
Epoch 382/1000
71/71 0s 4ms/step - loss:
0.0581 - mae: 0.1797
Epoch 383/1000
71/71 0s 3ms/step - loss:
0.0581 - mae: 0.1796
Epoch 384/1000
71/71 0s 3ms/step - loss:
0.0581 - mae: 0.1800

Epoch 385/1000
71/71 0s 3ms/step - loss:
0.0581 - mae: 0.1796
Epoch 386/1000
71/71 0s 3ms/step - loss:
0.0579 - mae: 0.1792
Epoch 387/1000
71/71 0s 3ms/step - loss:
0.0579 - mae: 0.1794
Epoch 388/1000
71/71 0s 3ms/step - loss:
0.0578 - mae: 0.1797
Epoch 389/1000
71/71 0s 3ms/step - loss:
0.0577 - mae: 0.1789
Epoch 390/1000
71/71 0s 3ms/step - loss:
0.0578 - mae: 0.1792
Epoch 391/1000
71/71 0s 3ms/step - loss:
0.0577 - mae: 0.1792
Epoch 392/1000
71/71 0s 3ms/step - loss:
0.0577 - mae: 0.1788
Epoch 393/1000
71/71 0s 3ms/step - loss:
0.0576 - mae: 0.1789
Epoch 394/1000
71/71 0s 3ms/step - loss:
0.0576 - mae: 0.1790
Epoch 395/1000
71/71 0s 3ms/step - loss:
0.0574 - mae: 0.1788
Epoch 396/1000
71/71 0s 3ms/step - loss:
0.0575 - mae: 0.1786
Epoch 397/1000
71/71 0s 3ms/step - loss:
0.0575 - mae: 0.1787
Epoch 398/1000
71/71 0s 3ms/step - loss:
0.0574 - mae: 0.1789
Epoch 399/1000
71/71 0s 3ms/step - loss:
0.0574 - mae: 0.1786
Epoch 400/1000
71/71 0s 3ms/step - loss:
0.0573 - mae: 0.1784

Epoch 401/1000
71/71 0s 3ms/step - loss:
0.0572 - mae: 0.1782
Epoch 402/1000
71/71 0s 3ms/step - loss:
0.0571 - mae: 0.1781
Epoch 403/1000
71/71 0s 3ms/step - loss:
0.0572 - mae: 0.1785
Epoch 404/1000
71/71 0s 3ms/step - loss:
0.0571 - mae: 0.1779
Epoch 405/1000
71/71 0s 4ms/step - loss:
0.0571 - mae: 0.1783
Epoch 406/1000
71/71 0s 3ms/step - loss:
0.0570 - mae: 0.1781
Epoch 407/1000
71/71 0s 3ms/step - loss:
0.0568 - mae: 0.1776
Epoch 408/1000
71/71 0s 3ms/step - loss:
0.0570 - mae: 0.1780
Epoch 409/1000
71/71 0s 3ms/step - loss:
0.0569 - mae: 0.1779
Epoch 410/1000
71/71 0s 3ms/step - loss:
0.0567 - mae: 0.1773
Epoch 411/1000
71/71 0s 3ms/step - loss:
0.0568 - mae: 0.1778
Epoch 412/1000
71/71 0s 3ms/step - loss:
0.0568 - mae: 0.1774
Epoch 413/1000
71/71 0s 3ms/step - loss:
0.0567 - mae: 0.1778
Epoch 414/1000
71/71 0s 3ms/step - loss:
0.0566 - mae: 0.1773
Epoch 415/1000
71/71 0s 3ms/step - loss:
0.0566 - mae: 0.1771
Epoch 416/1000
71/71 0s 4ms/step - loss:
0.0565 - mae: 0.1769

Epoch 417/1000
71/71 0s 3ms/step - loss:
0.0565 - mae: 0.1774
Epoch 418/1000
71/71 0s 3ms/step - loss:
0.0565 - mae: 0.1770
Epoch 419/1000
71/71 0s 3ms/step - loss:
0.0565 - mae: 0.1772
Epoch 420/1000
71/71 0s 3ms/step - loss:
0.0563 - mae: 0.1767
Epoch 421/1000
71/71 0s 3ms/step - loss:
0.0563 - mae: 0.1766
Epoch 422/1000
71/71 0s 3ms/step - loss:
0.0563 - mae: 0.1768
Epoch 423/1000
71/71 0s 3ms/step - loss:
0.0562 - mae: 0.1765
Epoch 424/1000
71/71 0s 3ms/step - loss:
0.0563 - mae: 0.1764
Epoch 425/1000
71/71 0s 3ms/step - loss:
0.0562 - mae: 0.1768
Epoch 426/1000
71/71 0s 3ms/step - loss:
0.0562 - mae: 0.1766
Epoch 427/1000
71/71 0s 4ms/step - loss:
0.0560 - mae: 0.1763
Epoch 428/1000
71/71 0s 3ms/step - loss:
0.0560 - mae: 0.1763
Epoch 429/1000
71/71 0s 3ms/step - loss:
0.0560 - mae: 0.1762
Epoch 430/1000
71/71 0s 3ms/step - loss:
0.0560 - mae: 0.1760
Epoch 431/1000
71/71 0s 3ms/step - loss:
0.0560 - mae: 0.1762
Epoch 432/1000
71/71 0s 3ms/step - loss:
0.0558 - mae: 0.1759

Epoch 433/1000
71/71 0s 3ms/step - loss:
0.0558 - mae: 0.1760
Epoch 434/1000
71/71 0s 3ms/step - loss:
0.0559 - mae: 0.1760
Epoch 435/1000
71/71 0s 3ms/step - loss:
0.0558 - mae: 0.1760
Epoch 436/1000
71/71 0s 3ms/step - loss:
0.0557 - mae: 0.1760
Epoch 437/1000
71/71 0s 3ms/step - loss:
0.0556 - mae: 0.1754
Epoch 438/1000
71/71 0s 4ms/step - loss:
0.0557 - mae: 0.1756
Epoch 439/1000
71/71 0s 3ms/step - loss:
0.0556 - mae: 0.1753
Epoch 440/1000
71/71 0s 3ms/step - loss:
0.0556 - mae: 0.1757
Epoch 441/1000
71/71 0s 3ms/step - loss:
0.0555 - mae: 0.1753
Epoch 442/1000
71/71 0s 3ms/step - loss:
0.0554 - mae: 0.1752
Epoch 443/1000
71/71 0s 3ms/step - loss:
0.0553 - mae: 0.1754
Epoch 444/1000
71/71 0s 3ms/step - loss:
0.0554 - mae: 0.1753
Epoch 445/1000
71/71 0s 3ms/step - loss:
0.0554 - mae: 0.1751
Epoch 446/1000
71/71 0s 3ms/step - loss:
0.0553 - mae: 0.1749
Epoch 447/1000
71/71 0s 3ms/step - loss:
0.0551 - mae: 0.1747
Epoch 448/1000
71/71 0s 3ms/step - loss:
0.0552 - mae: 0.1749

Epoch 449/1000
71/71 0s 4ms/step - loss:
0.0552 - mae: 0.1750
Epoch 450/1000
71/71 0s 3ms/step - loss:
0.0551 - mae: 0.1749
Epoch 451/1000
71/71 0s 3ms/step - loss:
0.0551 - mae: 0.1747
Epoch 452/1000
71/71 0s 3ms/step - loss:
0.0551 - mae: 0.1746
Epoch 453/1000
71/71 0s 3ms/step - loss:
0.0549 - mae: 0.1744
Epoch 454/1000
71/71 0s 3ms/step - loss:
0.0549 - mae: 0.1745
Epoch 455/1000
71/71 0s 3ms/step - loss:
0.0549 - mae: 0.1745
Epoch 456/1000
71/71 0s 3ms/step - loss:
0.0549 - mae: 0.1745
Epoch 457/1000
71/71 0s 3ms/step - loss:
0.0549 - mae: 0.1744
Epoch 458/1000
71/71 0s 3ms/step - loss:
0.0548 - mae: 0.1742
Epoch 459/1000
71/71 0s 3ms/step - loss:
0.0549 - mae: 0.1743
Epoch 460/1000
71/71 0s 4ms/step - loss:
0.0547 - mae: 0.1740
Epoch 461/1000
71/71 0s 3ms/step - loss:
0.0547 - mae: 0.1740
Epoch 462/1000
71/71 0s 3ms/step - loss:
0.0547 - mae: 0.1740
Epoch 463/1000
71/71 0s 3ms/step - loss:
0.0545 - mae: 0.1734
Epoch 464/1000
71/71 0s 3ms/step - loss:
0.0547 - mae: 0.1738

Epoch 465/1000
71/71 0s 3ms/step - loss:
0.0545 - mae: 0.1734
Epoch 466/1000
71/71 0s 3ms/step - loss:
0.0546 - mae: 0.1738
Epoch 467/1000
71/71 0s 3ms/step - loss:
0.0545 - mae: 0.1737
Epoch 468/1000
71/71 0s 3ms/step - loss:
0.0544 - mae: 0.1735
Epoch 469/1000
71/71 0s 3ms/step - loss:
0.0544 - mae: 0.1735
Epoch 470/1000
71/71 0s 3ms/step - loss:
0.0544 - mae: 0.1733
Epoch 471/1000
71/71 0s 4ms/step - loss:
0.0542 - mae: 0.1732
Epoch 472/1000
71/71 0s 3ms/step - loss:
0.0543 - mae: 0.1734
Epoch 473/1000
71/71 0s 3ms/step - loss:
0.0543 - mae: 0.1735
Epoch 474/1000
71/71 0s 3ms/step - loss:
0.0543 - mae: 0.1735
Epoch 475/1000
71/71 0s 3ms/step - loss:
0.0542 - mae: 0.1730
Epoch 476/1000
71/71 0s 3ms/step - loss:
0.0541 - mae: 0.1725
Epoch 477/1000
71/71 0s 3ms/step - loss:
0.0541 - mae: 0.1729
Epoch 478/1000
71/71 0s 3ms/step - loss:
0.0539 - mae: 0.1727
Epoch 479/1000
71/71 0s 3ms/step - loss:
0.0539 - mae: 0.1730
Epoch 480/1000
71/71 0s 3ms/step - loss:
0.0538 - mae: 0.1722

Epoch 481/1000
71/71 0s 3ms/step - loss:
0.0539 - mae: 0.1728
Epoch 482/1000
71/71 0s 4ms/step - loss:
0.0539 - mae: 0.1726
Epoch 483/1000
71/71 0s 3ms/step - loss:
0.0539 - mae: 0.1725
Epoch 484/1000
71/71 0s 3ms/step - loss:
0.0539 - mae: 0.1725
Epoch 485/1000
71/71 0s 3ms/step - loss:
0.0538 - mae: 0.1724
Epoch 486/1000
71/71 0s 3ms/step - loss:
0.0538 - mae: 0.1723
Epoch 487/1000
71/71 0s 3ms/step - loss:
0.0537 - mae: 0.1724
Epoch 488/1000
71/71 0s 3ms/step - loss:
0.0537 - mae: 0.1724
Epoch 489/1000
71/71 0s 3ms/step - loss:
0.0536 - mae: 0.1721
Epoch 490/1000
71/71 0s 3ms/step - loss:
0.0537 - mae: 0.1724
Epoch 491/1000
71/71 0s 3ms/step - loss:
0.0535 - mae: 0.1720
Epoch 492/1000
71/71 0s 3ms/step - loss:
0.0535 - mae: 0.1719
Epoch 493/1000
71/71 0s 4ms/step - loss:
0.0535 - mae: 0.1716
Epoch 494/1000
71/71 0s 3ms/step - loss:
0.0535 - mae: 0.1720
Epoch 495/1000
71/71 0s 3ms/step - loss:
0.0535 - mae: 0.1717
Epoch 496/1000
71/71 0s 3ms/step - loss:
0.0534 - mae: 0.1719

Epoch 497/1000
71/71 0s 3ms/step - loss:
0.0535 - mae: 0.1717
Epoch 498/1000
71/71 0s 3ms/step - loss:
0.0534 - mae: 0.1717
Epoch 499/1000
71/71 0s 3ms/step - loss:
0.0534 - mae: 0.1719
Epoch 500/1000
71/71 0s 3ms/step - loss:
0.0532 - mae: 0.1715
Epoch 501/1000
71/71 0s 3ms/step - loss:
0.0532 - mae: 0.1715
Epoch 502/1000
71/71 0s 3ms/step - loss:
0.0531 - mae: 0.1710
Epoch 503/1000
71/71 0s 3ms/step - loss:
0.0532 - mae: 0.1714
Epoch 504/1000
71/71 0s 4ms/step - loss:
0.0532 - mae: 0.1712
Epoch 505/1000
71/71 0s 3ms/step - loss:
0.0531 - mae: 0.1714
Epoch 506/1000
71/71 0s 3ms/step - loss:
0.0531 - mae: 0.1710
Epoch 507/1000
71/71 0s 3ms/step - loss:
0.0530 - mae: 0.1712
Epoch 508/1000
71/71 0s 3ms/step - loss:
0.0529 - mae: 0.1712
Epoch 509/1000
71/71 0s 3ms/step - loss:
0.0530 - mae: 0.1711
Epoch 510/1000
71/71 0s 3ms/step - loss:
0.0530 - mae: 0.1712
Epoch 511/1000
71/71 0s 3ms/step - loss:
0.0528 - mae: 0.1707
Epoch 512/1000
71/71 0s 3ms/step - loss:
0.0529 - mae: 0.1711

Epoch 513/1000
71/71 0s 3ms/step - loss:
0.0528 - mae: 0.1707
Epoch 514/1000
71/71 0s 3ms/step - loss:
0.0527 - mae: 0.1707
Epoch 515/1000
71/71 0s 4ms/step - loss:
0.0528 - mae: 0.1705
Epoch 516/1000
71/71 0s 3ms/step - loss:
0.0528 - mae: 0.1703
Epoch 517/1000
71/71 0s 3ms/step - loss:
0.0528 - mae: 0.1706
Epoch 518/1000
71/71 0s 3ms/step - loss:
0.0526 - mae: 0.1704
Epoch 519/1000
71/71 0s 3ms/step - loss:
0.0526 - mae: 0.1701
Epoch 520/1000
71/71 0s 3ms/step - loss:
0.0526 - mae: 0.1702
Epoch 521/1000
71/71 0s 3ms/step - loss:
0.0525 - mae: 0.1702
Epoch 522/1000
71/71 0s 3ms/step - loss:
0.0525 - mae: 0.1702
Epoch 523/1000
71/71 0s 3ms/step - loss:
0.0525 - mae: 0.1702
Epoch 524/1000
71/71 0s 3ms/step - loss:
0.0524 - mae: 0.1700
Epoch 525/1000
71/71 0s 3ms/step - loss:
0.0524 - mae: 0.1702
Epoch 526/1000
71/71 0s 4ms/step - loss:
0.0524 - mae: 0.1700
Epoch 527/1000
71/71 0s 3ms/step - loss:
0.0523 - mae: 0.1699
Epoch 528/1000
71/71 0s 3ms/step - loss:
0.0524 - mae: 0.1700

Epoch 529/1000
 71/71 0s 3ms/step - loss:
 0.0522 - mae: 0.1699
 Epoch 530/1000
 71/71 0s 3ms/step - loss:
 0.0522 - mae: 0.1697
 Epoch 531/1000
 71/71 0s 3ms/step - loss:
 0.0523 - mae: 0.1697
 Epoch 532/1000
 71/71 0s 3ms/step - loss:
 0.0522 - mae: 0.1697
 Epoch 533/1000
 71/71 0s 3ms/step - loss:
 0.0522 - mae: 0.1696
 Epoch 534/1000
 71/71 0s 3ms/step - loss:
 0.0521 - mae: 0.1694
 Epoch 535/1000
 71/71 0s 3ms/step - loss:
 0.0522 - mae: 0.1697
 Epoch 536/1000
 71/71 0s 3ms/step - loss:
 0.0521 - mae: 0.1692
 Epoch 537/1000
 71/71 0s 4ms/step - loss:
 0.0519 - mae: 0.1694
 Epoch 538/1000
 71/71 0s 3ms/step - loss:
 0.0520 - mae: 0.1693
 Epoch 539/1000
 71/71 0s 3ms/step - loss:
 0.0520 - mae: 0.1695
 Epoch 540/1000
 71/71 0s 3ms/step - loss:
 0.0520 - mae: 0.1693
 Epoch 541/1000
 71/71 0s 3ms/step - loss:
 0.0518 - mae: 0.1689
 Epoch 542/1000
 71/71 0s 3ms/step - loss:
 0.0520 - mae: 0.1692
 Epoch 543/1000
 71/71 0s 3ms/step - loss:
 0.0518 - mae: 0.1691
 Epoch 544/1000
 71/71 0s 3ms/step - loss:
 0.0518 - mae: 0.1688

Epoch 545/1000
71/71 0s 3ms/step - loss:
0.0518 - mae: 0.1689
Epoch 546/1000
71/71 0s 3ms/step - loss:
0.0516 - mae: 0.1684
Epoch 547/1000
71/71 0s 3ms/step - loss:
0.0516 - mae: 0.1688
Epoch 548/1000
71/71 0s 4ms/step - loss:
0.0517 - mae: 0.1686
Epoch 549/1000
71/71 0s 3ms/step - loss:
0.0516 - mae: 0.1684
Epoch 550/1000
71/71 0s 3ms/step - loss:
0.0517 - mae: 0.1688
Epoch 551/1000
71/71 0s 3ms/step - loss:
0.0517 - mae: 0.1685
Epoch 552/1000
71/71 0s 3ms/step - loss:
0.0516 - mae: 0.1686
Epoch 553/1000
71/71 0s 3ms/step - loss:
0.0515 - mae: 0.1684
Epoch 554/1000
71/71 0s 3ms/step - loss:
0.0516 - mae: 0.1684
Epoch 555/1000
71/71 0s 3ms/step - loss:
0.0514 - mae: 0.1680
Epoch 556/1000
71/71 0s 3ms/step - loss:
0.0514 - mae: 0.1686
Epoch 557/1000
71/71 0s 3ms/step - loss:
0.0514 - mae: 0.1686
Epoch 558/1000
71/71 0s 3ms/step - loss:
0.0513 - mae: 0.1679
Epoch 559/1000
71/71 0s 4ms/step - loss:
0.0514 - mae: 0.1681
Epoch 560/1000
71/71 0s 3ms/step - loss:
0.0513 - mae: 0.1682

Epoch 561/1000
71/71 0s 3ms/step - loss:
0.0512 - mae: 0.1680
Epoch 562/1000
71/71 0s 3ms/step - loss:
0.0512 - mae: 0.1677
Epoch 563/1000
71/71 0s 3ms/step - loss:
0.0511 - mae: 0.1678
Epoch 564/1000
71/71 0s 3ms/step - loss:
0.0512 - mae: 0.1677
Epoch 565/1000
71/71 0s 3ms/step - loss:
0.0512 - mae: 0.1678
Epoch 566/1000
71/71 0s 3ms/step - loss:
0.0511 - mae: 0.1677
Epoch 567/1000
71/71 0s 3ms/step - loss:
0.0511 - mae: 0.1676
Epoch 568/1000
71/71 0s 3ms/step - loss:
0.0511 - mae: 0.1678
Epoch 569/1000
71/71 0s 3ms/step - loss:
0.0511 - mae: 0.1674
Epoch 570/1000
71/71 0s 4ms/step - loss:
0.0510 - mae: 0.1676
Epoch 571/1000
71/71 0s 3ms/step - loss:
0.0510 - mae: 0.1674
Epoch 572/1000
71/71 0s 3ms/step - loss:
0.0510 - mae: 0.1676
Epoch 573/1000
71/71 0s 3ms/step - loss:
0.0509 - mae: 0.1671
Epoch 574/1000
71/71 0s 3ms/step - loss:
0.0509 - mae: 0.1674
Epoch 575/1000
71/71 0s 3ms/step - loss:
0.0509 - mae: 0.1672
Epoch 576/1000
71/71 0s 3ms/step - loss:
0.0508 - mae: 0.1672

Epoch 577/1000
71/71 0s 3ms/step - loss:
0.0508 - mae: 0.1671
Epoch 578/1000
71/71 0s 3ms/step - loss:
0.0507 - mae: 0.1669
Epoch 579/1000
71/71 0s 3ms/step - loss:
0.0507 - mae: 0.1668
Epoch 580/1000
71/71 0s 3ms/step - loss:
0.0507 - mae: 0.1670
Epoch 581/1000
71/71 0s 4ms/step - loss:
0.0507 - mae: 0.1668
Epoch 582/1000
71/71 0s 3ms/step - loss:
0.0507 - mae: 0.1670
Epoch 583/1000
71/71 0s 3ms/step - loss:
0.0506 - mae: 0.1666
Epoch 584/1000
71/71 0s 3ms/step - loss:
0.0507 - mae: 0.1671
Epoch 585/1000
71/71 0s 3ms/step - loss:
0.0506 - mae: 0.1665
Epoch 586/1000
71/71 0s 3ms/step - loss:
0.0505 - mae: 0.1667
Epoch 587/1000
71/71 0s 3ms/step - loss:
0.0505 - mae: 0.1665
Epoch 588/1000
71/71 0s 3ms/step - loss:
0.0503 - mae: 0.1662
Epoch 589/1000
71/71 0s 3ms/step - loss:
0.0505 - mae: 0.1667
Epoch 590/1000
71/71 0s 3ms/step - loss:
0.0503 - mae: 0.1659
Epoch 591/1000
71/71 0s 3ms/step - loss:
0.0504 - mae: 0.1663
Epoch 592/1000
71/71 0s 4ms/step - loss:
0.0502 - mae: 0.1659

Epoch 593/1000
71/71 0s 3ms/step - loss:
0.0503 - mae: 0.1661
Epoch 594/1000
71/71 0s 3ms/step - loss:
0.0504 - mae: 0.1663
Epoch 595/1000
71/71 0s 3ms/step - loss:
0.0503 - mae: 0.1661
Epoch 596/1000
71/71 0s 3ms/step - loss:
0.0503 - mae: 0.1663
Epoch 597/1000
71/71 0s 3ms/step - loss:
0.0501 - mae: 0.1661
Epoch 598/1000
71/71 0s 3ms/step - loss:
0.0502 - mae: 0.1659
Epoch 599/1000
71/71 0s 3ms/step - loss:
0.0500 - mae: 0.1655
Epoch 600/1000
71/71 0s 3ms/step - loss:
0.0502 - mae: 0.1663
Epoch 601/1000
71/71 0s 3ms/step - loss:
0.0500 - mae: 0.1658
Epoch 602/1000
71/71 0s 3ms/step - loss:
0.0501 - mae: 0.1659
Epoch 603/1000
71/71 0s 4ms/step - loss:
0.0500 - mae: 0.1658
Epoch 604/1000
71/71 0s 3ms/step - loss:
0.0500 - mae: 0.1658
Epoch 605/1000
71/71 0s 3ms/step - loss:
0.0500 - mae: 0.1654
Epoch 606/1000
71/71 0s 3ms/step - loss:
0.0499 - mae: 0.1656
Epoch 607/1000
71/71 0s 3ms/step - loss:
0.0499 - mae: 0.1658
Epoch 608/1000
71/71 0s 3ms/step - loss:
0.0500 - mae: 0.1655

Epoch 609/1000
71/71 0s 3ms/step - loss:
0.0499 - mae: 0.1655
Epoch 610/1000
71/71 0s 3ms/step - loss:
0.0498 - mae: 0.1647
Epoch 611/1000
71/71 0s 3ms/step - loss:
0.0498 - mae: 0.1654
Epoch 612/1000
71/71 0s 3ms/step - loss:
0.0497 - mae: 0.1650
Epoch 613/1000
71/71 0s 3ms/step - loss:
0.0497 - mae: 0.1652
Epoch 614/1000
71/71 0s 4ms/step - loss:
0.0497 - mae: 0.1651
Epoch 615/1000
71/71 0s 3ms/step - loss:
0.0498 - mae: 0.1654
Epoch 616/1000
71/71 0s 3ms/step - loss:
0.0497 - mae: 0.1650
Epoch 617/1000
71/71 0s 3ms/step - loss:
0.0497 - mae: 0.1648
Epoch 618/1000
71/71 0s 3ms/step - loss:
0.0496 - mae: 0.1649
Epoch 619/1000
71/71 0s 3ms/step - loss:
0.0496 - mae: 0.1647
Epoch 620/1000
71/71 0s 3ms/step - loss:
0.0496 - mae: 0.1650
Epoch 621/1000
71/71 0s 3ms/step - loss:
0.0495 - mae: 0.1645
Epoch 622/1000
71/71 0s 3ms/step - loss:
0.0495 - mae: 0.1647
Epoch 623/1000
71/71 0s 3ms/step - loss:
0.0495 - mae: 0.1650
Epoch 624/1000
71/71 0s 3ms/step - loss:
0.0495 - mae: 0.1646

Epoch 625/1000
71/71 0s 3ms/step - loss:
0.0494 - mae: 0.1646
Epoch 626/1000
71/71 0s 3ms/step - loss:
0.0494 - mae: 0.1647
Epoch 627/1000
71/71 0s 3ms/step - loss:
0.0494 - mae: 0.1645
Epoch 628/1000
71/71 0s 3ms/step - loss:
0.0494 - mae: 0.1646
Epoch 629/1000
71/71 0s 3ms/step - loss:
0.0493 - mae: 0.1643
Epoch 630/1000
71/71 0s 3ms/step - loss:
0.0493 - mae: 0.1642
Epoch 631/1000
71/71 0s 3ms/step - loss:
0.0492 - mae: 0.1642
Epoch 632/1000
71/71 0s 3ms/step - loss:
0.0491 - mae: 0.1641
Epoch 633/1000
71/71 0s 3ms/step - loss:
0.0491 - mae: 0.1640
Epoch 634/1000
71/71 0s 3ms/step - loss:
0.0491 - mae: 0.1640
Epoch 635/1000
71/71 0s 3ms/step - loss:
0.0492 - mae: 0.1639
Epoch 636/1000
71/71 0s 4ms/step - loss:
0.0491 - mae: 0.1639
Epoch 637/1000
71/71 0s 3ms/step - loss:
0.0491 - mae: 0.1642
Epoch 638/1000
71/71 0s 3ms/step - loss:
0.0491 - mae: 0.1643
Epoch 639/1000
71/71 0s 3ms/step - loss:
0.0490 - mae: 0.1638
Epoch 640/1000
71/71 0s 3ms/step - loss:
0.0490 - mae: 0.1636

Epoch 641/1000
71/71 0s 3ms/step - loss:
0.0491 - mae: 0.1639
Epoch 642/1000
71/71 0s 3ms/step - loss:
0.0490 - mae: 0.1639
Epoch 643/1000
71/71 0s 3ms/step - loss:
0.0489 - mae: 0.1635
Epoch 644/1000
71/71 0s 3ms/step - loss:
0.0489 - mae: 0.1634
Epoch 645/1000
71/71 0s 3ms/step - loss:
0.0489 - mae: 0.1637
Epoch 646/1000
71/71 0s 3ms/step - loss:
0.0489 - mae: 0.1636
Epoch 647/1000
71/71 0s 4ms/step - loss:
0.0487 - mae: 0.1632
Epoch 648/1000
71/71 0s 3ms/step - loss:
0.0488 - mae: 0.1634
Epoch 649/1000
71/71 0s 3ms/step - loss:
0.0488 - mae: 0.1636
Epoch 650/1000
71/71 0s 3ms/step - loss:
0.0488 - mae: 0.1633
Epoch 651/1000
71/71 0s 3ms/step - loss:
0.0487 - mae: 0.1632
Epoch 652/1000
71/71 0s 3ms/step - loss:
0.0487 - mae: 0.1633
Epoch 653/1000
71/71 0s 3ms/step - loss:
0.0486 - mae: 0.1633
Epoch 654/1000
71/71 0s 3ms/step - loss:
0.0486 - mae: 0.1632
Epoch 655/1000
71/71 0s 3ms/step - loss:
0.0486 - mae: 0.1629
Epoch 656/1000
71/71 0s 3ms/step - loss:
0.0486 - mae: 0.1630

Epoch 657/1000
71/71 0s 3ms/step - loss:
0.0486 - mae: 0.1631
Epoch 658/1000
71/71 0s 4ms/step - loss:
0.0485 - mae: 0.1631
Epoch 659/1000
71/71 0s 3ms/step - loss:
0.0485 - mae: 0.1629
Epoch 660/1000
71/71 0s 3ms/step - loss:
0.0485 - mae: 0.1627
Epoch 661/1000
71/71 0s 3ms/step - loss:
0.0485 - mae: 0.1626
Epoch 662/1000
71/71 0s 3ms/step - loss:
0.0484 - mae: 0.1629
Epoch 663/1000
71/71 0s 3ms/step - loss:
0.0485 - mae: 0.1627
Epoch 664/1000
71/71 0s 3ms/step - loss:
0.0482 - mae: 0.1626
Epoch 665/1000
71/71 0s 3ms/step - loss:
0.0485 - mae: 0.1626
Epoch 666/1000
71/71 0s 3ms/step - loss:
0.0484 - mae: 0.1626
Epoch 667/1000
71/71 0s 3ms/step - loss:
0.0484 - mae: 0.1628
Epoch 668/1000
71/71 0s 3ms/step - loss:
0.0483 - mae: 0.1627
Epoch 669/1000
71/71 0s 4ms/step - loss:
0.0482 - mae: 0.1623
Epoch 670/1000
71/71 0s 3ms/step - loss:
0.0482 - mae: 0.1627
Epoch 671/1000
71/71 0s 3ms/step - loss:
0.0482 - mae: 0.1622
Epoch 672/1000
71/71 0s 3ms/step - loss:
0.0481 - mae: 0.1620

Epoch 673/1000
71/71 0s 3ms/step - loss:
0.0481 - mae: 0.1624
Epoch 674/1000
71/71 0s 3ms/step - loss:
0.0482 - mae: 0.1621
Epoch 675/1000
71/71 0s 3ms/step - loss:
0.0481 - mae: 0.1624
Epoch 676/1000
71/71 0s 3ms/step - loss:
0.0480 - mae: 0.1622
Epoch 677/1000
71/71 0s 3ms/step - loss:
0.0480 - mae: 0.1624
Epoch 678/1000
71/71 0s 3ms/step - loss:
0.0480 - mae: 0.1621
Epoch 679/1000
71/71 0s 3ms/step - loss:
0.0480 - mae: 0.1619
Epoch 680/1000
71/71 0s 4ms/step - loss:
0.0479 - mae: 0.1623
Epoch 681/1000
71/71 0s 3ms/step - loss:
0.0479 - mae: 0.1619
Epoch 682/1000
71/71 0s 3ms/step - loss:
0.0479 - mae: 0.1617
Epoch 683/1000
71/71 0s 3ms/step - loss:
0.0479 - mae: 0.1621
Epoch 684/1000
71/71 0s 3ms/step - loss:
0.0479 - mae: 0.1616
Epoch 685/1000
71/71 0s 3ms/step - loss:
0.0478 - mae: 0.1616
Epoch 686/1000
71/71 0s 3ms/step - loss:
0.0478 - mae: 0.1615
Epoch 687/1000
71/71 0s 3ms/step - loss:
0.0478 - mae: 0.1618
Epoch 688/1000
71/71 0s 3ms/step - loss:
0.0477 - mae: 0.1617

Epoch 689/1000
71/71 0s 3ms/step - loss:
0.0478 - mae: 0.1618
Epoch 690/1000
71/71 0s 3ms/step - loss:
0.0478 - mae: 0.1617
Epoch 691/1000
71/71 0s 4ms/step - loss:
0.0477 - mae: 0.1613
Epoch 692/1000
71/71 0s 3ms/step - loss:
0.0477 - mae: 0.1616
Epoch 693/1000
71/71 0s 3ms/step - loss:
0.0476 - mae: 0.1614
Epoch 694/1000
71/71 0s 3ms/step - loss:
0.0476 - mae: 0.1614
Epoch 695/1000
71/71 0s 3ms/step - loss:
0.0476 - mae: 0.1614
Epoch 696/1000
71/71 0s 3ms/step - loss:
0.0476 - mae: 0.1610
Epoch 697/1000
71/71 0s 3ms/step - loss:
0.0476 - mae: 0.1613
Epoch 698/1000
71/71 0s 3ms/step - loss:
0.0475 - mae: 0.1613
Epoch 699/1000
71/71 0s 3ms/step - loss:
0.0476 - mae: 0.1610
Epoch 700/1000
71/71 0s 3ms/step - loss:
0.0475 - mae: 0.1611
Epoch 701/1000
71/71 0s 3ms/step - loss:
0.0475 - mae: 0.1612
Epoch 702/1000
71/71 0s 4ms/step - loss:
0.0475 - mae: 0.1610
Epoch 703/1000
71/71 0s 3ms/step - loss:
0.0474 - mae: 0.1611
Epoch 704/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1605

Epoch 705/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1607
Epoch 706/1000
71/71 0s 3ms/step - loss:
0.0474 - mae: 0.1606
Epoch 707/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1608
Epoch 708/1000
71/71 0s 3ms/step - loss:
0.0473 - mae: 0.1606
Epoch 709/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1608
Epoch 710/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1606
Epoch 711/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1604
Epoch 712/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1603
Epoch 713/1000
71/71 0s 4ms/step - loss:
0.0472 - mae: 0.1603
Epoch 714/1000
71/71 0s 3ms/step - loss:
0.0471 - mae: 0.1604
Epoch 715/1000
71/71 0s 3ms/step - loss:
0.0470 - mae: 0.1602
Epoch 716/1000
71/71 0s 3ms/step - loss:
0.0472 - mae: 0.1606
Epoch 717/1000
71/71 0s 3ms/step - loss:
0.0471 - mae: 0.1602
Epoch 718/1000
71/71 0s 3ms/step - loss:
0.0470 - mae: 0.1601
Epoch 719/1000
71/71 0s 3ms/step - loss:
0.0469 - mae: 0.1599
Epoch 720/1000
71/71 0s 3ms/step - loss:
0.0470 - mae: 0.1603

Epoch 721/1000
71/71 0s 3ms/step - loss:
0.0470 - mae: 0.1604
Epoch 722/1000
71/71 0s 3ms/step - loss:
0.0469 - mae: 0.1601
Epoch 723/1000
71/71 0s 3ms/step - loss:
0.0469 - mae: 0.1601
Epoch 724/1000
71/71 0s 4ms/step - loss:
0.0469 - mae: 0.1602
Epoch 725/1000
71/71 0s 3ms/step - loss:
0.0468 - mae: 0.1595
Epoch 726/1000
71/71 0s 3ms/step - loss:
0.0468 - mae: 0.1596
Epoch 727/1000
71/71 0s 3ms/step - loss:
0.0468 - mae: 0.1597
Epoch 728/1000
71/71 0s 3ms/step - loss:
0.0468 - mae: 0.1598
Epoch 729/1000
71/71 0s 3ms/step - loss:
0.0467 - mae: 0.1599
Epoch 730/1000
71/71 0s 3ms/step - loss:
0.0468 - mae: 0.1598
Epoch 731/1000
71/71 0s 3ms/step - loss:
0.0467 - mae: 0.1597
Epoch 732/1000
71/71 0s 3ms/step - loss:
0.0467 - mae: 0.1598
Epoch 733/1000
71/71 0s 3ms/step - loss:
0.0468 - mae: 0.1598
Epoch 734/1000
71/71 0s 3ms/step - loss:
0.0466 - mae: 0.1594
Epoch 735/1000
71/71 0s 4ms/step - loss:
0.0465 - mae: 0.1592
Epoch 736/1000
71/71 0s 3ms/step - loss:
0.0466 - mae: 0.1593

Epoch 737/1000
71/71 0s 3ms/step - loss:
0.0466 - mae: 0.1593
Epoch 738/1000
71/71 0s 3ms/step - loss:
0.0465 - mae: 0.1593
Epoch 739/1000
71/71 0s 3ms/step - loss:
0.0465 - mae: 0.1593
Epoch 740/1000
71/71 0s 3ms/step - loss:
0.0465 - mae: 0.1594
Epoch 741/1000
71/71 0s 3ms/step - loss:
0.0465 - mae: 0.1591
Epoch 742/1000
71/71 0s 3ms/step - loss:
0.0466 - mae: 0.1597
Epoch 743/1000
71/71 0s 3ms/step - loss:
0.0465 - mae: 0.1591
Epoch 744/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1585
Epoch 745/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1589
Epoch 746/1000
71/71 0s 4ms/step - loss:
0.0463 - mae: 0.1591
Epoch 747/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1589
Epoch 748/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1592
Epoch 749/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1588
Epoch 750/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1591
Epoch 751/1000
71/71 0s 3ms/step - loss:
0.0462 - mae: 0.1587
Epoch 752/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1592

Epoch 753/1000
71/71 0s 3ms/step - loss:
0.0463 - mae: 0.1588
Epoch 754/1000
71/71 0s 3ms/step - loss:
0.0462 - mae: 0.1586
Epoch 755/1000
71/71 0s 3ms/step - loss:
0.0462 - mae: 0.1590
Epoch 756/1000
71/71 0s 3ms/step - loss:
0.0461 - mae: 0.1587
Epoch 757/1000
71/71 0s 3ms/step - loss:
0.0461 - mae: 0.1586
Epoch 758/1000
71/71 0s 3ms/step - loss:
0.0462 - mae: 0.1585
Epoch 759/1000
71/71 0s 3ms/step - loss:
0.0461 - mae: 0.1587
Epoch 760/1000
71/71 0s 3ms/step - loss:
0.0460 - mae: 0.1584
Epoch 761/1000
71/71 0s 3ms/step - loss:
0.0460 - mae: 0.1582
Epoch 762/1000
71/71 0s 3ms/step - loss:
0.0460 - mae: 0.1582
Epoch 763/1000
71/71 0s 3ms/step - loss:
0.0460 - mae: 0.1584
Epoch 764/1000
71/71 0s 3ms/step - loss:
0.0459 - mae: 0.1583
Epoch 765/1000
71/71 0s 3ms/step - loss:
0.0459 - mae: 0.1578
Epoch 766/1000
71/71 0s 3ms/step - loss:
0.0459 - mae: 0.1586
Epoch 767/1000
71/71 0s 3ms/step - loss:
0.0459 - mae: 0.1584
Epoch 768/1000
71/71 0s 4ms/step - loss:
0.0458 - mae: 0.1579

Epoch 769/1000
71/71 0s 3ms/step - loss:
0.0458 - mae: 0.1582
Epoch 770/1000
71/71 0s 3ms/step - loss:
0.0458 - mae: 0.1578
Epoch 771/1000
71/71 0s 3ms/step - loss:
0.0458 - mae: 0.1580
Epoch 772/1000
71/71 0s 3ms/step - loss:
0.0459 - mae: 0.1579
Epoch 773/1000
71/71 0s 3ms/step - loss:
0.0459 - mae: 0.1582
Epoch 774/1000
71/71 0s 3ms/step - loss:
0.0458 - mae: 0.1579
Epoch 775/1000
71/71 0s 3ms/step - loss:
0.0457 - mae: 0.1575
Epoch 776/1000
71/71 0s 3ms/step - loss:
0.0456 - mae: 0.1578
Epoch 777/1000
71/71 0s 3ms/step - loss:
0.0457 - mae: 0.1580
Epoch 778/1000
71/71 0s 3ms/step - loss:
0.0457 - mae: 0.1578
Epoch 779/1000
71/71 0s 4ms/step - loss:
0.0456 - mae: 0.1576
Epoch 780/1000
71/71 0s 3ms/step - loss:
0.0457 - mae: 0.1577
Epoch 781/1000
71/71 0s 3ms/step - loss:
0.0455 - mae: 0.1576
Epoch 782/1000
71/71 0s 3ms/step - loss:
0.0456 - mae: 0.1574
Epoch 783/1000
71/71 0s 3ms/step - loss:
0.0454 - mae: 0.1572
Epoch 784/1000
71/71 0s 3ms/step - loss:
0.0455 - mae: 0.1574

Epoch 785/1000
71/71 0s 3ms/step - loss:
0.0455 - mae: 0.1580
Epoch 786/1000
71/71 0s 3ms/step - loss:
0.0454 - mae: 0.1574
Epoch 787/1000
71/71 0s 3ms/step - loss:
0.0454 - mae: 0.1575
Epoch 788/1000
71/71 0s 3ms/step - loss:
0.0454 - mae: 0.1574
Epoch 789/1000
71/71 0s 3ms/step - loss:
0.0454 - mae: 0.1573
Epoch 790/1000
71/71 0s 4ms/step - loss:
0.0454 - mae: 0.1574
Epoch 791/1000
71/71 0s 3ms/step - loss:
0.0454 - mae: 0.1572
Epoch 792/1000
71/71 0s 3ms/step - loss:
0.0454 - mae: 0.1572
Epoch 793/1000
71/71 0s 3ms/step - loss:
0.0453 - mae: 0.1570
Epoch 794/1000
71/71 0s 3ms/step - loss:
0.0453 - mae: 0.1574
Epoch 795/1000
71/71 0s 3ms/step - loss:
0.0453 - mae: 0.1573
Epoch 796/1000
71/71 0s 3ms/step - loss:
0.0453 - mae: 0.1571
Epoch 797/1000
71/71 0s 3ms/step - loss:
0.0450 - mae: 0.1569
Epoch 798/1000
71/71 0s 3ms/step - loss:
0.0451 - mae: 0.1572
Epoch 799/1000
71/71 0s 3ms/step - loss:
0.0452 - mae: 0.1568
Epoch 800/1000
71/71 0s 3ms/step - loss:
0.0451 - mae: 0.1570

Epoch 801/1000
71/71 0s 4ms/step - loss:
0.0452 - mae: 0.1568
Epoch 802/1000
71/71 0s 3ms/step - loss:
0.0450 - mae: 0.1565
Epoch 803/1000
71/71 0s 3ms/step - loss:
0.0451 - mae: 0.1566
Epoch 804/1000
71/71 0s 3ms/step - loss:
0.0451 - mae: 0.1567
Epoch 805/1000
71/71 0s 3ms/step - loss:
0.0451 - mae: 0.1568
Epoch 806/1000
71/71 0s 3ms/step - loss:
0.0451 - mae: 0.1567
Epoch 807/1000
71/71 0s 3ms/step - loss:
0.0450 - mae: 0.1563
Epoch 808/1000
71/71 0s 3ms/step - loss:
0.0449 - mae: 0.1567
Epoch 809/1000
71/71 0s 3ms/step - loss:
0.0449 - mae: 0.1566
Epoch 810/1000
71/71 0s 3ms/step - loss:
0.0450 - mae: 0.1565
Epoch 811/1000
71/71 0s 3ms/step - loss:
0.0450 - mae: 0.1565
Epoch 812/1000
71/71 0s 4ms/step - loss:
0.0449 - mae: 0.1564
Epoch 813/1000
71/71 0s 3ms/step - loss:
0.0450 - mae: 0.1565
Epoch 814/1000
71/71 0s 3ms/step - loss:
0.0449 - mae: 0.1562
Epoch 815/1000
71/71 0s 3ms/step - loss:
0.0448 - mae: 0.1561
Epoch 816/1000
71/71 0s 3ms/step - loss:
0.0449 - mae: 0.1566

Epoch 817/1000
71/71 0s 3ms/step - loss:
0.0448 - mae: 0.1561
Epoch 818/1000
71/71 0s 3ms/step - loss:
0.0447 - mae: 0.1559
Epoch 819/1000
71/71 0s 3ms/step - loss:
0.0448 - mae: 0.1562
Epoch 820/1000
71/71 0s 3ms/step - loss:
0.0447 - mae: 0.1559
Epoch 821/1000
71/71 0s 3ms/step - loss:
0.0447 - mae: 0.1558
Epoch 822/1000
71/71 0s 3ms/step - loss:
0.0447 - mae: 0.1559
Epoch 823/1000
71/71 0s 4ms/step - loss:
0.0447 - mae: 0.1556
Epoch 824/1000
71/71 0s 3ms/step - loss:
0.0447 - mae: 0.1556
Epoch 825/1000
71/71 0s 3ms/step - loss:
0.0446 - mae: 0.1555
Epoch 826/1000
71/71 0s 3ms/step - loss:
0.0447 - mae: 0.1559
Epoch 827/1000
71/71 0s 3ms/step - loss:
0.0446 - mae: 0.1554
Epoch 828/1000
71/71 0s 3ms/step - loss:
0.0446 - mae: 0.1559
Epoch 829/1000
71/71 0s 3ms/step - loss:
0.0447 - mae: 0.1560
Epoch 830/1000
71/71 0s 3ms/step - loss:
0.0444 - mae: 0.1554
Epoch 831/1000
71/71 0s 3ms/step - loss:
0.0445 - mae: 0.1558
Epoch 832/1000
71/71 0s 3ms/step - loss:
0.0445 - mae: 0.1555

Epoch 833/1000
71/71 0s 4ms/step - loss:
0.0444 - mae: 0.1558
Epoch 834/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1552
Epoch 835/1000
71/71 0s 3ms/step - loss:
0.0444 - mae: 0.1553
Epoch 836/1000
71/71 0s 3ms/step - loss:
0.0445 - mae: 0.1554
Epoch 837/1000
71/71 0s 3ms/step - loss:
0.0444 - mae: 0.1554
Epoch 838/1000
71/71 0s 3ms/step - loss:
0.0445 - mae: 0.1552
Epoch 839/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1555
Epoch 840/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1551
Epoch 841/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1553
Epoch 842/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1552
Epoch 843/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1554
Epoch 844/1000
71/71 0s 3ms/step - loss:
0.0441 - mae: 0.1551
Epoch 845/1000
71/71 0s 3ms/step - loss:
0.0442 - mae: 0.1550
Epoch 846/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1550
Epoch 847/1000
71/71 0s 3ms/step - loss:
0.0443 - mae: 0.1551
Epoch 848/1000
71/71 0s 3ms/step - loss:
0.0442 - mae: 0.1549

Epoch 849/1000
71/71 0s 3ms/step - loss:
0.0442 - mae: 0.1549
Epoch 850/1000
71/71 0s 3ms/step - loss:
0.0442 - mae: 0.1551
Epoch 851/1000
71/71 0s 3ms/step - loss:
0.0441 - mae: 0.1550
Epoch 852/1000
71/71 0s 3ms/step - loss:
0.0442 - mae: 0.1549
Epoch 853/1000
71/71 0s 3ms/step - loss:
0.0441 - mae: 0.1548
Epoch 854/1000
71/71 0s 3ms/step - loss:
0.0440 - mae: 0.1549
Epoch 855/1000
71/71 0s 3ms/step - loss:
0.0440 - mae: 0.1546
Epoch 856/1000
71/71 0s 4ms/step - loss:
0.0441 - mae: 0.1544
Epoch 857/1000
71/71 0s 3ms/step - loss:
0.0439 - mae: 0.1544
Epoch 858/1000
71/71 0s 3ms/step - loss:
0.0440 - mae: 0.1543
Epoch 859/1000
71/71 0s 3ms/step - loss:
0.0438 - mae: 0.1542
Epoch 860/1000
71/71 0s 3ms/step - loss:
0.0439 - mae: 0.1543
Epoch 861/1000
71/71 0s 3ms/step - loss:
0.0439 - mae: 0.1543
Epoch 862/1000
71/71 0s 3ms/step - loss:
0.0438 - mae: 0.1542
Epoch 863/1000
71/71 0s 3ms/step - loss:
0.0438 - mae: 0.1539
Epoch 864/1000
71/71 0s 3ms/step - loss:
0.0437 - mae: 0.1542

Epoch 865/1000
71/71 0s 3ms/step - loss:
0.0438 - mae: 0.1540
Epoch 866/1000
71/71 0s 3ms/step - loss:
0.0438 - mae: 0.1542
Epoch 867/1000
71/71 0s 4ms/step - loss:
0.0438 - mae: 0.1544
Epoch 868/1000
71/71 0s 3ms/step - loss:
0.0438 - mae: 0.1542
Epoch 869/1000
71/71 0s 3ms/step - loss:
0.0438 - mae: 0.1542
Epoch 870/1000
71/71 0s 3ms/step - loss:
0.0437 - mae: 0.1544
Epoch 871/1000
71/71 0s 3ms/step - loss:
0.0437 - mae: 0.1538
Epoch 872/1000
71/71 0s 3ms/step - loss:
0.0436 - mae: 0.1543
Epoch 873/1000
71/71 0s 3ms/step - loss:
0.0437 - mae: 0.1538
Epoch 874/1000
71/71 0s 3ms/step - loss:
0.0436 - mae: 0.1541
Epoch 875/1000
71/71 0s 3ms/step - loss:
0.0436 - mae: 0.1537
Epoch 876/1000
71/71 0s 3ms/step - loss:
0.0436 - mae: 0.1543
Epoch 877/1000
71/71 0s 3ms/step - loss:
0.0436 - mae: 0.1539
Epoch 878/1000
71/71 0s 4ms/step - loss:
0.0436 - mae: 0.1537
Epoch 879/1000
71/71 0s 3ms/step - loss:
0.0437 - mae: 0.1540
Epoch 880/1000
71/71 0s 3ms/step - loss:
0.0434 - mae: 0.1532

Epoch 881/1000
71/71 0s 3ms/step - loss:
0.0435 - mae: 0.1537
Epoch 882/1000
71/71 0s 3ms/step - loss:
0.0434 - mae: 0.1536
Epoch 883/1000
71/71 0s 3ms/step - loss:
0.0435 - mae: 0.1535
Epoch 884/1000
71/71 0s 3ms/step - loss:
0.0434 - mae: 0.1536
Epoch 885/1000
71/71 0s 3ms/step - loss:
0.0435 - mae: 0.1537
Epoch 886/1000
71/71 0s 3ms/step - loss:
0.0434 - mae: 0.1534
Epoch 887/1000
71/71 0s 3ms/step - loss:
0.0434 - mae: 0.1535
Epoch 888/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1534
Epoch 889/1000
71/71 0s 4ms/step - loss:
0.0434 - mae: 0.1534
Epoch 890/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1533
Epoch 891/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1535
Epoch 892/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1534
Epoch 893/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1532
Epoch 894/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1533
Epoch 895/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1531
Epoch 896/1000
71/71 0s 3ms/step - loss:
0.0433 - mae: 0.1534

Epoch 897/1000
71/71 0s 3ms/step - loss:
0.0432 - mae: 0.1532
Epoch 898/1000
71/71 0s 3ms/step - loss:
0.0432 - mae: 0.1531
Epoch 899/1000
71/71 0s 4ms/step - loss:
0.0431 - mae: 0.1529
Epoch 900/1000
71/71 0s 3ms/step - loss:
0.0432 - mae: 0.1529
Epoch 901/1000
71/71 0s 3ms/step - loss:
0.0432 - mae: 0.1530
Epoch 902/1000
71/71 0s 3ms/step - loss:
0.0431 - mae: 0.1529
Epoch 903/1000
71/71 0s 3ms/step - loss:
0.0432 - mae: 0.1531
Epoch 904/1000
71/71 0s 3ms/step - loss:
0.0431 - mae: 0.1529
Epoch 905/1000
71/71 0s 3ms/step - loss:
0.0430 - mae: 0.1527
Epoch 906/1000
71/71 0s 3ms/step - loss:
0.0430 - mae: 0.1526
Epoch 907/1000
71/71 0s 3ms/step - loss:
0.0430 - mae: 0.1529
Epoch 908/1000
71/71 0s 3ms/step - loss:
0.0430 - mae: 0.1525
Epoch 909/1000
71/71 0s 3ms/step - loss:
0.0430 - mae: 0.1525
Epoch 910/1000
71/71 0s 3ms/step - loss:
0.0429 - mae: 0.1528
Epoch 911/1000
71/71 0s 3ms/step - loss:
0.0430 - mae: 0.1527
Epoch 912/1000
71/71 0s 3ms/step - loss:
0.0429 - mae: 0.1522

Epoch 913/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1527
Epoch 914/1000
71/71 0s 3ms/step - loss:
0.0430 - mae: 0.1527
Epoch 915/1000
71/71 0s 3ms/step - loss:
0.0429 - mae: 0.1525
Epoch 916/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1520
Epoch 917/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1520
Epoch 918/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1525
Epoch 919/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1523
Epoch 920/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1519
Epoch 921/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1525
Epoch 922/1000
71/71 0s 4ms/step - loss:
0.0427 - mae: 0.1522
Epoch 923/1000
71/71 0s 3ms/step - loss:
0.0426 - mae: 0.1524
Epoch 924/1000
71/71 0s 3ms/step - loss:
0.0427 - mae: 0.1522
Epoch 925/1000
71/71 0s 3ms/step - loss:
0.0427 - mae: 0.1524
Epoch 926/1000
71/71 0s 3ms/step - loss:
0.0428 - mae: 0.1521
Epoch 927/1000
71/71 0s 3ms/step - loss:
0.0426 - mae: 0.1519
Epoch 928/1000
71/71 0s 3ms/step - loss:
0.0425 - mae: 0.1517

Epoch 929/1000
71/71 0s 3ms/step - loss:
0.0426 - mae: 0.1519
Epoch 930/1000
71/71 0s 3ms/step - loss:
0.0426 - mae: 0.1518
Epoch 931/1000
71/71 0s 3ms/step - loss:
0.0426 - mae: 0.1518
Epoch 932/1000
71/71 0s 3ms/step - loss:
0.0426 - mae: 0.1519
Epoch 933/1000
71/71 0s 4ms/step - loss:
0.0426 - mae: 0.1516
Epoch 934/1000
71/71 0s 3ms/step - loss:
0.0425 - mae: 0.1517
Epoch 935/1000
71/71 0s 3ms/step - loss:
0.0424 - mae: 0.1517
Epoch 936/1000
71/71 0s 3ms/step - loss:
0.0425 - mae: 0.1518
Epoch 937/1000
71/71 0s 3ms/step - loss:
0.0425 - mae: 0.1518
Epoch 938/1000
71/71 0s 3ms/step - loss:
0.0424 - mae: 0.1514
Epoch 939/1000
71/71 0s 3ms/step - loss:
0.0425 - mae: 0.1520
Epoch 940/1000
71/71 0s 3ms/step - loss:
0.0424 - mae: 0.1512
Epoch 941/1000
71/71 0s 3ms/step - loss:
0.0424 - mae: 0.1514
Epoch 942/1000
71/71 0s 3ms/step - loss:
0.0424 - mae: 0.1516
Epoch 943/1000
71/71 0s 3ms/step - loss:
0.0424 - mae: 0.1515
Epoch 944/1000
71/71 0s 4ms/step - loss:
0.0423 - mae: 0.1513

Epoch 945/1000
71/71 0s 3ms/step - loss:
0.0423 - mae: 0.1515
Epoch 946/1000
71/71 0s 3ms/step - loss:
0.0423 - mae: 0.1514
Epoch 947/1000
71/71 0s 3ms/step - loss:
0.0423 - mae: 0.1516
Epoch 948/1000
71/71 0s 3ms/step - loss:
0.0423 - mae: 0.1509
Epoch 949/1000
71/71 0s 3ms/step - loss:
0.0422 - mae: 0.1511
Epoch 950/1000
71/71 0s 3ms/step - loss:
0.0423 - mae: 0.1513
Epoch 951/1000
71/71 0s 3ms/step - loss:
0.0423 - mae: 0.1516
Epoch 952/1000
71/71 0s 3ms/step - loss:
0.0422 - mae: 0.1510
Epoch 953/1000
71/71 0s 3ms/step - loss:
0.0420 - mae: 0.1505
Epoch 954/1000
71/71 0s 3ms/step - loss:
0.0421 - mae: 0.1515
Epoch 955/1000
71/71 0s 4ms/step - loss:
0.0422 - mae: 0.1510
Epoch 956/1000
71/71 0s 3ms/step - loss:
0.0422 - mae: 0.1512
Epoch 957/1000
71/71 0s 3ms/step - loss:
0.0421 - mae: 0.1508
Epoch 958/1000
71/71 0s 3ms/step - loss:
0.0421 - mae: 0.1511
Epoch 959/1000
71/71 0s 3ms/step - loss:
0.0421 - mae: 0.1508
Epoch 960/1000
71/71 0s 3ms/step - loss:
0.0421 - mae: 0.1511

Epoch 961/1000
71/71 0s 3ms/step - loss:
0.0420 - mae: 0.1510
Epoch 962/1000
71/71 0s 3ms/step - loss:
0.0419 - mae: 0.1506
Epoch 963/1000
71/71 0s 3ms/step - loss:
0.0420 - mae: 0.1507
Epoch 964/1000
71/71 0s 3ms/step - loss:
0.0420 - mae: 0.1509
Epoch 965/1000
71/71 0s 3ms/step - loss:
0.0418 - mae: 0.1504
Epoch 966/1000
71/71 0s 4ms/step - loss:
0.0420 - mae: 0.1508
Epoch 967/1000
71/71 0s 3ms/step - loss:
0.0420 - mae: 0.1504
Epoch 968/1000
71/71 0s 3ms/step - loss:
0.0419 - mae: 0.1505
Epoch 969/1000
71/71 0s 3ms/step - loss:
0.0419 - mae: 0.1504
Epoch 970/1000
71/71 0s 3ms/step - loss:
0.0419 - mae: 0.1505
Epoch 971/1000
71/71 0s 3ms/step - loss:
0.0418 - mae: 0.1507
Epoch 972/1000
71/71 0s 3ms/step - loss:
0.0418 - mae: 0.1502
Epoch 973/1000
71/71 0s 3ms/step - loss:
0.0418 - mae: 0.1509
Epoch 974/1000
71/71 0s 3ms/step - loss:
0.0418 - mae: 0.1504
Epoch 975/1000
71/71 0s 3ms/step - loss:
0.0419 - mae: 0.1505
Epoch 976/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1502

Epoch 977/1000
71/71 0s 4ms/step - loss:
0.0418 - mae: 0.1504
Epoch 978/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1501
Epoch 979/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1501
Epoch 980/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1500
Epoch 981/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1499
Epoch 982/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1502
Epoch 983/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1504
Epoch 984/1000
71/71 0s 3ms/step - loss:
0.0416 - mae: 0.1497
Epoch 985/1000
71/71 0s 3ms/step - loss:
0.0416 - mae: 0.1500
Epoch 986/1000
71/71 0s 3ms/step - loss:
0.0417 - mae: 0.1501
Epoch 987/1000
71/71 0s 3ms/step - loss:
0.0416 - mae: 0.1501
Epoch 988/1000
71/71 0s 4ms/step - loss:
0.0416 - mae: 0.1497
Epoch 989/1000
71/71 0s 3ms/step - loss:
0.0416 - mae: 0.1501
Epoch 990/1000
71/71 0s 3ms/step - loss:
0.0416 - mae: 0.1498
Epoch 991/1000
71/71 0s 3ms/step - loss:
0.0415 - mae: 0.1499
Epoch 992/1000
71/71 0s 3ms/step - loss:
0.0415 - mae: 0.1498

```

Epoch 993/1000
71/71          0s 3ms/step - loss:
0.0414 - mae: 0.1495
Epoch 994/1000
71/71          0s 3ms/step - loss:
0.0414 - mae: 0.1497
Epoch 995/1000
71/71          0s 3ms/step - loss:
0.0415 - mae: 0.1496
Epoch 996/1000
71/71          0s 3ms/step - loss:
0.0415 - mae: 0.1499
Epoch 997/1000
71/71          0s 3ms/step - loss:
0.0414 - mae: 0.1493
Epoch 998/1000
71/71          0s 3ms/step - loss:
0.0414 - mae: 0.1499
Epoch 999/1000
71/71          0s 4ms/step - loss:
0.0414 - mae: 0.1497
Epoch 1000/1000
71/71          0s 3ms/step - loss:
0.0415 - mae: 0.1496
Epoch 1/1000
55/57          0s 2ms/step - loss:
0.0414 - mae: 0.1499

```

```

C:\Users\barre\my_new_env2\Lib\site-
packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran
out of data; interrupting training. Make sure that your dataset or generator can
generate at least `steps_per_epoch * epochs` batches. You may need to use the
`.repeat()` function when building your dataset.

```

```

    self._interrupted_warning()

57/57          0s 7ms/step - loss:
0.0416 - mae: 0.1489 - val_loss: 0.0404 - val_mae: 0.1511
Epoch 2/1000
57/57          0s 4ms/step - loss:
0.0415 - mae: 0.1498 - val_loss: 0.0411 - val_mae: 0.1498
Epoch 3/1000
57/57          0s 4ms/step - loss:
0.0414 - mae: 0.1491 - val_loss: 0.0408 - val_mae: 0.1496
Epoch 4/1000
57/57          0s 4ms/step - loss:
0.0416 - mae: 0.1491 - val_loss: 0.0405 - val_mae: 0.1496
Epoch 5/1000
57/57          0s 4ms/step - loss:
0.0416 - mae: 0.1492 - val_loss: 0.0405 - val_mae: 0.1498

```


Epoch 6/1000
57/57 0s 4ms/step - loss:
0.0415 - mae: 0.1493 - val_loss: 0.0408 - val_mae: 0.1498
Epoch 7/1000
57/57 0s 4ms/step - loss:
0.0414 - mae: 0.1490 - val_loss: 0.0405 - val_mae: 0.1500
Epoch 8/1000
57/57 0s 4ms/step - loss:
0.0414 - mae: 0.1490 - val_loss: 0.0405 - val_mae: 0.1504
Epoch 9/1000
57/57 0s 4ms/step - loss:
0.0415 - mae: 0.1493 - val_loss: 0.0405 - val_mae: 0.1500
Epoch 10/1000
57/57 0s 4ms/step - loss:
0.0414 - mae: 0.1492 - val_loss: 0.0405 - val_mae: 0.1504
Epoch 11/1000
57/57 0s 4ms/step - loss:
0.0414 - mae: 0.1493 - val_loss: 0.0407 - val_mae: 0.1501
Epoch 12/1000
57/57 0s 4ms/step - loss:
0.0414 - mae: 0.1489 - val_loss: 0.0405 - val_mae: 0.1506
Epoch 13/1000
57/57 0s 4ms/step - loss:
0.0413 - mae: 0.1489 - val_loss: 0.0405 - val_mae: 0.1502
Epoch 14/1000
57/57 0s 4ms/step - loss:
0.0414 - mae: 0.1488 - val_loss: 0.0408 - val_mae: 0.1522
Epoch 15/1000
57/57 0s 4ms/step - loss:
0.0414 - mae: 0.1494 - val_loss: 0.0405 - val_mae: 0.1503
Epoch 16/1000
57/57 0s 4ms/step - loss:
0.0413 - mae: 0.1491 - val_loss: 0.0412 - val_mae: 0.1506
Epoch 17/1000
57/57 0s 4ms/step - loss:
0.0413 - mae: 0.1491 - val_loss: 0.0410 - val_mae: 0.1504
Epoch 18/1000
57/57 0s 4ms/step - loss:
0.0413 - mae: 0.1488 - val_loss: 0.0405 - val_mae: 0.1510
Epoch 19/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1486 - val_loss: 0.0406 - val_mae: 0.1514
Epoch 20/1000
57/57 0s 4ms/step - loss:
0.0413 - mae: 0.1490 - val_loss: 0.0407 - val_mae: 0.1520
Epoch 21/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1490 - val_loss: 0.0411 - val_mae: 0.1506

Epoch 22/1000
57/57 0s 4ms/step - loss:
0.0413 - mae: 0.1490 - val_loss: 0.0411 - val_mae: 0.1506
Epoch 23/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1485 - val_loss: 0.0406 - val_mae: 0.1513
Epoch 24/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1485 - val_loss: 0.0407 - val_mae: 0.1519
Epoch 25/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1489 - val_loss: 0.0406 - val_mae: 0.1512
Epoch 26/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1488 - val_loss: 0.0406 - val_mae: 0.1507
Epoch 27/1000
57/57 0s 4ms/step - loss:
0.0411 - mae: 0.1489 - val_loss: 0.0407 - val_mae: 0.1505
Epoch 28/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1487 - val_loss: 0.0406 - val_mae: 0.1506
Epoch 29/1000
57/57 0s 4ms/step - loss:
0.0411 - mae: 0.1486 - val_loss: 0.0406 - val_mae: 0.1507
Epoch 30/1000
57/57 0s 4ms/step - loss:
0.0410 - mae: 0.1486 - val_loss: 0.0407 - val_mae: 0.1505
Epoch 31/1000
57/57 0s 4ms/step - loss:
0.0411 - mae: 0.1488 - val_loss: 0.0408 - val_mae: 0.1505
Epoch 32/1000
57/57 0s 4ms/step - loss:
0.0412 - mae: 0.1488 - val_loss: 0.0408 - val_mae: 0.1506
Epoch 33/1000
57/57 0s 4ms/step - loss:
0.0411 - mae: 0.1485 - val_loss: 0.0406 - val_mae: 0.1507
Epoch 34/1000
57/57 0s 4ms/step - loss:
0.0410 - mae: 0.1486 - val_loss: 0.0408 - val_mae: 0.1506
Epoch 35/1000
57/57 0s 4ms/step - loss:
0.0409 - mae: 0.1480 - val_loss: 0.0406 - val_mae: 0.1512
Epoch 36/1000
57/57 0s 4ms/step - loss:
0.0410 - mae: 0.1483 - val_loss: 0.0412 - val_mae: 0.1539
Epoch 37/1000
57/57 0s 4ms/step - loss:
0.0411 - mae: 0.1489 - val_loss: 0.0408 - val_mae: 0.1506

```

Epoch 38/1000
57/57          0s 4ms/step - loss:
0.0412 - mae: 0.1488 - val_loss: 0.0408 - val_mae: 0.1506
Epoch 39/1000
57/57          0s 4ms/step - loss:
0.0410 - mae: 0.1484 - val_loss: 0.0407 - val_mae: 0.1506
Epoch 40/1000
57/57          0s 4ms/step - loss:
0.0410 - mae: 0.1483 - val_loss: 0.0407 - val_mae: 0.1507
Epoch 41/1000
57/57          0s 4ms/step - loss:
0.0409 - mae: 0.1483 - val_loss: 0.0408 - val_mae: 0.1506
Epoch 42/1000
57/57          0s 4ms/step - loss:
0.0409 - mae: 0.1482 - val_loss: 0.0406 - val_mae: 0.1513
Epoch 43/1000
57/57          0s 4ms/step - loss:
0.0409 - mae: 0.1483 - val_loss: 0.0406 - val_mae: 0.1510
Epoch 44/1000
57/57          0s 4ms/step - loss:
0.0410 - mae: 0.1484 - val_loss: 0.0407 - val_mae: 0.1506
Epoch 45/1000
57/57          0s 4ms/step - loss:
0.0408 - mae: 0.1478 - val_loss: 0.0410 - val_mae: 0.1529
Epoch 46/1000
57/57          0s 4ms/step - loss:
0.0409 - mae: 0.1487 - val_loss: 0.0409 - val_mae: 0.1507
Epoch 47/1000
57/57          0s 4ms/step - loss:
0.0409 - mae: 0.1482 - val_loss: 0.0406 - val_mae: 0.1511
Epoch 48/1000
57/57          0s 4ms/step - loss:
0.0409 - mae: 0.1482 - val_loss: 0.0407 - val_mae: 0.1512
Epoch 49/1000
57/57          0s 4ms/step - loss:
0.0409 - mae: 0.1485 - val_loss: 0.0408 - val_mae: 0.1506
Epoch 50/1000
57/57          0s 4ms/step - loss:
0.0408 - mae: 0.1479 - val_loss: 0.0406 - val_mae: 0.1510
Epoch 50: early stopping
Restoring model weights from the end of the best epoch: 1.
Raw evaluate results: []

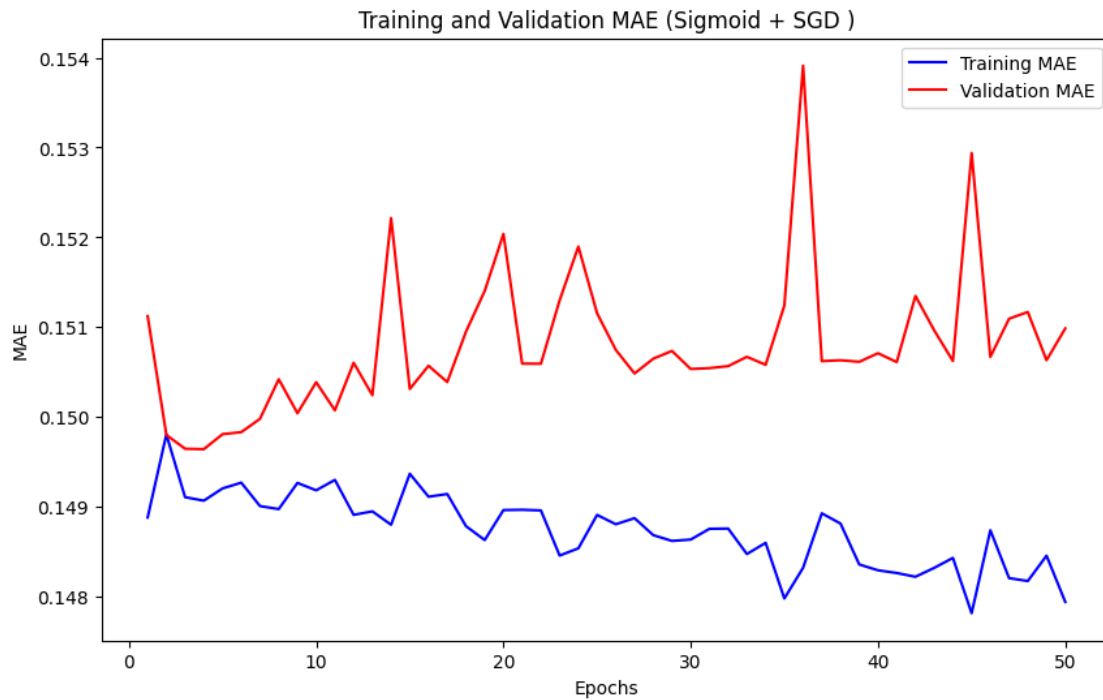
```

```

C:\Users\barre\my_new_env2\Lib\site-
packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran
out of data; interrupting training. Make sure that your dataset or generator can
generate at least `steps_per_epoch * epochs` batches. You may need to use the
`.repeat()` function when building your dataset.

```

```
self._interrupted_warning()
```



```
[7]: print("\n--- Training Model with ReLu and Adam ---")
model_relu = Sequential()
model_relu.add(Dense(256, input_dim=train_features.shape[1], activation='relu'))
model_relu.add(Dense(128, activation='relu'))
model_relu.add(Dense(64, activation='relu'))
model_relu.add(Dense(1, activation='linear'))

model_relu.compile(optimizer=Adam(learning_rate=0.001), loss="mse",
    ↪metrics=["mae"])

# Define early stopping
early_stop_relu = EarlyStopping(monitor='val_loss', patience=50, verbose=1,
    ↪restore_best_weights=True)

model_relu.fit(train_features, train_target,
               epochs=1000,
               batch_size=32,
               validation_data=(test_features, test_target),
               callbacks=[early_stop_relu],
               verbose=1)

loss_relu = model_relu.evaluate(test_features, test_target, verbose=0)
```

```

history_relu = model_relu.fit(train_features, train_target,
                              epochs=1000,
                              batch_size=32,
                              validation_split=0.2,
                              callbacks=[early_stop_tanh],
                              verbose=1)

relu_results = model_relu.evaluate(test_features, test_target, verbose=0)
print("Raw evaluate results:", relu_results)

# Plot the training and validation MAE
mae_relu = history_relu.history['mae']
val_mae_relu = history_relu.history['val_mae']
epochs = range(1, len(mae_relu) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs, mae_relu, 'b', label='Training MAE')
plt.plot(epochs, val_mae_relu, 'r', label='Validation MAE')
plt.title('Training and Validation MAE (Sigmoid + SGD)')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()

```

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

--- Training Model with ReLu and Adam ---

Epoch 1/1000

71/71 2s 5ms/step - loss:

8.4954 - mae: 2.0538

Epoch 2/1000

71/71 0s 2ms/step - loss:

0.6219 - mae: 0.6007

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran out of data; interrupting training. Make sure that your dataset or generator can generate at least `steps\_per\_epoch \* epochs` batches. You may need to use the `.repeat()` function when building your dataset.

```
self._interrupted_warning()
```

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\callbacks\early_stopping.py:99: UserWarning: Early stopping conditioned on metric `val\_loss` which is not available. Available metrics are: loss,mae

```

    current = self.get_monitor_value(logs)

71/71          0s 3ms/step - loss:
0.3666 - mae: 0.4480
Epoch 3/1000
71/71          0s 3ms/step - loss:
0.1182 - mae: 0.2645
Epoch 4/1000
71/71          0s 3ms/step - loss:
0.0839 - mae: 0.2216
Epoch 5/1000
71/71          0s 3ms/step - loss:
0.0705 - mae: 0.2012
Epoch 6/1000
71/71          0s 3ms/step - loss:
0.0614 - mae: 0.1882
Epoch 7/1000
71/71          0s 3ms/step - loss:
0.0576 - mae: 0.1810
Epoch 8/1000
71/71          0s 3ms/step - loss:
0.0528 - mae: 0.1722
Epoch 9/1000
71/71          0s 3ms/step - loss:
0.0552 - mae: 0.1769
Epoch 10/1000
71/71          0s 3ms/step - loss:
0.0482 - mae: 0.1637
Epoch 11/1000
71/71          0s 3ms/step - loss:
0.0427 - mae: 0.1543
Epoch 12/1000
71/71          0s 3ms/step - loss:
0.0428 - mae: 0.1559
Epoch 13/1000
71/71          0s 3ms/step - loss:
0.0399 - mae: 0.1473
Epoch 14/1000
71/71          0s 3ms/step - loss:
0.0389 - mae: 0.1450
Epoch 15/1000
71/71          0s 3ms/step - loss:
0.0424 - mae: 0.1542
Epoch 16/1000
71/71          0s 3ms/step - loss:
0.0385 - mae: 0.1492
Epoch 17/1000
71/71          0s 3ms/step - loss:

```

```

0.0365 - mae: 0.1433
Epoch 18/1000
71/71          0s 3ms/step - loss:
0.0344 - mae: 0.1342
Epoch 19/1000
71/71          0s 3ms/step - loss:
0.0329 - mae: 0.1340
Epoch 20/1000
71/71          0s 3ms/step - loss:
0.0328 - mae: 0.1357
Epoch 21/1000
71/71          0s 4ms/step - loss:
0.0298 - mae: 0.1249
Epoch 22/1000
71/71          0s 3ms/step - loss:
0.0298 - mae: 0.1300
Epoch 23/1000
71/71          0s 3ms/step - loss:
0.0327 - mae: 0.1360
Epoch 24/1000
71/71          0s 3ms/step - loss:
0.0307 - mae: 0.1311
Epoch 25/1000
71/71          0s 3ms/step - loss:
0.0317 - mae: 0.1342
Epoch 26/1000
71/71          0s 3ms/step - loss:
0.0250 - mae: 0.1175
Epoch 27/1000
71/71          0s 3ms/step - loss:
0.0255 - mae: 0.1159
Epoch 28/1000
71/71          0s 3ms/step - loss:
0.0245 - mae: 0.1173
Epoch 29/1000
71/71          0s 3ms/step - loss:
0.0246 - mae: 0.1155
Epoch 30/1000
71/71          0s 3ms/step - loss:
0.0256 - mae: 0.1194
Epoch 31/1000
71/71          0s 3ms/step - loss:
0.0263 - mae: 0.1216
Epoch 32/1000
71/71          0s 4ms/step - loss:
0.0322 - mae: 0.1350
Epoch 33/1000
71/71          0s 3ms/step - loss:

```

0.0244 - mae: 0.1156
Epoch 34/1000
71/71 0s 3ms/step - loss:
0.0221 - mae: 0.1113
Epoch 35/1000
71/71 0s 3ms/step - loss:
0.0239 - mae: 0.1158
Epoch 36/1000
71/71 0s 3ms/step - loss:
0.0191 - mae: 0.1021
Epoch 37/1000
71/71 0s 3ms/step - loss:
0.0237 - mae: 0.1168
Epoch 38/1000
71/71 0s 3ms/step - loss:
0.0216 - mae: 0.1109
Epoch 39/1000
71/71 0s 3ms/step - loss:
0.0211 - mae: 0.1079
Epoch 40/1000
71/71 0s 3ms/step - loss:
0.0221 - mae: 0.1127
Epoch 41/1000
71/71 0s 3ms/step - loss:
0.0227 - mae: 0.1132
Epoch 42/1000
71/71 0s 3ms/step - loss:
0.0251 - mae: 0.1206
Epoch 43/1000
71/71 0s 4ms/step - loss:
0.0201 - mae: 0.1061
Epoch 44/1000
71/71 0s 3ms/step - loss:
0.0232 - mae: 0.1129
Epoch 45/1000
71/71 0s 3ms/step - loss:
0.0162 - mae: 0.0928
Epoch 46/1000
71/71 0s 3ms/step - loss:
0.0174 - mae: 0.0986
Epoch 47/1000
71/71 0s 3ms/step - loss:
0.0185 - mae: 0.1017
Epoch 48/1000
71/71 0s 3ms/step - loss:
0.0222 - mae: 0.1114
Epoch 49/1000
71/71 0s 3ms/step - loss:

0.0231 - mae: 0.1154
Epoch 50/1000
71/71 0s 3ms/step - loss:
0.0213 - mae: 0.1107
Epoch 51/1000
71/71 0s 3ms/step - loss:
0.0169 - mae: 0.0966
Epoch 52/1000
71/71 0s 3ms/step - loss:
0.0169 - mae: 0.0978
Epoch 53/1000
71/71 0s 3ms/step - loss:
0.0212 - mae: 0.1099
Epoch 54/1000
71/71 0s 4ms/step - loss:
0.0187 - mae: 0.1029
Epoch 55/1000
71/71 0s 3ms/step - loss:
0.0189 - mae: 0.1038
Epoch 56/1000
71/71 0s 3ms/step - loss:
0.0216 - mae: 0.1118
Epoch 57/1000
71/71 0s 3ms/step - loss:
0.0167 - mae: 0.0975
Epoch 58/1000
71/71 0s 3ms/step - loss:
0.0145 - mae: 0.0894
Epoch 59/1000
71/71 0s 3ms/step - loss:
0.0147 - mae: 0.0911
Epoch 60/1000
71/71 0s 3ms/step - loss:
0.0174 - mae: 0.1002
Epoch 61/1000
71/71 0s 3ms/step - loss:
0.0195 - mae: 0.1085
Epoch 62/1000
71/71 0s 3ms/step - loss:
0.0178 - mae: 0.0994
Epoch 63/1000
71/71 0s 3ms/step - loss:
0.0196 - mae: 0.1078
Epoch 64/1000
71/71 0s 3ms/step - loss:
0.0161 - mae: 0.0971
Epoch 65/1000
71/71 0s 4ms/step - loss:

```

0.0150 - mae: 0.0937
Epoch 66/1000
71/71          0s 3ms/step - loss:
0.0150 - mae: 0.0924
Epoch 67/1000
71/71          0s 3ms/step - loss:
0.0127 - mae: 0.0842
Epoch 68/1000
71/71          0s 3ms/step - loss:
0.0146 - mae: 0.0927
Epoch 69/1000
71/71          0s 3ms/step - loss:
0.0132 - mae: 0.0858
Epoch 70/1000
71/71          0s 3ms/step - loss:
0.0152 - mae: 0.0925
Epoch 71/1000
71/71          0s 3ms/step - loss:
0.0223 - mae: 0.1136
Epoch 72/1000
71/71          0s 3ms/step - loss:
0.0258 - mae: 0.1246
Epoch 73/1000
71/71          0s 3ms/step - loss:
0.0185 - mae: 0.1059
Epoch 74/1000
71/71          0s 3ms/step - loss:
0.0232 - mae: 0.1193
Epoch 75/1000
71/71          0s 3ms/step - loss:
0.0171 - mae: 0.0994
Epoch 76/1000
71/71          0s 4ms/step - loss:
0.0167 - mae: 0.0978
Epoch 77/1000
71/71          0s 3ms/step - loss:
0.0130 - mae: 0.0842
Epoch 78/1000
71/71          0s 3ms/step - loss:
0.0190 - mae: 0.1045
Epoch 79/1000
71/71          0s 3ms/step - loss:
0.0164 - mae: 0.0968
Epoch 80/1000
71/71          0s 3ms/step - loss:
0.0103 - mae: 0.0747
Epoch 81/1000
71/71          0s 3ms/step - loss:

```

0.0135 - mae: 0.0890
 Epoch 82/1000
 71/71 0s 3ms/step - loss:
 0.0200 - mae: 0.1056
 Epoch 83/1000
 71/71 0s 3ms/step - loss:
 0.0195 - mae: 0.1049
 Epoch 84/1000
 71/71 0s 3ms/step - loss:
 0.0114 - mae: 0.0815
 Epoch 85/1000
 71/71 0s 3ms/step - loss:
 0.0132 - mae: 0.0867
 Epoch 86/1000
 71/71 0s 3ms/step - loss:
 0.0131 - mae: 0.0877
 Epoch 87/1000
 71/71 0s 3ms/step - loss:
 0.0180 - mae: 0.1037
 Epoch 88/1000
 71/71 0s 3ms/step - loss:
 0.0119 - mae: 0.0830
 Epoch 89/1000
 71/71 0s 3ms/step - loss:
 0.0105 - mae: 0.0760
 Epoch 90/1000
 71/71 0s 3ms/step - loss:
 0.0083 - mae: 0.0669
 Epoch 91/1000
 71/71 0s 3ms/step - loss:
 0.0119 - mae: 0.0842
 Epoch 92/1000
 71/71 0s 3ms/step - loss:
 0.0104 - mae: 0.0771
 Epoch 93/1000
 71/71 0s 3ms/step - loss:
 0.0167 - mae: 0.0988
 Epoch 94/1000
 71/71 0s 3ms/step - loss:
 0.0140 - mae: 0.0916
 Epoch 95/1000
 71/71 0s 3ms/step - loss:
 0.0125 - mae: 0.0844
 Epoch 96/1000
 71/71 0s 3ms/step - loss:
 0.0125 - mae: 0.0856
 Epoch 97/1000
 71/71 0s 3ms/step - loss:

0.0155 - mae: 0.0955
Epoch 98/1000
71/71 0s 4ms/step - loss:
0.0148 - mae: 0.0943
Epoch 99/1000
71/71 0s 3ms/step - loss:
0.0156 - mae: 0.0977
Epoch 100/1000
71/71 0s 3ms/step - loss:
0.0143 - mae: 0.0913
Epoch 101/1000
71/71 0s 3ms/step - loss:
0.0102 - mae: 0.0759
Epoch 102/1000
71/71 0s 3ms/step - loss:
0.0125 - mae: 0.0861
Epoch 103/1000
71/71 0s 3ms/step - loss:
0.0124 - mae: 0.0860
Epoch 104/1000
71/71 0s 3ms/step - loss:
0.0123 - mae: 0.0846
Epoch 105/1000
71/71 0s 3ms/step - loss:
0.0094 - mae: 0.0743
Epoch 106/1000
71/71 0s 3ms/step - loss:
0.0097 - mae: 0.0749
Epoch 107/1000
71/71 0s 3ms/step - loss:
0.0115 - mae: 0.0820
Epoch 108/1000
71/71 0s 3ms/step - loss:
0.0097 - mae: 0.0760
Epoch 109/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0793
Epoch 110/1000
71/71 0s 3ms/step - loss:
0.0163 - mae: 0.0955
Epoch 111/1000
71/71 0s 3ms/step - loss:
0.0166 - mae: 0.0994
Epoch 112/1000
71/71 0s 3ms/step - loss:
0.0170 - mae: 0.1007
Epoch 113/1000
71/71 0s 3ms/step - loss:

0.0085 - mae: 0.0696
Epoch 114/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0785
Epoch 115/1000
71/71 0s 3ms/step - loss:
0.0114 - mae: 0.0831
Epoch 116/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0660
Epoch 117/1000
71/71 0s 3ms/step - loss:
0.0089 - mae: 0.0730
Epoch 118/1000
71/71 0s 3ms/step - loss:
0.0139 - mae: 0.0930
Epoch 119/1000
71/71 0s 3ms/step - loss:
0.0109 - mae: 0.0809
Epoch 120/1000
71/71 0s 4ms/step - loss:
0.0098 - mae: 0.0752
Epoch 121/1000
71/71 0s 3ms/step - loss:
0.0099 - mae: 0.0765
Epoch 122/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0645
Epoch 123/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0663
Epoch 124/1000
71/71 0s 3ms/step - loss:
0.0172 - mae: 0.1021
Epoch 125/1000
71/71 0s 3ms/step - loss:
0.0115 - mae: 0.0839
Epoch 126/1000
71/71 0s 3ms/step - loss:
0.0098 - mae: 0.0757
Epoch 127/1000
71/71 0s 3ms/step - loss:
0.0087 - mae: 0.0717
Epoch 128/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0670
Epoch 129/1000
71/71 0s 3ms/step - loss:

0.0103 - mae: 0.0759
Epoch 130/1000
71/71 0s 3ms/step - loss:
0.0119 - mae: 0.0795
Epoch 131/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0643
Epoch 132/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0812
Epoch 133/1000
71/71 0s 3ms/step - loss:
0.0155 - mae: 0.0977
Epoch 134/1000
71/71 0s 3ms/step - loss:
0.0101 - mae: 0.0784
Epoch 135/1000
71/71 0s 3ms/step - loss:
0.0087 - mae: 0.0717
Epoch 136/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0687
Epoch 137/1000
71/71 0s 3ms/step - loss:
0.0087 - mae: 0.0710
Epoch 138/1000
71/71 0s 3ms/step - loss:
0.0075 - mae: 0.0657
Epoch 139/1000
71/71 0s 3ms/step - loss:
0.0091 - mae: 0.0744
Epoch 140/1000
71/71 0s 3ms/step - loss:
0.0093 - mae: 0.0749
Epoch 141/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0577
Epoch 142/1000
71/71 0s 4ms/step - loss:
0.0056 - mae: 0.0577
Epoch 143/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0661
Epoch 144/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0608
Epoch 145/1000
71/71 0s 3ms/step - loss:

0.0054 - mae: 0.0572
Epoch 146/1000
71/71 0s 3ms/step - loss:
0.0084 - mae: 0.0718
Epoch 147/1000
71/71 0s 3ms/step - loss:
0.0092 - mae: 0.0752
Epoch 148/1000
71/71 0s 3ms/step - loss:
0.0101 - mae: 0.0783
Epoch 149/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0700
Epoch 150/1000
71/71 0s 3ms/step - loss:
0.0082 - mae: 0.0703
Epoch 151/1000
71/71 0s 3ms/step - loss:
0.0133 - mae: 0.0874
Epoch 152/1000
71/71 0s 3ms/step - loss:
0.0136 - mae: 0.0919
Epoch 153/1000
71/71 0s 4ms/step - loss:
0.0096 - mae: 0.0758
Epoch 154/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0664
Epoch 155/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0597
Epoch 156/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0558
Epoch 157/1000
71/71 0s 3ms/step - loss:
0.0083 - mae: 0.0710
Epoch 158/1000
71/71 0s 3ms/step - loss:
0.0077 - mae: 0.0687
Epoch 159/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0579
Epoch 160/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0572
Epoch 161/1000
71/71 0s 3ms/step - loss:

0.0088 - mae: 0.0729
Epoch 162/1000
71/71 0s 3ms/step - loss:
0.0084 - mae: 0.0709
Epoch 163/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0584
Epoch 164/1000
71/71 0s 4ms/step - loss:
0.0073 - mae: 0.0661
Epoch 165/1000
71/71 0s 3ms/step - loss:
0.0069 - mae: 0.0643
Epoch 166/1000
71/71 0s 3ms/step - loss:
0.0099 - mae: 0.0762
Epoch 167/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0658
Epoch 168/1000
71/71 0s 3ms/step - loss:
0.0126 - mae: 0.0868
Epoch 169/1000
71/71 0s 3ms/step - loss:
0.0094 - mae: 0.0724
Epoch 170/1000
71/71 0s 3ms/step - loss:
0.0136 - mae: 0.0826
Epoch 171/1000
71/71 0s 3ms/step - loss:
0.0117 - mae: 0.0809
Epoch 172/1000
71/71 0s 3ms/step - loss:
0.0157 - mae: 0.0974
Epoch 173/1000
71/71 0s 3ms/step - loss:
0.0115 - mae: 0.0813
Epoch 174/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0607
Epoch 175/1000
71/71 0s 4ms/step - loss:
0.0053 - mae: 0.0557
Epoch 176/1000
71/71 0s 3ms/step - loss:
0.0062 - mae: 0.0607
Epoch 177/1000
71/71 0s 3ms/step - loss:

0.0072 - mae: 0.0655
Epoch 178/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0498
Epoch 179/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0572
Epoch 180/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0604
Epoch 181/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0530
Epoch 182/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0536
Epoch 183/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0650
Epoch 184/1000
71/71 0s 3ms/step - loss:
0.0076 - mae: 0.0683
Epoch 185/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0650
Epoch 186/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0638
Epoch 187/1000
71/71 0s 3ms/step - loss:
0.0138 - mae: 0.0934
Epoch 188/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0634
Epoch 189/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0613
Epoch 190/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0552
Epoch 191/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0521
Epoch 192/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0554
Epoch 193/1000
71/71 0s 3ms/step - loss:

0.0054 - mae: 0.0568
Epoch 194/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0604
Epoch 195/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0591
Epoch 196/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0555
Epoch 197/1000
71/71 0s 4ms/step - loss:
0.0071 - mae: 0.0647
Epoch 198/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0622
Epoch 199/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0562
Epoch 200/1000
71/71 0s 3ms/step - loss:
0.0091 - mae: 0.0752
Epoch 201/1000
71/71 0s 3ms/step - loss:
0.0107 - mae: 0.0814
Epoch 202/1000
71/71 0s 3ms/step - loss:
0.0134 - mae: 0.0891
Epoch 203/1000
71/71 0s 3ms/step - loss:
0.0092 - mae: 0.0694
Epoch 204/1000
71/71 0s 3ms/step - loss:
0.0091 - mae: 0.0723
Epoch 205/1000
71/71 0s 3ms/step - loss:
0.0092 - mae: 0.0732
Epoch 206/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0523
Epoch 207/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0497
Epoch 208/1000
71/71 0s 4ms/step - loss:
0.0058 - mae: 0.0589
Epoch 209/1000
71/71 0s 3ms/step - loss:

0.0046 - mae: 0.0523
Epoch 210/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0542
Epoch 211/1000
71/71 0s 3ms/step - loss:
0.0061 - mae: 0.0615
Epoch 212/1000
71/71 0s 3ms/step - loss:
0.0073 - mae: 0.0673
Epoch 213/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0682
Epoch 214/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0611
Epoch 215/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0567
Epoch 216/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0537
Epoch 217/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0561
Epoch 218/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0533
Epoch 219/1000
71/71 0s 4ms/step - loss:
0.0042 - mae: 0.0490
Epoch 220/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0525
Epoch 221/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0550
Epoch 222/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0492
Epoch 223/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0561
Epoch 224/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0696
Epoch 225/1000
71/71 0s 3ms/step - loss:

0.0071 - mae: 0.0649
Epoch 226/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0589
Epoch 227/1000
71/71 0s 3ms/step - loss:
0.0097 - mae: 0.0701
Epoch 228/1000
71/71 0s 3ms/step - loss:
0.0101 - mae: 0.0735
Epoch 229/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0577
Epoch 230/1000
71/71 0s 4ms/step - loss:
0.0063 - mae: 0.0608
Epoch 231/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0556
Epoch 232/1000
71/71 0s 3ms/step - loss:
0.0072 - mae: 0.0666
Epoch 233/1000
71/71 0s 3ms/step - loss:
0.0107 - mae: 0.0790
Epoch 234/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0517
Epoch 235/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0468
Epoch 236/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0481
Epoch 237/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0460
Epoch 238/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0615
Epoch 239/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0812
Epoch 240/1000
71/71 0s 3ms/step - loss:
0.0092 - mae: 0.0753
Epoch 241/1000
71/71 0s 4ms/step - loss:

0.0076 - mae: 0.0679
Epoch 242/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0480
Epoch 243/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0458
Epoch 244/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0555
Epoch 245/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0539
Epoch 246/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0573
Epoch 247/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0482
Epoch 248/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0547
Epoch 249/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0511
Epoch 250/1000
71/71 0s 3ms/step - loss:
0.0063 - mae: 0.0620
Epoch 251/1000
71/71 0s 3ms/step - loss:
0.0064 - mae: 0.0627
Epoch 252/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0482
Epoch 253/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0428
Epoch 254/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0437
Epoch 255/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0469
Epoch 256/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0518
Epoch 257/1000
71/71 0s 3ms/step - loss:

0.0043 - mae: 0.0513
Epoch 258/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0645
Epoch 259/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0599
Epoch 260/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0587
Epoch 261/1000
71/71 0s 3ms/step - loss:
0.0075 - mae: 0.0674
Epoch 262/1000
71/71 0s 3ms/step - loss:
0.0067 - mae: 0.0611
Epoch 263/1000
71/71 0s 4ms/step - loss:
0.0062 - mae: 0.0618
Epoch 264/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0581
Epoch 265/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0483
Epoch 266/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0548
Epoch 267/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0515
Epoch 268/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0484
Epoch 269/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0461
Epoch 270/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0437
Epoch 271/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0495
Epoch 272/1000
71/71 0s 3ms/step - loss:
0.0082 - mae: 0.0709
Epoch 273/1000
71/71 0s 3ms/step - loss:

0.0065 - mae: 0.0624
Epoch 274/1000
71/71 0s 4ms/step - loss:
0.0054 - mae: 0.0574
Epoch 275/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0820
Epoch 276/1000
71/71 0s 3ms/step - loss:
0.0112 - mae: 0.0772
Epoch 277/1000
71/71 0s 3ms/step - loss:
0.0099 - mae: 0.0734
Epoch 278/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0643
Epoch 279/1000
71/71 0s 3ms/step - loss:
0.0080 - mae: 0.0692
Epoch 280/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0550
Epoch 281/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0433
Epoch 282/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0428
Epoch 283/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0392
Epoch 284/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0515
Epoch 285/1000
71/71 0s 4ms/step - loss:
0.0038 - mae: 0.0477
Epoch 286/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0487
Epoch 287/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0535
Epoch 288/1000
71/71 0s 3ms/step - loss:
0.0055 - mae: 0.0574
Epoch 289/1000
71/71 0s 3ms/step - loss:

0.0070 - mae: 0.0660
Epoch 290/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0479
Epoch 291/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0439
Epoch 292/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0594
Epoch 293/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0562
Epoch 294/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0492
Epoch 295/1000
71/71 0s 3ms/step - loss:
0.0051 - mae: 0.0571
Epoch 296/1000
71/71 0s 3ms/step - loss:
0.0104 - mae: 0.0805
Epoch 297/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0585
Epoch 298/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0480
Epoch 299/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0434
Epoch 300/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0573
Epoch 301/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0567
Epoch 302/1000
71/71 0s 3ms/step - loss:
0.0086 - mae: 0.0727
Epoch 303/1000
71/71 0s 3ms/step - loss:
0.0066 - mae: 0.0633
Epoch 304/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0535
Epoch 305/1000
71/71 0s 3ms/step - loss:

0.0041 - mae: 0.0500
Epoch 306/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0532
Epoch 307/1000
71/71 0s 4ms/step - loss:
0.0030 - mae: 0.0424
Epoch 308/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0501
Epoch 309/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0554
Epoch 310/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0594
Epoch 311/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0606
Epoch 312/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0460
Epoch 313/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0434
Epoch 314/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0492
Epoch 315/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0490
Epoch 316/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0473
Epoch 317/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0466
Epoch 318/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0398
Epoch 319/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0394
Epoch 320/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0464
Epoch 321/1000
71/71 0s 3ms/step - loss:

0.0037 - mae: 0.0481
Epoch 322/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0483
Epoch 323/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0486
Epoch 324/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0498
Epoch 325/1000
71/71 0s 3ms/step - loss:
0.0064 - mae: 0.0629
Epoch 326/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0565
Epoch 327/1000
71/71 0s 3ms/step - loss:
0.0043 - mae: 0.0501
Epoch 328/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0511
Epoch 329/1000
71/71 0s 4ms/step - loss:
0.0035 - mae: 0.0461
Epoch 330/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0470
Epoch 331/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0523
Epoch 332/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0526
Epoch 333/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0583
Epoch 334/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0424
Epoch 335/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0444
Epoch 336/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0484
Epoch 337/1000
71/71 0s 3ms/step - loss:

```

0.0056 - mae: 0.0577
Epoch 338/1000
71/71          0s 3ms/step - loss:
0.0045 - mae: 0.0513
Epoch 339/1000
71/71          0s 3ms/step - loss:
0.0045 - mae: 0.0515
Epoch 340/1000
71/71          0s 4ms/step - loss:
0.0037 - mae: 0.0471
Epoch 341/1000
71/71          0s 3ms/step - loss:
0.0037 - mae: 0.0479
Epoch 342/1000
71/71          0s 3ms/step - loss:
0.0048 - mae: 0.0538
Epoch 343/1000
71/71          0s 3ms/step - loss:
0.0038 - mae: 0.0474
Epoch 344/1000
71/71          0s 3ms/step - loss:
0.0042 - mae: 0.0495
Epoch 345/1000
71/71          0s 3ms/step - loss:
0.0033 - mae: 0.0449
Epoch 346/1000
71/71          0s 3ms/step - loss:
0.0030 - mae: 0.0416
Epoch 347/1000
71/71          0s 3ms/step - loss:
0.0032 - mae: 0.0437
Epoch 348/1000
71/71          0s 3ms/step - loss:
0.0038 - mae: 0.0479
Epoch 349/1000
71/71          0s 3ms/step - loss:
0.0042 - mae: 0.0510
Epoch 350/1000
71/71          0s 3ms/step - loss:
0.0036 - mae: 0.0460
Epoch 351/1000
71/71          0s 4ms/step - loss:
0.0040 - mae: 0.0501
Epoch 352/1000
71/71          0s 3ms/step - loss:
0.0040 - mae: 0.0492
Epoch 353/1000
71/71          0s 3ms/step - loss:

```

0.0065 - mae: 0.0631
Epoch 354/1000
71/71 0s 3ms/step - loss:
0.0104 - mae: 0.0779
Epoch 355/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0675
Epoch 356/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0553
Epoch 357/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0463
Epoch 358/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0417
Epoch 359/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0377
Epoch 360/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0428
Epoch 361/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0383
Epoch 362/1000
71/71 0s 4ms/step - loss:
0.0032 - mae: 0.0442
Epoch 363/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0389
Epoch 364/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0348
Epoch 365/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0353
Epoch 366/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0427
Epoch 367/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0487
Epoch 368/1000
71/71 0s 3ms/step - loss:
0.0071 - mae: 0.0661
Epoch 369/1000
71/71 0s 3ms/step - loss:

0.0047 - mae: 0.0537
Epoch 370/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0594
Epoch 371/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0517
Epoch 372/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0530
Epoch 373/1000
71/71 0s 4ms/step - loss:
0.0025 - mae: 0.0378
Epoch 374/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0514
Epoch 375/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0482
Epoch 376/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0480
Epoch 377/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0444
Epoch 378/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0495
Epoch 379/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0401
Epoch 380/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0432
Epoch 381/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0483
Epoch 382/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0427
Epoch 383/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0445
Epoch 384/1000
71/71 0s 4ms/step - loss:
0.0040 - mae: 0.0488
Epoch 385/1000
71/71 0s 3ms/step - loss:

0.0074 - mae: 0.0666
Epoch 386/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0539
Epoch 387/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0439
Epoch 388/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0365
Epoch 389/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0368
Epoch 390/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0393
Epoch 391/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0459
Epoch 392/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0393
Epoch 393/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0463
Epoch 394/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0488
Epoch 395/1000
71/71 0s 4ms/step - loss:
0.0054 - mae: 0.0556
Epoch 396/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0542
Epoch 397/1000
71/71 0s 3ms/step - loss:
0.0085 - mae: 0.0663
Epoch 398/1000
71/71 0s 3ms/step - loss:
0.0083 - mae: 0.0687
Epoch 399/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0459
Epoch 400/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0514
Epoch 401/1000
71/71 0s 3ms/step - loss:

0.0024 - mae: 0.0382
Epoch 402/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0433
Epoch 403/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0542
Epoch 404/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0410
Epoch 405/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0391
Epoch 406/1000
71/71 0s 4ms/step - loss:
0.0023 - mae: 0.0377
Epoch 407/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0299
Epoch 408/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0328
Epoch 409/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0444
Epoch 410/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0357
Epoch 411/1000
71/71 0s 3ms/step - loss:
0.0046 - mae: 0.0508
Epoch 412/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0638
Epoch 413/1000
71/71 0s 3ms/step - loss:
0.0074 - mae: 0.0669
Epoch 414/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0453
Epoch 415/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0378
Epoch 416/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0363
Epoch 417/1000
71/71 0s 4ms/step - loss:

0.0025 - mae: 0.0390
Epoch 418/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0473
Epoch 419/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0505
Epoch 420/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0439
Epoch 421/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0365
Epoch 422/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0419
Epoch 423/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0420
Epoch 424/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0407
Epoch 425/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0400
Epoch 426/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0425
Epoch 427/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0440
Epoch 428/1000
71/71 0s 4ms/step - loss:
0.0023 - mae: 0.0383
Epoch 429/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0361
Epoch 430/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0385
Epoch 431/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0370
Epoch 432/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0399
Epoch 433/1000
71/71 0s 3ms/step - loss:

0.0035 - mae: 0.0454
Epoch 434/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0516
Epoch 435/1000
71/71 0s 3ms/step - loss:
0.0047 - mae: 0.0525
Epoch 436/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0485
Epoch 437/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0480
Epoch 438/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0510
Epoch 439/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0400
Epoch 440/1000
71/71 0s 3ms/step - loss:
0.0070 - mae: 0.0644
Epoch 441/1000
71/71 0s 3ms/step - loss:
0.0052 - mae: 0.0555
Epoch 442/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0403
Epoch 443/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0381
Epoch 444/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0361
Epoch 445/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0501
Epoch 446/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0466
Epoch 447/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0427
Epoch 448/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0449
Epoch 449/1000
71/71 0s 3ms/step - loss:

0.0035 - mae: 0.0460
Epoch 450/1000
71/71 0s 4ms/step - loss:
0.0035 - mae: 0.0457
Epoch 451/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0347
Epoch 452/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0392
Epoch 453/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0356
Epoch 454/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0394
Epoch 455/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0399
Epoch 456/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0334
Epoch 457/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0308
Epoch 458/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0372
Epoch 459/1000
71/71 0s 3ms/step - loss:
0.0036 - mae: 0.0466
Epoch 460/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0436
Epoch 461/1000
71/71 0s 4ms/step - loss:
0.0059 - mae: 0.0584
Epoch 462/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0495
Epoch 463/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0497
Epoch 464/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0418
Epoch 465/1000
71/71 0s 3ms/step - loss:

0.0025 - mae: 0.0393
Epoch 466/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0327
Epoch 467/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0309
Epoch 468/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0324
Epoch 469/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0339
Epoch 470/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0359
Epoch 471/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0431
Epoch 472/1000
71/71 0s 4ms/step - loss:
0.0032 - mae: 0.0433
Epoch 473/1000
71/71 0s 3ms/step - loss:
0.0056 - mae: 0.0569
Epoch 474/1000
71/71 0s 3ms/step - loss:
0.0053 - mae: 0.0543
Epoch 475/1000
71/71 0s 3ms/step - loss:
0.0058 - mae: 0.0594
Epoch 476/1000
71/71 0s 3ms/step - loss:
0.0110 - mae: 0.0788
Epoch 477/1000
71/71 0s 3ms/step - loss:
0.0108 - mae: 0.0786
Epoch 478/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0566
Epoch 479/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0409
Epoch 480/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0442
Epoch 481/1000
71/71 0s 3ms/step - loss:

0.0025 - mae: 0.0392
Epoch 482/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0307
Epoch 483/1000
71/71 0s 4ms/step - loss:
0.0011 - mae: 0.0255
Epoch 484/1000
71/71 0s 3ms/step - loss:
9.8044e-04 - mae: 0.0237
Epoch 485/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0284
Epoch 486/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0318
Epoch 487/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0337
Epoch 488/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0299
Epoch 489/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0294
Epoch 490/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0346
Epoch 491/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0361
Epoch 492/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0482
Epoch 493/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0519
Epoch 494/1000
71/71 0s 4ms/step - loss:
0.0024 - mae: 0.0379
Epoch 495/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0399
Epoch 496/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0403
Epoch 497/1000
71/71 0s 3ms/step - loss:

0.0036 - mae: 0.0460
Epoch 498/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0360
Epoch 499/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0378
Epoch 500/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0368
Epoch 501/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0416
Epoch 502/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0409
Epoch 503/1000
71/71 0s 3ms/step - loss:
0.0057 - mae: 0.0589
Epoch 504/1000
71/71 0s 3ms/step - loss:
0.0059 - mae: 0.0601
Epoch 505/1000
71/71 0s 4ms/step - loss:
0.0035 - mae: 0.0466
Epoch 506/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0401
Epoch 507/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0352
Epoch 508/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0321
Epoch 509/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0335
Epoch 510/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0412
Epoch 511/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0398
Epoch 512/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0408
Epoch 513/1000
71/71 0s 3ms/step - loss:

0.0044 - mae: 0.0519
 Epoch 514/1000
 71/71 0s 3ms/step - loss:
 0.0032 - mae: 0.0439
 Epoch 515/1000
 71/71 0s 3ms/step - loss:
 0.0029 - mae: 0.0414
 Epoch 516/1000
 71/71 0s 4ms/step - loss:
 0.0021 - mae: 0.0356
 Epoch 517/1000
 71/71 0s 3ms/step - loss:
 0.0037 - mae: 0.0465
 Epoch 518/1000
 71/71 0s 3ms/step - loss:
 0.0029 - mae: 0.0407
 Epoch 519/1000
 71/71 0s 3ms/step - loss:
 0.0031 - mae: 0.0440
 Epoch 520/1000
 71/71 0s 3ms/step - loss:
 0.0027 - mae: 0.0405
 Epoch 521/1000
 71/71 0s 3ms/step - loss:
 0.0021 - mae: 0.0356
 Epoch 522/1000
 71/71 0s 3ms/step - loss:
 0.0017 - mae: 0.0319
 Epoch 523/1000
 71/71 0s 3ms/step - loss:
 0.0022 - mae: 0.0358
 Epoch 524/1000
 71/71 0s 3ms/step - loss:
 0.0030 - mae: 0.0432
 Epoch 525/1000
 71/71 0s 3ms/step - loss:
 0.0024 - mae: 0.0379
 Epoch 526/1000
 71/71 0s 3ms/step - loss:
 0.0016 - mae: 0.0313
 Epoch 527/1000
 71/71 0s 4ms/step - loss:
 0.0023 - mae: 0.0372
 Epoch 528/1000
 71/71 0s 3ms/step - loss:
 0.0042 - mae: 0.0512
 Epoch 529/1000
 71/71 0s 3ms/step - loss:

0.0052 - mae: 0.0572
Epoch 530/1000
71/71 0s 3ms/step - loss:
0.0034 - mae: 0.0451
Epoch 531/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0446
Epoch 532/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0393
Epoch 533/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0387
Epoch 534/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0368
Epoch 535/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0329
Epoch 536/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0390
Epoch 537/1000
71/71 0s 3ms/step - loss:
0.0030 - mae: 0.0432
Epoch 538/1000
71/71 0s 4ms/step - loss:
0.0021 - mae: 0.0354
Epoch 539/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0310
Epoch 540/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0277
Epoch 541/1000
71/71 0s 3ms/step - loss:
0.0042 - mae: 0.0498
Epoch 542/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0479
Epoch 543/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0398
Epoch 544/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0336
Epoch 545/1000
71/71 0s 3ms/step - loss:

0.0050 - mae: 0.0558
Epoch 546/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0524
Epoch 547/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0415
Epoch 548/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0358
Epoch 549/1000
71/71 0s 4ms/step - loss:
0.0022 - mae: 0.0360
Epoch 550/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0364
Epoch 551/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0355
Epoch 552/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0277
Epoch 553/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0282
Epoch 554/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0345
Epoch 555/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0281
Epoch 556/1000
71/71 0s 3ms/step - loss:
0.0041 - mae: 0.0486
Epoch 557/1000
71/71 0s 3ms/step - loss:
0.0117 - mae: 0.0817
Epoch 558/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0537
Epoch 559/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0409
Epoch 560/1000
71/71 0s 4ms/step - loss:
0.0024 - mae: 0.0388
Epoch 561/1000
71/71 0s 3ms/step - loss:

0.0025 - mae: 0.0387
Epoch 562/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0354
Epoch 563/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0397
Epoch 564/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0370
Epoch 565/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0415
Epoch 566/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0363
Epoch 567/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0362
Epoch 568/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0327
Epoch 569/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0314
Epoch 570/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0284
Epoch 571/1000
71/71 0s 4ms/step - loss:
0.0021 - mae: 0.0359
Epoch 572/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0343
Epoch 573/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0337
Epoch 574/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0362
Epoch 575/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0367
Epoch 576/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0453
Epoch 577/1000
71/71 0s 3ms/step - loss:

0.0029 - mae: 0.0421
Epoch 578/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0441
Epoch 579/1000
71/71 0s 3ms/step - loss:
0.0050 - mae: 0.0537
Epoch 580/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0478
Epoch 581/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0437
Epoch 582/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0352
Epoch 583/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0338
Epoch 584/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0292
Epoch 585/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0323
Epoch 586/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0289
Epoch 587/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0270
Epoch 588/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0274
Epoch 589/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0371
Epoch 590/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0392
Epoch 591/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0412
Epoch 592/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0353
Epoch 593/1000
71/71 0s 4ms/step - loss:

0.0029 - mae: 0.0419
Epoch 594/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0390
Epoch 595/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0427
Epoch 596/1000
71/71 0s 3ms/step - loss:
0.0068 - mae: 0.0644
Epoch 597/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0505
Epoch 598/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0370
Epoch 599/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0312
Epoch 600/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0272
Epoch 601/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0341
Epoch 602/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0317
Epoch 603/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0354
Epoch 604/1000
71/71 0s 4ms/step - loss:
0.0019 - mae: 0.0337
Epoch 605/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0306
Epoch 606/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0283
Epoch 607/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0363
Epoch 608/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0608
Epoch 609/1000
71/71 0s 3ms/step - loss:

0.0077 - mae: 0.0686
Epoch 610/1000
71/71 0s 3ms/step - loss:
0.0049 - mae: 0.0552
Epoch 611/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0470
Epoch 612/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0367
Epoch 613/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0456
Epoch 614/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0390
Epoch 615/1000
71/71 0s 4ms/step - loss:
0.0019 - mae: 0.0343
Epoch 616/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0358
Epoch 617/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0294
Epoch 618/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0331
Epoch 619/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0265
Epoch 620/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0251
Epoch 621/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0289
Epoch 622/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0368
Epoch 623/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0400
Epoch 624/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0442
Epoch 625/1000
71/71 0s 3ms/step - loss:

0.0029 - mae: 0.0416
Epoch 626/1000
71/71 0s 4ms/step - loss:
0.0029 - mae: 0.0413
Epoch 627/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0400
Epoch 628/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0364
Epoch 629/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0382
Epoch 630/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0363
Epoch 631/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0303
Epoch 632/1000
71/71 0s 3ms/step - loss:
0.0011 - mae: 0.0254
Epoch 633/1000
71/71 0s 3ms/step - loss:
0.0011 - mae: 0.0258
Epoch 634/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0283
Epoch 635/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0352
Epoch 636/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0308
Epoch 637/1000
71/71 0s 4ms/step - loss:
0.0016 - mae: 0.0310
Epoch 638/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0273
Epoch 639/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0304
Epoch 640/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0345
Epoch 641/1000
71/71 0s 3ms/step - loss:

```

0.0028 - mae: 0.0410
Epoch 642/1000
71/71          0s 3ms/step - loss:
0.0026 - mae: 0.0393
Epoch 643/1000
71/71          0s 3ms/step - loss:
0.0060 - mae: 0.0615
Epoch 644/1000
71/71          0s 3ms/step - loss:
0.0038 - mae: 0.0471
Epoch 645/1000
71/71          0s 3ms/step - loss:
0.0022 - mae: 0.0362
Epoch 646/1000
71/71          0s 3ms/step - loss:
0.0020 - mae: 0.0345
Epoch 647/1000
71/71          0s 3ms/step - loss:
0.0024 - mae: 0.0380
Epoch 648/1000
71/71          0s 4ms/step - loss:
0.0027 - mae: 0.0406
Epoch 649/1000
71/71          0s 3ms/step - loss:
0.0022 - mae: 0.0362
Epoch 650/1000
71/71          0s 3ms/step - loss:
0.0019 - mae: 0.0341
Epoch 651/1000
71/71          0s 3ms/step - loss:
0.0018 - mae: 0.0329
Epoch 652/1000
71/71          0s 3ms/step - loss:
0.0014 - mae: 0.0290
Epoch 653/1000
71/71          0s 3ms/step - loss:
0.0017 - mae: 0.0315
Epoch 654/1000
71/71          0s 3ms/step - loss:
0.0017 - mae: 0.0318
Epoch 655/1000
71/71          0s 3ms/step - loss:
0.0038 - mae: 0.0480
Epoch 656/1000
71/71          0s 3ms/step - loss:
0.0038 - mae: 0.0479
Epoch 657/1000
71/71          0s 3ms/step - loss:

```

0.0044 - mae: 0.0513
Epoch 658/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0382
Epoch 659/1000
71/71 0s 4ms/step - loss:
0.0016 - mae: 0.0303
Epoch 660/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0261
Epoch 661/1000
71/71 0s 3ms/step - loss:
0.0060 - mae: 0.0599
Epoch 662/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0443
Epoch 663/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0320
Epoch 664/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0262
Epoch 665/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0328
Epoch 666/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0341
Epoch 667/1000
71/71 0s 3ms/step - loss:
9.4968e-04 - mae: 0.0236
Epoch 668/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0271
Epoch 669/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0292
Epoch 670/1000
71/71 0s 4ms/step - loss:
0.0015 - mae: 0.0303
Epoch 671/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0270
Epoch 672/1000
71/71 0s 3ms/step - loss:
0.0033 - mae: 0.0445
Epoch 673/1000
71/71 0s 3ms/step - loss:

0.0023 - mae: 0.0372
Epoch 674/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0371
Epoch 675/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0335
Epoch 676/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0365
Epoch 677/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0333
Epoch 678/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0375
Epoch 679/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0379
Epoch 680/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0354
Epoch 681/1000
71/71 0s 4ms/step - loss:
0.0019 - mae: 0.0340
Epoch 682/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0342
Epoch 683/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0359
Epoch 684/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0373
Epoch 685/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0441
Epoch 686/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0431
Epoch 687/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0415
Epoch 688/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0385
Epoch 689/1000
71/71 0s 3ms/step - loss:

0.0015 - mae: 0.0300
Epoch 690/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0388
Epoch 691/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0317
Epoch 692/1000
71/71 0s 4ms/step - loss:
0.0015 - mae: 0.0304
Epoch 693/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0338
Epoch 694/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0317
Epoch 695/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0315
Epoch 696/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0292
Epoch 697/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0306
Epoch 698/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0563
Epoch 699/1000
71/71 0s 3ms/step - loss:
0.0054 - mae: 0.0568
Epoch 700/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0391
Epoch 701/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0406
Epoch 702/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0378
Epoch 703/1000
71/71 0s 4ms/step - loss:
0.0029 - mae: 0.0422
Epoch 704/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0318
Epoch 705/1000
71/71 0s 3ms/step - loss:

0.0018 - mae: 0.0324
Epoch 706/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0307
Epoch 707/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0265
Epoch 708/1000
71/71 0s 3ms/step - loss:
9.8951e-04 - mae: 0.0240
Epoch 709/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0284
Epoch 710/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0388
Epoch 711/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0457
Epoch 712/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0454
Epoch 713/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0348
Epoch 714/1000
71/71 0s 4ms/step - loss:
0.0029 - mae: 0.0419
Epoch 715/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0388
Epoch 716/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0327
Epoch 717/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0311
Epoch 718/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0401
Epoch 719/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0381
Epoch 720/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0293
Epoch 721/1000
71/71 0s 3ms/step - loss:

0.0015 - mae: 0.0301
Epoch 722/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0285
Epoch 723/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0303
Epoch 724/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0350
Epoch 725/1000
71/71 0s 4ms/step - loss:
0.0017 - mae: 0.0324
Epoch 726/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0391
Epoch 727/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0278
Epoch 728/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0268
Epoch 729/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0293
Epoch 730/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0295
Epoch 731/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0376
Epoch 732/1000
71/71 0s 3ms/step - loss:
0.0044 - mae: 0.0518
Epoch 733/1000
71/71 0s 3ms/step - loss:
0.0037 - mae: 0.0473
Epoch 734/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0377
Epoch 735/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0269
Epoch 736/1000
71/71 0s 4ms/step - loss:
0.0013 - mae: 0.0282
Epoch 737/1000
71/71 0s 3ms/step - loss:

0.0041 - mae: 0.0503
Epoch 738/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0382
Epoch 739/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0369
Epoch 740/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0379
Epoch 741/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0349
Epoch 742/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0330
Epoch 743/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0363
Epoch 744/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0394
Epoch 745/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0315
Epoch 746/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0267
Epoch 747/1000
71/71 0s 4ms/step - loss:
0.0011 - mae: 0.0261
Epoch 748/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0249
Epoch 749/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0269
Epoch 750/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0318
Epoch 751/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0248
Epoch 752/1000
71/71 0s 3ms/step - loss:
9.8531e-04 - mae: 0.0246
Epoch 753/1000
71/71 0s 3ms/step - loss:

```

0.0021 - mae: 0.0339
Epoch 754/1000
71/71          0s 3ms/step - loss:
0.0020 - mae: 0.0336
Epoch 755/1000
71/71          0s 3ms/step - loss:
0.0022 - mae: 0.0363
Epoch 756/1000
71/71          0s 3ms/step - loss:
0.0026 - mae: 0.0392
Epoch 757/1000
71/71          0s 3ms/step - loss:
0.0023 - mae: 0.0367
Epoch 758/1000
71/71          0s 4ms/step - loss:
0.0021 - mae: 0.0350
Epoch 759/1000
71/71          0s 3ms/step - loss:
0.0015 - mae: 0.0296
Epoch 760/1000
71/71          0s 3ms/step - loss:
0.0017 - mae: 0.0315
Epoch 761/1000
71/71          0s 3ms/step - loss:
0.0020 - mae: 0.0343
Epoch 762/1000
71/71          0s 3ms/step - loss:
0.0027 - mae: 0.0403
Epoch 763/1000
71/71          0s 3ms/step - loss:
0.0020 - mae: 0.0350
Epoch 764/1000
71/71          0s 3ms/step - loss:
0.0023 - mae: 0.0376
Epoch 765/1000
71/71          0s 3ms/step - loss:
0.0017 - mae: 0.0332
Epoch 766/1000
71/71          0s 3ms/step - loss:
0.0014 - mae: 0.0286
Epoch 767/1000
71/71          0s 3ms/step - loss:
0.0014 - mae: 0.0291
Epoch 768/1000
71/71          0s 3ms/step - loss:
0.0022 - mae: 0.0365
Epoch 769/1000
71/71          0s 4ms/step - loss:

```

0.0025 - mae: 0.0388
Epoch 770/1000
71/71 0s 3ms/step - loss:
0.0039 - mae: 0.0474
Epoch 771/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0337
Epoch 772/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0386
Epoch 773/1000
71/71 0s 3ms/step - loss:
0.0027 - mae: 0.0405
Epoch 774/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0308
Epoch 775/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0288
Epoch 776/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0358
Epoch 777/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0304
Epoch 778/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0367
Epoch 779/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0264
Epoch 780/1000
71/71 0s 4ms/step - loss:
0.0066 - mae: 0.0592
Epoch 781/1000
71/71 0s 3ms/step - loss:
0.0206 - mae: 0.1114
Epoch 782/1000
71/71 0s 3ms/step - loss:
0.0078 - mae: 0.0643
Epoch 783/1000
71/71 0s 3ms/step - loss:
0.0031 - mae: 0.0411
Epoch 784/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0309
Epoch 785/1000
71/71 0s 3ms/step - loss:

0.0012 - mae: 0.0274
 Epoch 786/1000
 71/71 0s 3ms/step - loss:
 9.8025e-04 - mae: 0.0241
 Epoch 787/1000
 71/71 0s 3ms/step - loss:
 8.1321e-04 - mae: 0.0220
 Epoch 788/1000
 71/71 0s 3ms/step - loss:
 7.2157e-04 - mae: 0.0205
 Epoch 789/1000
 71/71 0s 3ms/step - loss:
 0.0011 - mae: 0.0257
 Epoch 790/1000
 71/71 0s 3ms/step - loss:
 9.2767e-04 - mae: 0.0236
 Epoch 791/1000
 71/71 0s 4ms/step - loss:
 8.3717e-04 - mae: 0.0224
 Epoch 792/1000
 71/71 0s 3ms/step - loss:
 7.9673e-04 - mae: 0.0215
 Epoch 793/1000
 71/71 0s 3ms/step - loss:
 8.3876e-04 - mae: 0.0223
 Epoch 794/1000
 71/71 0s 3ms/step - loss:
 9.8183e-04 - mae: 0.0241
 Epoch 795/1000
 71/71 0s 3ms/step - loss:
 0.0015 - mae: 0.0293
 Epoch 796/1000
 71/71 0s 3ms/step - loss:
 0.0011 - mae: 0.0252
 Epoch 797/1000
 71/71 0s 3ms/step - loss:
 0.0010 - mae: 0.0242
 Epoch 798/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0298
 Epoch 799/1000
 71/71 0s 3ms/step - loss:
 0.0017 - mae: 0.0320
 Epoch 800/1000
 71/71 0s 3ms/step - loss:
 0.0021 - mae: 0.0358
 Epoch 801/1000
 71/71 0s 3ms/step - loss:

0.0024 - mae: 0.0376
Epoch 802/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0546
Epoch 803/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0398
Epoch 804/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0277
Epoch 805/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0282
Epoch 806/1000
71/71 0s 3ms/step - loss:
0.0011 - mae: 0.0251
Epoch 807/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0276
Epoch 808/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0245
Epoch 809/1000
71/71 0s 3ms/step - loss:
8.8176e-04 - mae: 0.0226
Epoch 810/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0247
Epoch 811/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0357
Epoch 812/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0320
Epoch 813/1000
71/71 0s 4ms/step - loss:
0.0016 - mae: 0.0305
Epoch 814/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0365
Epoch 815/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0356
Epoch 816/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0334
Epoch 817/1000
71/71 0s 3ms/step - loss:

0.0021 - mae: 0.0349
Epoch 818/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0348
Epoch 819/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0392
Epoch 820/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0274
Epoch 821/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0279
Epoch 822/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0298
Epoch 823/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0345
Epoch 824/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0378
Epoch 825/1000
71/71 0s 3ms/step - loss:
0.0038 - mae: 0.0486
Epoch 826/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0365
Epoch 827/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0297
Epoch 828/1000
71/71 0s 3ms/step - loss:
0.0011 - mae: 0.0258
Epoch 829/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0274
Epoch 830/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0266
Epoch 831/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0310
Epoch 832/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0336
Epoch 833/1000
71/71 0s 3ms/step - loss:

0.0016 - mae: 0.0316
 Epoch 834/1000
 71/71 0s 3ms/step - loss:
 0.0021 - mae: 0.0355
 Epoch 835/1000
 71/71 0s 4ms/step - loss:
 0.0022 - mae: 0.0366
 Epoch 836/1000
 71/71 0s 3ms/step - loss:
 0.0026 - mae: 0.0387
 Epoch 837/1000
 71/71 0s 3ms/step - loss:
 0.0022 - mae: 0.0359
 Epoch 838/1000
 71/71 0s 3ms/step - loss:
 0.0047 - mae: 0.0541
 Epoch 839/1000
 71/71 0s 3ms/step - loss:
 0.0033 - mae: 0.0424
 Epoch 840/1000
 71/71 0s 3ms/step - loss:
 0.0019 - mae: 0.0337
 Epoch 841/1000
 71/71 0s 3ms/step - loss:
 0.0020 - mae: 0.0350
 Epoch 842/1000
 71/71 0s 3ms/step - loss:
 0.0017 - mae: 0.0318
 Epoch 843/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0291
 Epoch 844/1000
 71/71 0s 3ms/step - loss:
 9.8622e-04 - mae: 0.0245
 Epoch 845/1000
 71/71 0s 3ms/step - loss:
 8.7258e-04 - mae: 0.0230
 Epoch 846/1000
 71/71 0s 3ms/step - loss:
 0.0019 - mae: 0.0341
 Epoch 847/1000
 71/71 0s 3ms/step - loss:
 0.0012 - mae: 0.0263
 Epoch 848/1000
 71/71 0s 3ms/step - loss:
 0.0011 - mae: 0.0263
 Epoch 849/1000
 71/71 0s 3ms/step - loss:

0.0017 - mae: 0.0325
 Epoch 850/1000
 71/71 0s 3ms/step - loss:
 0.0020 - mae: 0.0346
 Epoch 851/1000
 71/71 0s 3ms/step - loss:
 0.0019 - mae: 0.0340
 Epoch 852/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0286
 Epoch 853/1000
 71/71 0s 3ms/step - loss:
 9.7237e-04 - mae: 0.0239
 Epoch 854/1000
 71/71 0s 3ms/step - loss:
 0.0018 - mae: 0.0337
 Epoch 855/1000
 71/71 0s 3ms/step - loss:
 0.0019 - mae: 0.0342
 Epoch 856/1000
 71/71 0s 3ms/step - loss:
 0.0038 - mae: 0.0473
 Epoch 857/1000
 71/71 0s 4ms/step - loss:
 0.0023 - mae: 0.0366
 Epoch 858/1000
 71/71 0s 3ms/step - loss:
 0.0020 - mae: 0.0336
 Epoch 859/1000
 71/71 0s 3ms/step - loss:
 0.0018 - mae: 0.0322
 Epoch 860/1000
 71/71 0s 3ms/step - loss:
 0.0012 - mae: 0.0274
 Epoch 861/1000
 71/71 0s 3ms/step - loss:
 0.0015 - mae: 0.0305
 Epoch 862/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0294
 Epoch 863/1000
 71/71 0s 3ms/step - loss:
 0.0011 - mae: 0.0265
 Epoch 864/1000
 71/71 0s 3ms/step - loss:
 0.0012 - mae: 0.0262
 Epoch 865/1000
 71/71 0s 3ms/step - loss:

0.0012 - mae: 0.0273
Epoch 866/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0249
Epoch 867/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0253
Epoch 868/1000
71/71 0s 4ms/step - loss:
9.5912e-04 - mae: 0.0240
Epoch 869/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0251
Epoch 870/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0246
Epoch 871/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0286
Epoch 872/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0388
Epoch 873/1000
71/71 0s 3ms/step - loss:
0.0026 - mae: 0.0380
Epoch 874/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0315
Epoch 875/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0340
Epoch 876/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0372
Epoch 877/1000
71/71 0s 3ms/step - loss:
0.0045 - mae: 0.0525
Epoch 878/1000
71/71 0s 3ms/step - loss:
0.0029 - mae: 0.0409
Epoch 879/1000
71/71 0s 4ms/step - loss:
0.0022 - mae: 0.0347
Epoch 880/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0305
Epoch 881/1000
71/71 0s 3ms/step - loss:

0.0014 - mae: 0.0287
 Epoch 882/1000
 71/71 0s 3ms/step - loss:
 0.0023 - mae: 0.0384
 Epoch 883/1000
 71/71 0s 3ms/step - loss:
 0.0015 - mae: 0.0297
 Epoch 884/1000
 71/71 0s 3ms/step - loss:
 0.0022 - mae: 0.0367
 Epoch 885/1000
 71/71 0s 3ms/step - loss:
 0.0018 - mae: 0.0328
 Epoch 886/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0286
 Epoch 887/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0292
 Epoch 888/1000
 71/71 0s 3ms/step - loss:
 0.0016 - mae: 0.0307
 Epoch 889/1000
 71/71 0s 3ms/step - loss:
 0.0015 - mae: 0.0297
 Epoch 890/1000
 71/71 0s 4ms/step - loss:
 0.0027 - mae: 0.0397
 Epoch 891/1000
 71/71 0s 3ms/step - loss:
 0.0027 - mae: 0.0394
 Epoch 892/1000
 71/71 0s 3ms/step - loss:
 9.5641e-04 - mae: 0.0235
 Epoch 893/1000
 71/71 0s 3ms/step - loss:
 8.6118e-04 - mae: 0.0225
 Epoch 894/1000
 71/71 0s 3ms/step - loss:
 9.8993e-04 - mae: 0.0246
 Epoch 895/1000
 71/71 0s 3ms/step - loss:
 0.0011 - mae: 0.0262
 Epoch 896/1000
 71/71 0s 3ms/step - loss:
 0.0013 - mae: 0.0280
 Epoch 897/1000
 71/71 0s 3ms/step - loss:

9.3103e-04 - mae: 0.0237
Epoch 898/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0246
Epoch 899/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0283
Epoch 900/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0344
Epoch 901/1000
71/71 0s 4ms/step - loss:
0.0024 - mae: 0.0371
Epoch 902/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0291
Epoch 903/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0307
Epoch 904/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0385
Epoch 905/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0245
Epoch 906/1000
71/71 0s 3ms/step - loss:
9.2147e-04 - mae: 0.0235
Epoch 907/1000
71/71 0s 3ms/step - loss:
9.5465e-04 - mae: 0.0239
Epoch 908/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0321
Epoch 909/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0289
Epoch 910/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0308
Epoch 911/1000
71/71 0s 3ms/step - loss:
0.0035 - mae: 0.0452
Epoch 912/1000
71/71 0s 5ms/step - loss:
0.0024 - mae: 0.0376
Epoch 913/1000
71/71 0s 3ms/step - loss:

0.0018 - mae: 0.0320
Epoch 914/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0363
Epoch 915/1000
71/71 0s 3ms/step - loss:
0.0032 - mae: 0.0442
Epoch 916/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0292
Epoch 917/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0246
Epoch 918/1000
71/71 0s 3ms/step - loss:
7.4506e-04 - mae: 0.0213
Epoch 919/1000
71/71 0s 3ms/step - loss:
7.4807e-04 - mae: 0.0212
Epoch 920/1000
71/71 0s 3ms/step - loss:
6.4202e-04 - mae: 0.0196
Epoch 921/1000
71/71 0s 3ms/step - loss:
8.2805e-04 - mae: 0.0219
Epoch 922/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0359
Epoch 923/1000
71/71 0s 4ms/step - loss:
0.0014 - mae: 0.0289
Epoch 924/1000
71/71 0s 3ms/step - loss:
8.5599e-04 - mae: 0.0225
Epoch 925/1000
71/71 0s 3ms/step - loss:
9.0057e-04 - mae: 0.0230
Epoch 926/1000
71/71 0s 3ms/step - loss:
0.0024 - mae: 0.0381
Epoch 927/1000
71/71 0s 3ms/step - loss:
0.0028 - mae: 0.0415
Epoch 928/1000
71/71 0s 3ms/step - loss:
0.0048 - mae: 0.0525
Epoch 929/1000
71/71 0s 3ms/step - loss:

0.0041 - mae: 0.0490
Epoch 930/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0338
Epoch 931/1000
71/71 0s 3ms/step - loss:
0.0040 - mae: 0.0492
Epoch 932/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0322
Epoch 933/1000
71/71 0s 3ms/step - loss:
9.5594e-04 - mae: 0.0242
Epoch 934/1000
71/71 0s 4ms/step - loss:
6.6351e-04 - mae: 0.0201
Epoch 935/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0301
Epoch 936/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0332
Epoch 937/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0298
Epoch 938/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0308
Epoch 939/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0248
Epoch 940/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0328
Epoch 941/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0288
Epoch 942/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0293
Epoch 943/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0251
Epoch 944/1000
71/71 0s 3ms/step - loss:
7.5236e-04 - mae: 0.0210
Epoch 945/1000
71/71 0s 4ms/step - loss:

7.6817e-04 - mae: 0.0213
 Epoch 946/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0295
 Epoch 947/1000
 71/71 0s 3ms/step - loss:
 0.0012 - mae: 0.0274
 Epoch 948/1000
 71/71 0s 3ms/step - loss:
 0.0015 - mae: 0.0302
 Epoch 949/1000
 71/71 0s 3ms/step - loss:
 0.0018 - mae: 0.0338
 Epoch 950/1000
 71/71 0s 3ms/step - loss:
 0.0020 - mae: 0.0342
 Epoch 951/1000
 71/71 0s 3ms/step - loss:
 0.0021 - mae: 0.0360
 Epoch 952/1000
 71/71 0s 3ms/step - loss:
 0.0027 - mae: 0.0406
 Epoch 953/1000
 71/71 0s 3ms/step - loss:
 0.0024 - mae: 0.0373
 Epoch 954/1000
 71/71 0s 3ms/step - loss:
 0.0022 - mae: 0.0360
 Epoch 955/1000
 71/71 0s 3ms/step - loss:
 0.0013 - mae: 0.0279
 Epoch 956/1000
 71/71 0s 4ms/step - loss:
 0.0015 - mae: 0.0292
 Epoch 957/1000
 71/71 0s 3ms/step - loss:
 0.0014 - mae: 0.0282
 Epoch 958/1000
 71/71 0s 3ms/step - loss:
 9.7743e-04 - mae: 0.0242
 Epoch 959/1000
 71/71 0s 3ms/step - loss:
 0.0011 - mae: 0.0254
 Epoch 960/1000
 71/71 0s 3ms/step - loss:
 0.0017 - mae: 0.0328
 Epoch 961/1000
 71/71 0s 3ms/step - loss:

0.0019 - mae: 0.0342
Epoch 962/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0270
Epoch 963/1000
71/71 0s 3ms/step - loss:
0.0011 - mae: 0.0261
Epoch 964/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0273
Epoch 965/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0391
Epoch 966/1000
71/71 0s 3ms/step - loss:
0.0022 - mae: 0.0369
Epoch 967/1000
71/71 0s 3ms/step - loss:
0.0025 - mae: 0.0384
Epoch 968/1000
71/71 0s 3ms/step - loss:
0.0010 - mae: 0.0248
Epoch 969/1000
71/71 0s 3ms/step - loss:
0.0012 - mae: 0.0274
Epoch 970/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0274
Epoch 971/1000
71/71 0s 3ms/step - loss:
0.0016 - mae: 0.0315
Epoch 972/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0324
Epoch 973/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0325
Epoch 974/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0345
Epoch 975/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0286
Epoch 976/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0319
Epoch 977/1000
71/71 0s 3ms/step - loss:

0.0020 - mae: 0.0347
Epoch 978/1000
71/71 0s 3ms/step - loss:
0.0021 - mae: 0.0354
Epoch 979/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0325
Epoch 980/1000
71/71 0s 3ms/step - loss:
0.0023 - mae: 0.0367
Epoch 981/1000
71/71 0s 3ms/step - loss:
0.0014 - mae: 0.0289
Epoch 982/1000
71/71 0s 3ms/step - loss:
8.7063e-04 - mae: 0.0231
Epoch 983/1000
71/71 0s 3ms/step - loss:
0.0011 - mae: 0.0257
Epoch 984/1000
71/71 0s 3ms/step - loss:
7.6442e-04 - mae: 0.0213
Epoch 985/1000
71/71 0s 3ms/step - loss:
7.5310e-04 - mae: 0.0213
Epoch 986/1000
71/71 0s 3ms/step - loss:
0.0020 - mae: 0.0352
Epoch 987/1000
71/71 0s 3ms/step - loss:
0.0017 - mae: 0.0325
Epoch 988/1000
71/71 0s 3ms/step - loss:
0.0019 - mae: 0.0336
Epoch 989/1000
71/71 0s 4ms/step - loss:
0.0032 - mae: 0.0427
Epoch 990/1000
71/71 0s 3ms/step - loss:
0.0018 - mae: 0.0326
Epoch 991/1000
71/71 0s 3ms/step - loss:
0.0015 - mae: 0.0304
Epoch 992/1000
71/71 0s 3ms/step - loss:
0.0013 - mae: 0.0279
Epoch 993/1000
71/71 0s 3ms/step - loss:

```

0.0019 - mae: 0.0346
Epoch 994/1000
71/71          0s 3ms/step - loss:
0.0016 - mae: 0.0308
Epoch 995/1000
71/71          0s 3ms/step - loss:
0.0014 - mae: 0.0291
Epoch 996/1000
71/71          0s 3ms/step - loss:
0.0012 - mae: 0.0272
Epoch 997/1000
71/71          0s 3ms/step - loss:
0.0043 - mae: 0.0514
Epoch 998/1000
71/71          0s 3ms/step - loss:
0.0024 - mae: 0.0368
Epoch 999/1000
71/71          0s 3ms/step - loss:
9.0108e-04 - mae: 0.0229
Epoch 1000/1000
71/71          0s 4ms/step - loss:
8.6267e-04 - mae: 0.0230
Epoch 1/1000
47/57          0s 2ms/step - loss:
7.7991e-04 - mae: 0.0220

C:\Users\barre\my_new_env2\Lib\site-
packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran
out of data; interrupting training. Make sure that your dataset or generator can
generate at least `steps_per_epoch * epochs` batches. You may need to use the
`.repeat()` function when building your dataset.
  self._interrupted_warning()

57/57          0s 7ms/step - loss:
8.8728e-04 - mae: 0.0230 - val_loss: 9.1117e-04 - val_mae: 0.0239
Epoch 2/1000
57/57          0s 4ms/step - loss:
7.8951e-04 - mae: 0.0216 - val_loss: 8.3138e-04 - val_mae: 0.0218
Epoch 3/1000
57/57          0s 4ms/step - loss:
7.6965e-04 - mae: 0.0216 - val_loss: 8.4899e-04 - val_mae: 0.0225
Epoch 4/1000
57/57          0s 4ms/step - loss:
0.0012 - mae: 0.0261 - val_loss: 0.0011 - val_mae: 0.0250
Epoch 5/1000
57/57          0s 4ms/step - loss:
0.0011 - mae: 0.0261 - val_loss: 8.2354e-04 - val_mae: 0.0218
Epoch 6/1000
57/57          0s 4ms/step - loss:

```

0.0013 - mae: 0.0283 - val_loss: 7.6647e-04 - val_mae: 0.0210
 Epoch 7/1000
 57/57 0s 4ms/step - loss:
 9.1730e-04 - mae: 0.0235 - val_loss: 0.0015 - val_mae: 0.0301
 Epoch 8/1000
 57/57 0s 4ms/step - loss:
 6.0636e-04 - mae: 0.0188 - val_loss: 8.9567e-04 - val_mae: 0.0234
 Epoch 9/1000
 57/57 0s 4ms/step - loss:
 0.0013 - mae: 0.0275 - val_loss: 0.0017 - val_mae: 0.0325
 Epoch 10/1000
 57/57 0s 4ms/step - loss:
 0.0017 - mae: 0.0317 - val_loss: 9.8047e-04 - val_mae: 0.0239
 Epoch 11/1000
 57/57 0s 4ms/step - loss:
 8.4458e-04 - mae: 0.0221 - val_loss: 0.0012 - val_mae: 0.0257
 Epoch 12/1000
 57/57 0s 4ms/step - loss:
 0.0016 - mae: 0.0310 - val_loss: 0.0012 - val_mae: 0.0260
 Epoch 13/1000
 57/57 0s 4ms/step - loss:
 0.0024 - mae: 0.0387 - val_loss: 0.0016 - val_mae: 0.0308
 Epoch 14/1000
 57/57 0s 4ms/step - loss:
 0.0030 - mae: 0.0423 - val_loss: 0.0017 - val_mae: 0.0339
 Epoch 15/1000
 57/57 0s 4ms/step - loss:
 0.0029 - mae: 0.0403 - val_loss: 0.0014 - val_mae: 0.0282
 Epoch 16/1000
 57/57 0s 4ms/step - loss:
 0.0017 - mae: 0.0312 - val_loss: 0.0013 - val_mae: 0.0281
 Epoch 17/1000
 57/57 0s 4ms/step - loss:
 0.0015 - mae: 0.0299 - val_loss: 0.0051 - val_mae: 0.0645
 Epoch 18/1000
 57/57 0s 4ms/step - loss:
 0.0048 - mae: 0.0547 - val_loss: 0.0028 - val_mae: 0.0438
 Epoch 19/1000
 57/57 0s 4ms/step - loss:
 0.0025 - mae: 0.0381 - val_loss: 0.0013 - val_mae: 0.0271
 Epoch 20/1000
 57/57 0s 4ms/step - loss:
 0.0014 - mae: 0.0278 - val_loss: 0.0015 - val_mae: 0.0288
 Epoch 21/1000
 57/57 0s 4ms/step - loss:
 0.0018 - mae: 0.0320 - val_loss: 0.0018 - val_mae: 0.0350
 Epoch 22/1000
 57/57 0s 4ms/step - loss:

0.0020 - mae: 0.0350 - val_loss: 0.0013 - val_mae: 0.0275
 Epoch 23/1000
 57/57 0s 4ms/step - loss:
 0.0010 - mae: 0.0248 - val_loss: 0.0011 - val_mae: 0.0246
 Epoch 24/1000
 57/57 0s 4ms/step - loss:
 7.9992e-04 - mae: 0.0218 - val_loss: 0.0012 - val_mae: 0.0262
 Epoch 25/1000
 57/57 0s 4ms/step - loss:
 5.6860e-04 - mae: 0.0186 - val_loss: 9.2450e-04 - val_mae: 0.0225
 Epoch 26/1000
 57/57 0s 4ms/step - loss:
 6.1076e-04 - mae: 0.0190 - val_loss: 0.0013 - val_mae: 0.0283
 Epoch 27/1000
 57/57 0s 4ms/step - loss:
 6.7225e-04 - mae: 0.0200 - val_loss: 8.6861e-04 - val_mae: 0.0221
 Epoch 28/1000
 57/57 0s 4ms/step - loss:
 5.0188e-04 - mae: 0.0171 - val_loss: 0.0015 - val_mae: 0.0305
 Epoch 29/1000
 57/57 0s 4ms/step - loss:
 5.9695e-04 - mae: 0.0191 - val_loss: 9.6878e-04 - val_mae: 0.0227
 Epoch 30/1000
 57/57 0s 4ms/step - loss:
 6.9006e-04 - mae: 0.0205 - val_loss: 0.0013 - val_mae: 0.0274
 Epoch 31/1000
 57/57 0s 4ms/step - loss:
 9.8937e-04 - mae: 0.0248 - val_loss: 0.0014 - val_mae: 0.0275
 Epoch 32/1000
 57/57 0s 4ms/step - loss:
 8.8011e-04 - mae: 0.0230 - val_loss: 0.0014 - val_mae: 0.0276
 Epoch 33/1000
 57/57 0s 4ms/step - loss:
 0.0013 - mae: 0.0283 - val_loss: 0.0012 - val_mae: 0.0253
 Epoch 34/1000
 57/57 0s 4ms/step - loss:
 0.0010 - mae: 0.0248 - val_loss: 0.0017 - val_mae: 0.0333
 Epoch 35/1000
 57/57 0s 4ms/step - loss:
 0.0019 - mae: 0.0347 - val_loss: 0.0015 - val_mae: 0.0301
 Epoch 36/1000
 57/57 0s 4ms/step - loss:
 0.0011 - mae: 0.0254 - val_loss: 0.0012 - val_mae: 0.0260
 Epoch 37/1000
 57/57 0s 4ms/step - loss:
 0.0010 - mae: 0.0250 - val_loss: 0.0011 - val_mae: 0.0244
 Epoch 38/1000
 57/57 0s 4ms/step - loss:

```

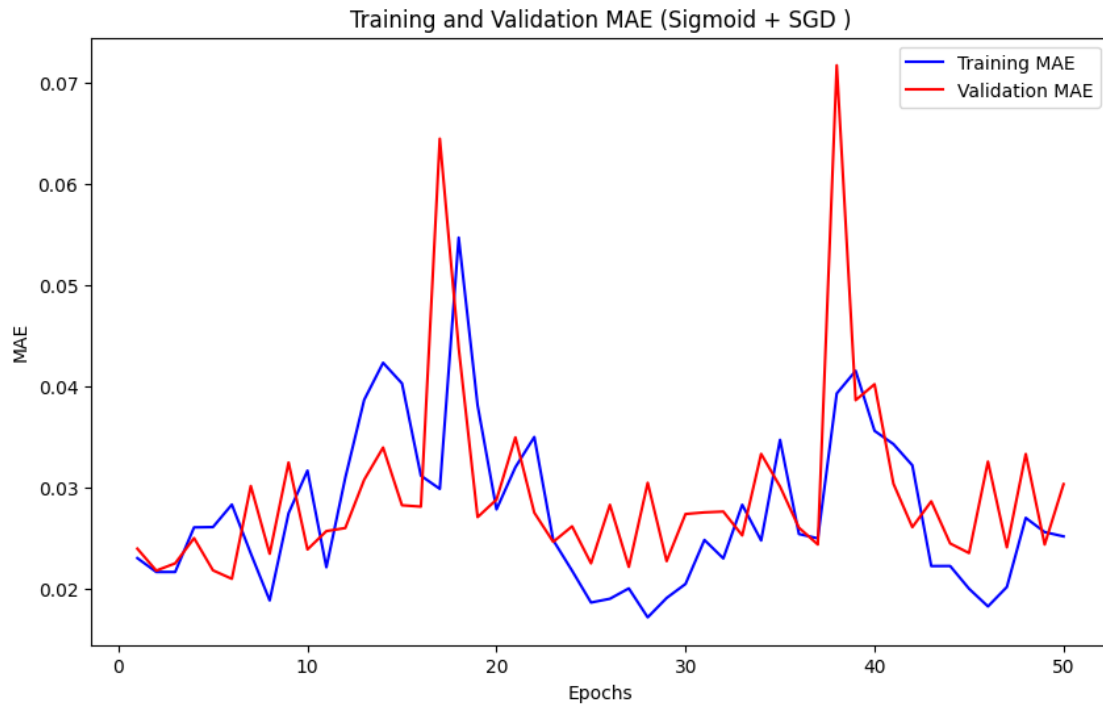
0.0028 - mae: 0.0393 - val_loss: 0.0073 - val_mae: 0.0718
Epoch 39/1000
57/57          0s 4ms/step - loss:
0.0029 - mae: 0.0415 - val_loss: 0.0022 - val_mae: 0.0386
Epoch 40/1000
57/57          0s 4ms/step - loss:
0.0022 - mae: 0.0356 - val_loss: 0.0024 - val_mae: 0.0402
Epoch 41/1000
57/57          0s 4ms/step - loss:
0.0019 - mae: 0.0343 - val_loss: 0.0015 - val_mae: 0.0304
Epoch 42/1000
57/57          0s 4ms/step - loss:
0.0016 - mae: 0.0322 - val_loss: 0.0012 - val_mae: 0.0261
Epoch 43/1000
57/57          0s 4ms/step - loss:
8.1576e-04 - mae: 0.0222 - val_loss: 0.0014 - val_mae: 0.0286
Epoch 44/1000
57/57          0s 4ms/step - loss:
8.1966e-04 - mae: 0.0222 - val_loss: 0.0011 - val_mae: 0.0245
Epoch 45/1000
57/57          0s 4ms/step - loss:
6.5776e-04 - mae: 0.0200 - val_loss: 0.0010 - val_mae: 0.0235
Epoch 46/1000
57/57          0s 4ms/step - loss:
5.5843e-04 - mae: 0.0182 - val_loss: 0.0018 - val_mae: 0.0326
Epoch 47/1000
57/57          0s 4ms/step - loss:
6.4755e-04 - mae: 0.0202 - val_loss: 0.0011 - val_mae: 0.0241
Epoch 48/1000
57/57          0s 4ms/step - loss:
0.0011 - mae: 0.0270 - val_loss: 0.0017 - val_mae: 0.0333
Epoch 49/1000
57/57          0s 4ms/step - loss:
0.0011 - mae: 0.0256 - val_loss: 0.0011 - val_mae: 0.0244
Epoch 50/1000
57/57          0s 4ms/step - loss:
0.0011 - mae: 0.0252 - val_loss: 0.0015 - val_mae: 0.0303
Epoch 50: early stopping
Restoring model weights from the end of the best epoch: 1.
Raw evaluate results: []

```

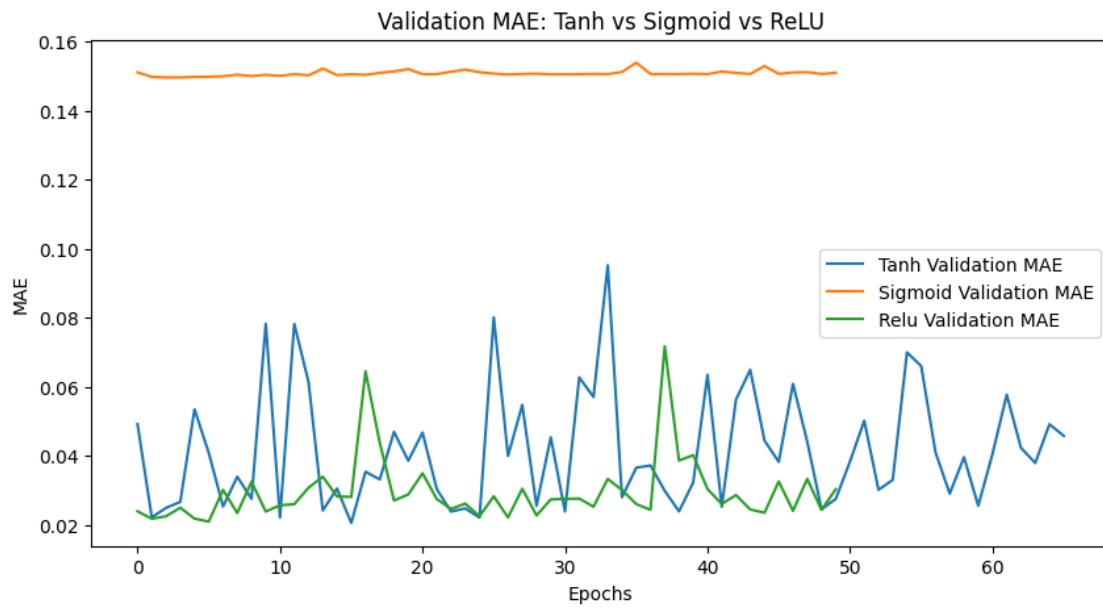
```

C:\Users\barre\my_new_env2\Lib\site-
packages\keras\src\trainers\epoch_iterator.py:164: UserWarning: Your input ran
out of data; interrupting training. Make sure that your dataset or generator can
generate at least `steps_per_epoch * epochs` batches. You may need to use the
`.repeat()` function when building your dataset.
  self._interrupted_warning()

```



```
[8]: plt.figure(figsize=(10,5))
plt.plot(history_tanh.history['val_mae'], label="Tanh Validation MAE")
plt.plot(history_sig.history['val_mae'], label="Sigmoid Validation MAE")
plt.plot(history_relu.history['val_mae'], label="Relu Validation MAE")
plt.title("Validation MAE: Tanh vs Sigmoid vs ReLU")
plt.xlabel("Epochs")
plt.ylabel("MAE")
plt.legend()
plt.show()
```

[]:

[]: